TFPT — Research Contracts for the Remaining Interfaces:

$v_{\rm geo}$, $G_{\rm net}$, $F_{\rm transfer}$

($U_{\rm wall}$): the parabolic flavor wall-selection and ($G_{\rm metric}$): the full quantum-gravity measure

Stefan Hamann Alessandro Rizzo

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The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\rm car} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\rm geo}, G_{\rm net}, F_{\rm transfer}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

You are reading companion R (Research Contracts).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1, 1)\rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Gravity is parameter-free. The classical metric-sector field equation is no longer only an $R+R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2| 2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit v_{geo}; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) is now *closed modulo a cited theorem*. The explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD = 1 (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package (h_s=16/16=1∈$\mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM μ=4/2^2=1; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed′` (`v469`); so this is *closed modulo a cited theorem*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76`/`v330`). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus F_{\text{transfer}}$, with the seam G_{net} closed modulo a cited theorem and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\text{car}} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1,1,2)$: $e_1, e_2, e_3 = 4,5,2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

↓

Physical bridges: v_{geo}, G_{net}, F_{transfer} *([C] bridges)*

↓

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

What this document is about

After the compiler closure the entire residual is

$$\boxed{\text{Rest} = (U_{\text{wall}}) \ \oplus \ (G_{\text{metric}}) \ \oplus \ (F_{\text{frontier}})}$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today. (F_{frontier}) (Koide, η_B, axion relic scale, m_p/m_e) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:** (U_{wall}) *first* (finite, algebraic, falsifiable), then (G_{metric}) (deep analytic programme).

Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity). The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\boxed{\text{Rest}_{\text{operativ}} = v_{\text{geo}} \ \oplus \ G_{\text{net}} \ \oplus \ F_{\text{transfer}}}$$

— $(U_{\text{wall}}) \to v_{\text{geo}}$: the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71), U_{point} collapses to the *one* dimensionful unit v_{geo} shared with $1/G$ (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where U_{point} still appears); $(G_{\text{metric}}) \to G_{\text{net}}$: holographically reduced to the single boundary-net statement *"the seam–Calderón inclusion has Jones index $4 = |\mu_4|$"* — holomorphy, $(E_8)_1$ and the unique 2D bulk then *follow* (v89/v92/`GATE.METRIC.05-07`); and the carrier-side QBL input now merges into the *same* statement: the tower carrier $\to SO(16)_1 \to (E_8)_1$ carries one and the same 16-fermion content, whose single quasi-free kernel (rank $= c$; Wick/Pfaffian exact) is the certified Calderón datum — closing G_{net} therefore also closes the carrier choice [E] ([`verification/v113_quasifree_kernel.py`]); $(F_{\text{frontier}}) \to F_{\text{transfer}}$: named QFT/cosmology *transfer* interfaces, never compiler powers — and *not* four unrelated interfaces but *one* discrete geodesic on the Sheet Diamond (the $K{\to}C{\to}F$ sheet axis $M(1,t)$, with the Koide source→pole as its 1-D Möbius projection), along which the external continuous solver merely evaluates ([`verification/v224_diamond_ftransfer_path.py`]).

The residual is certification, not construction ([`verification/v384_residual_certification.py`]). A ledger-CI audit classifies every remaining item into exactly three *external* kinds, with *zero* open *internal* TFPT mechanisms: [E] (A) an external math proof — `SEAM.EQUIV.01` continuum existence (the cited MMST theorem plus the crossed-product extension package, v336/v469, with AGT/AMT as second witness), `ALPHA.QUILLEN.EXACT.01` (v382, its level and normalisation sharpened by v470, the determinant-line/moduli bridge lemma exhibited at the finite level by v472 — the continuum ζ-det identification stays the open part) and the constructive `QG.AMB.01` measure (a [C] redundancy, v369); [E] (B) theorem-forbidden — v_{geo}, the one unit, by the No-Unit theorem (v153/v364); [E] (C) external physics — F_{transfer} (v371–v374, firewalled v187). The two load-bearing facts are checked (No-Unit dimensionlessness; the gap-decoupling margin $1.648 > 0$), so the only hard things left are theorems and external physics, not missing dynamics (`RESIDUAL.CERTIFICATION.01`).

The Coxeter-clock facet symmetry, resolved as a structural lemma (`RES.COXETER.SYMMETRY.01`, [`verification/v394_coxeter_atoms.py`]/[`verification/v409_coxeter_prime2_lemma.py`]). The order-30 clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$ has all three primes facet-typed, and they are exactly the three anchor atoms ($2{=}|Z_2|$, $3{=}N_{\mathrm{fam}}$, $5{=}g_{\mathrm{car}}$) wearing number-field hats ($\mathbb{Q}(i)/\mathbb{Q}(\sqrt{-3})/\mathbb{Q}(\sqrt{5})$, v390). The table is *not* symmetric: the seam (prime-2) appears as the *common factor* of both dynamic rates ($2/3 = |Z_2|/N_{\mathrm{fam}}$ and $(\varphi{+}2)/4 = (\varphi{+}2)/|\mu_4|$, $|\mu_4|{=}2^2$), not as an independent attractor. *The structural lemma (v409) closes the falsifiable corollary:* the sheet involution $\sigma{=}\operatorname{diag}(1,-1,-1)$ has spectrum $\{+1,-1\}$ and projector $(I{+}\sigma)/2$ spectrum $\{0,1\}$ — none in the open interval $(0,1)$, so prime-2 supplies a *parity/projection*, never a nonzero contraction rate; the genuine rates $2/3$ and $(1/3)^6$ lie in $(0,1)$ and carry prime-2 only as a *factor*. Hence prime-2 is *necessary* (admissible boundary transport is sheet-paired) but *not autonomous*: **no prime-2-only attractor exists** — dynamics begins only after coupling to family prime-3 or carrier prime-5 ($|\mu_4|{=}2^2$ the Galois bridge). This closes the corollary [E] and types the necessity [C], sharpening v383's "every sector is the same gapped operator" towards "the same *seam-driven* operator"; the full "appears in every gap" universality stays the v383 reading. The corollary is **machine-proved in Lean 4** (`TfptCarrier/CoxeterPrime2.lean`, `FORM.COXETER.PRIME2.01`: axiom-pure, `lake build` clean). [E]/[C].

QFT layer (this round). The boundary QFT is now *one relative object* $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$, closed modulo the *single* keystone `SEAM.EQUIV.01` (whose downstream conformal-deck face is `QGEO.SYM.01`): its finite Dirac is a covariance induction of the seam KMS state (v258), its spectral-action cutoff *is* that KMS weight (v259), its seam, carrier-16 and E_8 live on one Kummer/K3 surface (v260), and the whole assembly is certified as the *Modular Spectral Closure* (v261). The ambient quantum-gravity measure (G_{metric}) is gap-decoupled and kept separate by design — the QFT layer does not wait on it.

Research Contract 1 — (U_{wall})

Goal. Prove

$$\nabla^\star_F = \mathrm{Sel}_{D_4,\,\mathrm{cusp},\,\det R=8,\,\mathrm{Spec}(Q_+)=\{1,2,3\}}\big(\mathcal{M}_{\mathrm{par}}(\mathbb{P}^1, \mu_4, E, \alpha)\big),$$

with $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$ and $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$ at the four points of μ_4. The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed local monodromies (Mehta–Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one* D_4-symmetric, rank-3, four-point point that realises the selectors.

Lemma 1 (U0 — normalisation)**.** *The gate data are equivalent to the $SU(3)$ character variety $\mathcal{X}_{\mathrm{cusp}} = \{(M_1,\dots,M_4) \in C^4_{\mathrm{cusp}} : M_1M_2M_3M_4 = \mathbf{1}\}/SU(3)$ with $C_{\mathrm{cusp}} = \mathrm{Ad}_{SU(3)}\operatorname{diag}(1,\omega,\omega^2)$, $\omega = e^{2\pi i/3}$, and $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1,\dots,\gamma_4 \mid \prod \gamma_i = 1\rangle$. To show: $\mathcal{M}^{D_4}_{\mathrm{par}} \simeq \mathcal{X}^{D_4}_{\mathrm{cusp}}$ as polystable Mehta–Seshadri / tame Hodge realisation.* Machine: the group presentation and D_4-action are finite and codable (Sage/Wolfram).

Finite rigidity — now proven [E]. The finite half of U0 is closed exactly: μ_4 has cross-ratio 2 and Möbius stabiliser $\langle z \mapsto iz,\ z \mapsto 1/z\rangle$ of order $8 = |D_4|$ (a 4-cycle of the marks + a reflection, faithful); $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\mathrm{fam}} = 3$; and the forms $\omega_k = z^{k-1}dz/(z^4-1)$, $k = 1,2,3$, are μ_4-eigenforms ($z \mapsto iz :\ \omega_k \mapsto i^k\omega_k$) carrying the three nontrivial characters χ_1, χ_2, χ_3 with weights $(1,2,3) = \mathrm{Spec}(Q_+)$. Hence a genus-zero boundary with faithful D_4, four parabolic marks and this μ_4-decomposition is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$ with the $(1,2,3)$ grading (`QGEO.RIGID.01`, [E], [`verification/v168_qgeo_rigidity.py`]). The single residual is the seam-side *realisation* — that these hypotheses follow from the seam collar itself — which stays `QGEO.REALIZE.01` ([C]).

Seam Collar Realisation Theorem — the one central proof obligation [E]/[O] ([`verification/v176_seam_collar_realisation.py`])

Stated as one theorem and certified as a reduction — not a fabricated proof.

Theorem (target). *Given* (1) an oriented one-sided RP seam collar Σ; (2) a sheet-odd Calderón involution; (3) quasi-free one-particle seam data with gap $(2/3)^6$; (4) a faithful D_4 action on the boundary marks; (5) four parabolic marked points with μ_4-character decomposition; (6) an admissible $c{=}8$ boundary net — *then* the conformal boundary of Σ is Möbius-equivalent to $\mathbb{P}^1 \setminus \mu_4$, its H^1 character basis is canonically the generation basis with grading $(1,2,3)$, the seam Calderón contraction is the rank-8 K_Σ polarisation, and the index-4 simple-current extension is $(E_8)_1$.

The proof decomposes into six steps, five already [E]:

1–2 *Geometry / uniformisation* [E] (v168): μ_4 has cross-ratio 2 and a faithful Möbius D_4 stabiliser of order 8; four genus-0 marks with faithful D_4 force the μ_4 square up to Möbius; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4-1 = N_{\mathrm{fam}} = 3$.

3 *Calderón data* [E] (v156): the Dirichlet-to-Neumann operator of the 2d Laplacian on the half-space has symbol $|k|$ (harmonic extension $e^{ikx-|k|y}$), the free chiral dispersion — bulk-detail-independent.

4 *H^1 character = generation grading* [E] (v137/v168): $\omega_k = z^{k-1}dz/(z^4-1)$ carry μ_4 characters i^k with weights $(1,2,3)$ = the A_3 exponents = $\mathrm{Spec}(Q_+)$ — natural, not merely isomorphic.

5 *One-particle contraction* [E] (v162): a μ_4-equivariant operator is diagonal in the cusp basis ($\frac{1}{|\mu_4|}\sum_j U^j X U^{-j} = \mathrm{diag}\, X$ for $U = \mathrm{diag}(1,i,-1,-i)$); its spectrum is the cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}$, so the transfer eigenvalues $(1-\alpha)^{p_2}$ (p_2=6) are $\{1,(2/3)^6,(1/3)^6\}$, gap $(2/3)^6$ — forced, no free knob.

6 *AQFT closure* [E] (v154/v175): $(D_5)_1 \otimes (A_3)_1$ with the isotropic μ_4 glue has index $4 = |\mu_4|$, $c = 5+3 = 8$, μ-index $16/16 = 1 \Rightarrow$ holomorphic $\Rightarrow (E_8)_1$ ($248 = 120+128$); full-cone RP holds for all m via the CAR second-quantisation functor.

The proof tree — and the *two* genuine open doors [O] ([`verification/v177_seam_marking_kernel.py`]). Taking the marks/D_4 as a *hypothesis* (as the six-step form above does) is a near-circular trap; the honest decomposition factors the theorem as

$$\textsc{Realize} = \textsc{Marks} \cdot \textsc{Uniform} \cdot \textsc{Cohom} \cdot \textsc{Module} \cdot \textsc{Kernel} \cdot \textsc{Net},$$

of which *four nodes are now closed constructively*: Uniform [E] (an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4, and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 — so a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P}^1, \mu_4)$, strengthening v168's cross-ratio to the full uniformisation — and now **Lean-formalised** as `FORM.QGEO.02`: ρ^4=id, $\rho^2 \neq$id, σ^2=id, $\sigma\rho\sigma = \rho^{-1}$, the orbit scaling to μ_4, the μ_4 permutation, and the multiplier classification "order-4 $\Leftrightarrow \zeta = \pm i$" are machine-proved in `TfptCarrier/MobiusUniformisation.lean`, audit: pass); Cohom [E] ($\rho^*\omega_k = i^k\omega_k$, grading $(1,2,3)$ — also **Lean-formalised** as `FORM.QGEO.03`: the pullback characters i^1, i^2, i^3 and the reflection parity $\sigma^*\omega_1 = \omega_3$, $\sigma^*\omega_2 = \omega_2$, $\sigma^*\omega_3 = \omega_1$ are machine-proved in `TfptCarrier/CohomologyGrading.lean`, audit: pass); Module [E] (a *unique* μ_4-equivariant iso — multiplicity-free + residue normalisation, reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed; the parity is the Lean `FORM.QGEO.03` above, the *uniqueness* stays symbolic); and Net [E] (v154/v175). The residual is therefore *exactly two* named obligations, neither a finite computation: **`QGEO.MARKS.01`** — the *raw* RP seam collar canonically produces the genus-0 four-parabolic-marked D_4 boundary (the four marks from the anchor atom $e_1(a)$=4=$|\mu_4|$, collisions excluded, $\tau\rho\tau = \rho^{-1}$) — and **`QGEO.KERNEL.01`** — the *raw* seam Calderón operator equals the μ_4-equivariant free c=8 gapped one-particle contraction *as an operator* ($C_\Sigma = U^{-1}C_{\mu_4}U$), not merely the same spectral list. **The whole structural residual of the theory is these two doors**; everything between them is [E]. They need a human constructive-geometry/AQFT argument (or a proof assistant), not more arithmetic.

How far each door opens by computation ([`verification/v178_marks_kernel_attempt.py`]). Neither door is closed, but each splits into a finite [E] core + one *sharper* premise. *For* Marks*: given* a genus-0 double with an order-4 clock ρ (fixed points $\{0,\infty\}$) and reflection τ, the marks are forced to be a generic ρ-orbit $\{1,i,-1,-i\} = \mu_4$ (not the fixed points), distinct iff $b_1 = 3 = N_{\mathrm{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving faithful D_4 — so Marks reduces to the *single* topological premise "the seam double is genus-0 with an order-4 clock whose marks form one orbit". *For* Kernel*:* on the 3-dim multiplicity-free H^1 the μ_4 characters $(1,2,3)$ are distinct, so by Schur a μ_4-equivariant operator is forced *diagonal* and is *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ and the unique residue-normalised basis are *equal as operators*, not merely isospectral; the "spectrum vs operator" distinction *vanishes* on the finite block. So Kernel reduces to the *single* premise "the full DtN on L^2 reduces to this 3-dim gapped transfer block + the fixed rank-8 polarisation" (principal symbol $|k|$, Lee–Uhlmann v156; no drift v158). The two doors are thereby *milder*, but still open.

Marks without a geometric posit: a Lefschetz/character argument ([`verification/v195_marks_lefschetz_character.py`]). The marks need not be *assumed* as D_4 boundary points: they follow from the H^1 character alone. The order-4 clock ρ:$z \mapsto iz$ fixes exactly

$\{0,\infty\}$, so a puncture at a fixed point would contribute a *trivial* character to H^1; but the carrier datum is $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents $(1,2,3)$, with $\mathrm{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$), which has *no* trivial component — so no puncture sits at a fixed point. With rank $H^1 = n - 1 = 3 \Rightarrow n = 4$ punctures, all off the fixed locus and each in a free order-4 orbit, the only possibility is a *single* free μ_4 orbit (cross-ratio 2), and the closed-surface Lefschetz number $L(\rho) = 1 + 1 = 2 = |\{0,\infty\}|$ is consistent. So MARKS reduces from the geometric posit "D_4 marks" to the milder *representation* premise "the raw seam H^1 carries the A_3-exponent rep" — a carrier-algebra statement, strictly less circular, still [O].

And the two doors are in fact ONE (`[verification/v179_conformal_realisation.py]`**).** MARKS's residual is not even three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2|\, 2\pi\, \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$, and $\chi = 2 - 2g \Rightarrow g = 0$, so "genus-0" is the *same* $\chi = 2$ that supplies the "8" in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — it is the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* — the parabolic marks are not the two elliptic fixed points $\{0,\infty\}$, and a free order-4 orbit has exactly $4 = e_1(a)$ points. So MARKS reduces to "the carrier clock acts *conformally* on the genus-0 double", and KERNEL then *follows* from that same geometry (Lee–Uhlmann $|k|$ + the $c = 8$ rigidity v158 + the cusped-surface spectral split, residual $= H^1 = \mathbb{C}^3 = N_{\mathrm{fam}}$). **Hence the entire structural residual of the theory is a single geometric premise** `QGEO.CONF.01`: *the raw RP seam normal double is conformally* $(\mathbb{P}^1, \mu_4)$ *with the carrier clock as the order-4 Möbius automorphism* $z \mapsto iz$. Given it, everything — genus-0, the four D_4 marks, the $(1,2,3)$ grading, the DtN block, the rank-8 polarisation and the $(E_8)_1$ net — follows. This single premise is a constructive-geometry statement (the conformal class of the raw seam *is* the cusped sphere), *not* a finite computation, and is left honestly [O].

And even that conformal premise reduces to a milder isometry premise (`[verification/v180_clock_is_mobius.py]`**).** `QGEO.CONF.01` need not be assumed as a conformal-class identity: by *uniformisation* (Poincaré–Koebe) the genus-0 double (P1) with its RP-metric conformal structure *is* $\mathbb{P}^1$ — the target is *produced*, not assumed — and by *Kerékjártó* (1919; Constantin–Kolev 2003) a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)$=4) is topologically $z \mapsto iz$. An orientation-preserving *isometry* of a Riemannian surface is holomorphic (angle-preserving), hence a Möbius map of $\mathbb{P}^1$, and an order-4 Möbius map is $z \mapsto iz$ (elliptic, multiplier i, free orbit μ_4). So `QGEO.CONF.01` is *implied* by the strictly milder, non-circular premise `QGEO.ISO.01`: "the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric" — which does *not* name $\mathbb{P}^1/\mu_4$ and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). **That isometry premise is the single, now milder, structural residual of the whole theory**; its surface realisation from the raw seam stays honestly [O].

And one step further reaches bedrock (`[verification/v181_clock_is_conformal_symmetry.py]`**).** By *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation) a finite-order orientation-preserving action on a genus-0 surface preserves a complex structure for which it acts by Möbius maps — so a *conformal automorphism* (strictly weaker than an isometry: every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), i.e. the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure *by definition*, the residual collapses to the *definitional* identification `QGEO.SYM.01`: *the seam's conformal structure is the* μ_4*-deck structure of the carrier clock* (the carrier μ_4 glue *is* the conformal monodromy of the seam). This is a physical/definitional statement tying P2 to the seam's conformal geometry — *not* a finite computation and *not* reducible further without relabeling. **This is bedrock:** the chain `REALIZE` $\to$ theorem $\to$ MARKS+KERNEL $\to$ `CONF` $\to$ `ISO` $\to$ `SYM` (v175–v181) ends at a single definitional premise, left honestly [O]; everything above it is a theorem or an established citation.

A sharper, operator-checkable restatement: `QGEO.ENERGY.01` **(** `[verification/v192_qgeo_energy_clock.py]`**).** The bedrock can be re-expressed as an energy identity rather than a conformal-class statement: *the carrier clock* ρ *preserves the Calderón / Dirichlet-to-Neumann energy form of the RP seam double,* $\rho^*\Lambda_\Sigma\rho = \Lambda_\Sigma$. In two dimensions Dirichlet-energy invariance *is* conformal invariance, so this feeds the same chain (conformality $\to$ Möbius $\to z \mapsto iz \to \mu_4$). Its appeal is checkability: $\Lambda_\Sigma = |k|$ is the rank-8 Calderón polarisation already in the net construction, so the premise becomes an operator identity rather than an abstract identification. **Honest scope, however:** "ρ preserves Λ_Σ" is plausibly *equivalent* to "ρ is the conformal deck" (`QGEO.SYM.01`), so `QGEO.ENERGY.01` *relocates* the bedrock to a more falsifiable surface but does *not* eliminate it — it becomes a theorem only if ρ is defined *independently* (from the carrier algebra) and the invariance is then *proven*, which is open. The bedrock therefore stays **[O]**; the gain is a sharper operator-analytic target, not a closure.

The non-circular sharpening: `QGEO.ENERGY.02` ([`verification/v193_qgeo_energy_commutator.py`]). The way to make "ρ defined independently" precise is the *energy commutator* $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims: (1) ρ is the Coxeter element of $W(A_3) = S_4$ (order $4 = |\mu_4|$, v117), defined from the *carrier* algebra, not the seam; (2) *the non-circularity hinge* — Λ_Σ must be the DtN of the *raw* RP seam (definable from the RP data alone), not the μ_4 normal-form operator; (3) $[\rho, \Lambda_\Sigma] = 0$ forces the μ_4 character grading $(1, 2, 3)$ on H^1, already established (`QGEO.COHOM.01`). An exact consistency rider pins the direction: the induced EH coefficient $k = (\ln m)/(12\pi)$ (v150) reproduces $c_3/2$ *iff* $\ln m = q(A_3) = \frac{3}{4}$ (the *family* glue norm), while $q(D_5) = \frac{5}{4}$ (the carrier norm) misses by $g_{\text{car}}/N_{\text{fam}} = \frac{5}{3}$ — so the seam energy form must reproduce the *family* norm (gravity is family-geometry-induced). If sub-claim (2) holds, $[\rho, \Lambda_\Sigma] = 0$ turns `QGEO.SYM.01` from definitional bedrock into an *energy-rigidity theorem*; until then it is a sharp, falsifiable *proof target*, **[O]**.

Subclaim (2) tackled — one notch closer, not closed ([`verification/v194_raw_seam_dtn_rp.py`]). The non-circularity hinge is essentially *met* at the canonical level: Osterwalder–Schrader reconstruction yields a canonical transfer operator $T = e^{-H}$ from the RP state and reflection alone — no coordinates (v54; §(4) of `tfpt_2`) — on a *quasi-free* seam state (v155: the boundary marginal of a Gaussian bulk is the compression $P\Gamma P$, a contraction). So Λ_Σ is RP-definable without ever naming $\mathbb{P}^1 \setminus \mu_4$. The commutator $[\rho, \Lambda_\Sigma] = 0$ then *closes on the finite cohomology* H^1 **[E]**: $\rho : z \mapsto iz$ acts on $\langle w_1, w_2, w_3 \rangle$ ($w_k = z^{k-1}dz/(z^4 - 1)$) as $\rho^* w_k = i^k w_k$, distinct characters $i, -1, -i$, so any ρ-equivariant operator is diagonal there (v177). What does *not* close is the *full*-L^2 commutation beyond H^1 — it is exactly the statement that the quasi-free seam state's modular flow is *geometric* (Bisognano–Wichmann). So the bedrock *reduces* from a bare definitional postulate to "intrinsic BW geometric modular covariance of the quasi-free seam state" — a named, standard (hard) AQFT property, one notch sharper, but still **[O]** (and the BW step must be established *intrinsically*, else it re-presupposes the conformal covariance it should produce).

A concrete variational handle on the residual ([`verification/v196_seam_energy_functional.py`]). The open BW statement can be made a discretisable extremum problem via the failure functional $E_{\text{fail}}(\Sigma, \rho) = \|[\rho, \Lambda_\Sigma]\|^2_{HS} + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, whose zero locus is exactly the `QGEO.SYM.01` conditions (the energy commutator, the order-4 clock, and the RP-reflection twist). On the finite H^1 block $E_{\text{fail}} = 0$ at the μ_4 configuration ($\rho = \text{diag}(i, -1, -i)$, Λ μ_4-equivariant, Θ = conjugation) and > 0 for any μ_4-breaking perturbation, so μ_4 is a genuine minimiser **[E]**; minimising E_{fail} over the *full* raw-seam DtN is precisely the BW residual above. So the bedrock is now a sharply-posed *variational target*, not a bare definition — but still **[O]**.

The decisive crack: principal symbol exact + Tomita–Takesaki ([`verification/v198_modular_commutator_reduction.py`]). Two observations collapse the residual to a single non-circular state-invariance. *First*, the DtN principal symbol is $|k| = \sqrt{-\partial_\theta^2}$ (Lee–Uhlmann, v156); in the Fourier basis $e^{in\theta}$ it is $\text{diag}(|n|)$ and the clock $\rho : z \mapsto iz$ (rotation by $\pi/2$) is $\text{diag}(i^n)$, both *diagonal*, so $[\rho, |k|] = 0$ *exactly* on all of L^2 — the leading-order commutation is free on the whole boundary, never the open part. *Second*, by Tomita–Takesaki the modular operator Δ_ω is intrinsic to (M, Ω), so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with Δ and hence with $\log \Delta \sim \Lambda_\Sigma$: thus $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$ — using only the state and reflection (which RP supplies) and *no* conformal covariance, which **removes the BW circularity worry above**. Since the finite H^1 block is already μ_4-invariant (v177), the entire residual is now the single *state*-invariance "the raw quasi-free seam state is μ_4-invariant" ($\omega \circ \rho = \omega$, equivalently the Gaussian bulk measure is deck-invariant) — the maximally-operational, checkable form of `QGEO.SYM.01`. A foundational symmetry postulate cannot be derived from nothing (the role c = const plays in relativity); it is now in its sharpest, non-circular, falsifiable form, **[O]**.

The residual, fully localised and numerically probed ([`verification/v199_seam_state_invariance.py`], [`verification/v200_seam_variational_scan.py`]). Attacking $\omega \circ \rho = \omega$ on the *raw* seam (not the μ_4 normal form) confirms it is non-circular — ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$), an automorphism of the raw CAR algebra, defined entirely carrier-side — and reduces it, via the quasi-free covariance $C_\Sigma = \frac{1}{2}(1 + \text{sgn}\, H_1)$, to $[\rho, H_1] = 0$, i.e. H_1 *is block-diagonal in the four* μ_4*-character classes* $\{n \equiv r \bmod 4\}$. The principal symbol $|k| = \text{diag}(|n|)$ passes automatically, so the entire residual is the single bounded statement "the sub-principal symbol of the raw DtN has no off-character matrix elements". The companion variational scan minimises $E_{\text{fail}}(\theta) = \|\rho(\theta)^4 - 1\|^2 + (\text{faithfulness})$ over a free clock angle and finds its *unique* minimum at $\theta = \pi/2$ —

the μ_4 clock $z \mapsto iz$ (the only faithful order-4 grading $i, -1, -i$) — a numerical sign that the variational principle selects μ_4. This is the "next executable theorem" a working mathematician can take up; it does not close the postulate, **[O]**.

The bounded sub-principal residual is the image of the μ_4 marks, not a free postulate ([verification/v201_seam_subprincipal_marks.py]). One genuine step further: the seam DtN is $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature function $f(\theta)$ on the seam circle (the standard Steklov/DtN structure), so $\langle n|M_f|n'\rangle = f_{n-n'}$. Then M_f is μ_4-character-block-diagonal *iff* f has Fourier support only on modes $\equiv 0 \pmod 4$, i.e. iff f is $\mathbb{Z}_4$-invariant (the principal $|k|$ already commutes, v198). But a curvature sourced by the four marks, $f(\theta) = \sum_{j=0}^{3} g(\theta - 2\pi j/4)$ for *any* profile g, has $f_m = 4\, g_m\, [m \equiv 0 \bmod 4]$ — automatically $\mathbb{Z}_4$-invariant. So the μ_4-mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 equally-spaced marks ($\mathbb{Z}_3$) or 4 generic marks both break it (negative controls). The residual of $\omega \circ \rho = \omega$ thus collapses to "the DtN sub-principal symbol is *mark-local* (the seam is flat away from the μ_4 marks = the conformal-deck structure, v177)" — a locality/flatness property, structurally definitional, the narrowest form yet of `QGEO.SYM.01`, **[O]**.

Certified on realistic Steklov profiles, and at the state level ([verification/v210_mark_local_dtn.py]). The mode-level argument above is carried to concrete, smooth boundary curvature: a real von-Mises bump $g(\theta)$ localised at a puncture, summed over the four marks $f(\theta) = \sum_{j=0}^{3} g(\theta - j\pi/2)$, has *every* off-$(\bmod 4)$ Fourier coefficient vanishing to $< 10^{-16}$ (exact 4-fold cancellation), the assembled DtN $\Lambda = |D_\theta| + M_f$ has $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the *quasi-free state* it defines, with covariance $C = \frac{1}{2}(1 + \operatorname{sgn} H_1)$, is clock-invariant: $\rho\, C\, \rho^\dagger = C$ to $< 10^{-13}$, i.e. the actual $\omega \circ \rho = \omega$, not merely $[\rho, \Lambda] = 0$. A single off-centre bump, a $\mathbb{Z}_3$ (three-mark) source and four unequally-spaced marks each break both by $O(1)$, and the mark-local commutator stays $< 10^{-10}$ as the truncation grows ($N \in \{16, 32, 64\}$) — not an artefact. This *numerically certifies the implication* "mark-local DtN $\Rightarrow \omega \circ \rho = \omega$" for real profiles; the premise that the physical seam *is* mark-local stays **[O]** (it does not close `QGEO.SYM.01`).

The same residual, read dynamically: modular flow + a recovery Lindblad generator ([verification/v238_modular_lindblad_dynamics.py]). The whole `QGEO.SYM.01` discussion above is *static* — a commutator condition $[\rho, \Lambda_\Sigma] = 0$. Read as a *dynamics* it is the statement that the modular flow $\sigma_t = \Delta^{it}$ (the Tomita–Takesaki one-parameter group generated by the modular Hamiltonian $\Lambda_\Sigma = \log \Delta_\omega$) *conserves* the μ_4 clock: $\rho\, \sigma_t\, \rho^{-1} = \sigma_t\ \forall t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ — so the open premise is exactly "modular time conserves the carrier charge". The dissipative half is the recovery channel T of v221 read as a generator: $L = \log T$ is a doubly-stochastic (CPTP-classical) Markov generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time), and its dissipative gap is $-\log((2/3)^6) = 6 \log(3/2) = \Delta$, the OS mass gap of v64 — the static recovery bound and the dynamical mass gap are *one* number, one exponentiated. Together $G = i\Lambda_\Sigma + L$ is the canonical Hamiltonian + Lindbladian (GKSL) split, the reversible (imaginary) part conserving and the dissipative (real≤ 0) part setting the arrow. This is a finite-block exhibit of where the static compiler becomes a dynamics; the residual it leaves — that σ_t is the *geometric* modular flow (Connes–Rovelli "thermal time = physical time") on the full seam algebra — is the same `QGEO.SYM.01` bedrock, **[O]**.

From the dynamics to a QFT skeleton on the seam ([verification/v239_kms_thermal_time.py], [verification/v240_gns_os_reconstruction.py], [verification/v241_dhr_sectors.py], [verification/v242_spin_statistics_modular.py], [verification/v243_haag_ruelle_braiding.py], [verification/v244_spectral_action_contract.py]). With the static→dynamic flip in hand, the standard AQFT machinery assembles a finite-block QFT skeleton, each step typed honestly. *Thermal time* (v239): the modular flow is KMS at $\beta = 1$, and the seam unit $2\pi = 1/(4c_3)$ fixes the modular temperature as the horizon temperature $T_H = c_3/M$. *Reconstruction* (v240): GNS gives the Hilbert space and a cyclic vector from $(\mathcal{M}, \omega)$, and the OS Hamiltonian $H_{\mathrm{OS}} = -\log T = -L$ is positive with gap Δ — the v64 "$H \geq 0$ + mass gap" is the recovery generator read as Euclidean time. *Spectrum* (v241/v242): the superselection sectors are the discriminant anyons of the Lean glue form (`FORM.GLUE.01`); fusion is recovered from the modular S by Verlinde, the Gauss–Milgram sum returns $c = 8$, and the condensation tower $16 \to 4 \to 1$ (v235/v237) selects the holomorphic $(E_8)_1$. *Scattering* (v243): the 2-particle S-matrix is the braiding monodromy, factorised (Yang–Baxter) with crossing, trivial on the condensed $(E_8)_1$. The one genuinely 4d step is typed as a contract, `CONTRACT.QFT4D.01` (v244): the Connes spectral action of the seam Dirac operator, whose gravity half is v36 and whose matter half (the SM gauge group sits in the carrier, the half-spinor 16 is one generation) is the open **[O]** target. Its representation-theory core is now discharged ([verification/v245_spectral_matter_content.py]): the **16** is exactly one *anomaly-free* SM generation ($\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$) with the forced SM hypercharges and the spectral-action value $\sin^2\theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3}\,g_1$), so only the seam→Dirac realisation and the run-down couplings stay open. So the residual to a full 4d QFT is *one premise* (`QGEO.SYM.01`) plus *one contract* (`CONTRACT.QFT4D.01`), not a diffuse list.

The perturbative 4d S-matrix is constructible ([`verification/v269_spert_paqft_skeleton.py`]). The 4d scattering layer is typed precisely as three *distinct* objects: $S_{\rm top}$ (the 2d DHR braiding monodromy, $|M|$=1 statistics, v243), $S_{\rm pert}$ (the 4d Epstein–Glaser / BV S-matrix of the spectral action), and $S_{\rm phys}$ (LSZ on the OS-reconstructed Wightman functions, v240) — the 2d braiding is *not* the 4d cross-section S-matrix. [E] the spectral-action a_4 interaction is power-counting renormalizable (every operator mass dimension ≤ 4: 7 marginal $+3$ relevant, a *finite* counterterm basis), so [C] by the Epstein–Glaser theorem (causal perturbation theory; Brunetti–Fredenhagen pAQFT) the perturbative S-matrix $S(g) = T\exp(i\int g\,L_{\rm int})$ is built order by order by causal factorisation + finite local extension — no path integral, *no* UV divergences. [C] the gap $\Delta = 6\log(3/2)$ gives an IR-safe adiabatic limit. [C] this is *perturbative* only: the nonperturbative ambient measure `QG.AMB.01` is now a [C] redundancy (v369), a certification object rather than missing dynamics, and the canonical 4d reading remains boundary-only (v265); $S_{\rm pert}$ is the perturbative cross-section layer on top (`QFT4D.SPERT.01`).

A concrete one-loop EG number ([`verification/v271_eg_oneloop_quartic.py`]). The skeleton is taken from "exists" to an actual Epstein–Glaser renormalization: the order-1 product $T_1 = {:}L_{\rm int}{:}$ needs no extension, and the first one is the order-2 T_2, whose UV behaviour is the Steinmann *scaling degree.* [E] the massless propagator has sd $= 2$ in $d = 4$, the one-loop bubble (two propagators) has sd $= 4 = d$, so the singular order $\omega = \mathrm{sd} - d = 0 \Rightarrow$ *exactly one* logarithmic local counterterm ($C(\omega{+}d, d) = 1$ — finite, renormalizable, no infinite tower); the one-loop factor is $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly, the *same* $1/(16\pi^2)$ that normalises the spectral-action a_4 geometric quartic. [C] the ϕ^4 one-loop running is then $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry factor $3 =$ the s,t,u channels), with the EG initial condition fixed by the spectral action, $\lambda_\Phi(\mathrm{UV}) = 1/(16\pi^2)$. [O] this is order-2/one-loop, one diagram class; the all-order perturbative construction stays open (`QFT4D.SPERT.02`), while the nonperturbative measure is discharged as a [C] redundancy (below).

Red-team caveat, now discharged ([`verification/redteam/rt_F_qft4d.py`]). Power-counting renormalizability is *not* unitarity for the gravity sub-sector: the local R^2/Weyl^2 *truncation* gives a four-derivative propagator $1/(p^2(p^2 + M^2))$ whose massive spin-2 mode has a negative residue (the Stelle ghost). But that ghost is a Seeley–DeWitt *truncation artefact*: the entire seam KMS form factor $a(p^2){=}e^{p^2/M^2}$ keeps its only pole at $p^2{=}0$ (v304), the spin-2 sector is ghost-free via the Barnes–Rivers decomposition (v370), and the nearest truncation-zero modulus runs to infinity (v380), so *perturbative* graviton unitarity is established. The matter+gauge $S_{\rm pert}$ is unitary outright; the *nonperturbative* `QG.AMB.01` (asymptotic safety v207 / the boundary $(E_8)_1$ net) is then a [C] *redundancy* (v369; gap-decoupled from every TFPT readout, not a TFPT structural item, v335), a certification object rather than a new gap.

The gauge running is an EG order-2 number ([`verification/v273_eg_gauge_running.py`]). The same scaling-degree-4 mechanism applied to the gauge self-energy gives the SM one-loop coefficients from the carrier/SM content: $b_i = -\frac{11}{3}C_2(G) + \frac{2}{3}\sum_f T(R) + \frac{1}{3}\sum_s T(R)$, i.e. [E] $b_3 = -7$, $b_2 = -\frac{19}{6}$, $b_1 = \frac{41}{10}$ (GUT-normalised; the same 41 as the carrier algebra $10\,b_1 = g_{\rm car}\,2^{g_{\rm car}-2} + 1$, v159). [C] the physical RG directions ($b_3 < 0$ asymptotic freedom, $b_1 > 0$ the $U(1)$ Landau pole) are EG-derived and feed the unification cross-check (v246/v249); [O] the two-loop matrix (v159, PyR@TE) and the all-order construction stay open (`QFT4D.SPERT.03`).

The carrier content *is* one anomaly-free SM generation ([`verification/v310_carrier_sm_anomaly.py`]). Those one-loop coefficients sit on top of an exact spectrum/anomaly closure: the carrier half-spinor $16 = \Lambda^{\rm even}(\mathbb{C}^5) = 1{+}10{+}5$ decomposes under $SU(5) \subset SO(10)$ as $10{+}\bar{5}{+}1 =$ exactly one Standard-Model generation $(Q, u^c, d^c, L, e^c, \nu^c)$, and that content is [E] gauge-anomaly-free ($\sum Y = 0$, $\sum Y^3 = 0$, $[SU(2)]^2U(1) = [SU(3)]^2U(1) = 0$, exact) while reproducing $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family would shift b_1 off 41/10, so $N_{\rm fam}{=}3$ *pins* the RG. [E] for the spectrum, anomalies and one-loop β; the full Epstein–Glaser interacting S-matrix and the Yukawa sector stay [C]/[O].

The bridge to the physical S-matrix and one-loop unitarity ([`verification/v278_lsz_bridge_unitarity.py`]). The perturbative layer connects to the physical asymptotic S-matrix: [E] the EG time-ordered products of $S_{\rm pert}$ are the OS Wick-rotated correlators (v240), so amputating external legs and taking the on-shell residue gives $S_{\rm pert} \xrightarrow{\rm LSZ} S_{\rm phys}$ (the three-layer S-matrix $S_{\rm top}\,|\,S_{\rm pert} \xrightarrow{\rm LSZ} S_{\rm phys}$), valid because the gap $\Delta{=}6\log(3/2)$ gives Haag–Ruelle asymptotic states. [E] *one-loop unitarity* is the optical theorem: the s-channel bubble discontinuity

$\operatorname{Im} B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$ is *exactly* the two-body phase space (turning on at the physical threshold $s{=}4m^2$), so $2\operatorname{Im} M = \sum\int d\Pi\,|M|^2$ holds and the matter+gauge sector is unitary order by order. [C] the gravity sector's Stelle ghost is a truncation artefact (perturbative spin-2 unitarity established, v304/v370/v380); the nonperturbative `QG.AMB.01` is a [C] redundancy (v369) (`QFT4D.SPERT.04`).

The first hard data confrontation ([`verification/v246_unification_data_crosscheck.py`]). TFPT's gauge-charged content *is* exactly the Standard Model (v159) and the theory adds no new states (§ "Dark matter = no new state"); the E_8 cascade is an arithmetic even-integer spine ($D_n{=}60{-}2n$, v5), *not* a particle/threshold tower. So a "TFPT-adapted" RGE run *is* the SM run — and the SM does not unify. Tested against the *measured* couplings (v159 betas, plus the standard two-loop gauge matrix): at one loop the pairwise crossings spread over $\sim 10^{13}$–10^{17} GeV with residual $\Delta\alpha^{-1} \approx 9$ at $2{\times}10^{16}$ GeV; at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at $\mu \approx 1.7{\times}10^{14}$ GeV — the three couplings never meet (Fig. 2). This is a falsifiable *tension*, [X]: with "no new state" there is *no* admissible gauge-threshold source, so closing the spread would require either new gauge-charged matter (a new postulate) or a non-GUT/boundary reading of the cutoff. The spectral-action 4d-GUT route is thus in genuine tension — the same status as the Connes–Chamseddine NCG Standard Model — and is likely falsified unless the theory is extended; the AQFT skeleton, DHR structure and SM rep-core survive regardless as an emergent boundary QFT. Recorded as a kill-test, never as a confirmation.

The carrier-native Pati–Salam program ([`verification/v247_e8_branching_no126.py`], [`verification/v248_ps_algebra.py`], [`verification/v249_ps_unification.py`], [`verification/v250_dirac_triple_contract.py`]). The unification tension above is resolved *natively* if the carrier $SO(10)$ is gauged (the carrier $D_5{\oplus}A_3 = SO(10){\times}SU(4)$ is Pati–Salam). *First*, the E_8 audit hull *forbids* the **126**: the **248** branches as $(45,1){+}(1,15){+}(10,6){+}(16,4){+}(\overline{16},\bar 4)$, so the available $SO(10)$ reps are exactly $\{1,10,16,45\}$ — the renormalisable 126_H is algebraically absent, making the "16_H over 126_H" selection a theorem of the hull, not a fit (v247). *Second*, the Pati–Salam NCG algebra $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ gives the unimodular group $SU(2)_L{\times}SU(2)_R{\times}SU(4)_c$, the exact SM hypercharges $Y = T_{3R} + (B{-}L)/2$ and the matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W = 3/8$ (v248). *Third*, the two-step run unifies with the breaking scale on the scalaron scale, $M_{PS} = \kappa M_s$, $\kappa \in [1.0, 1.7]$ at one loop ($\to [1.0, 1.2]$ at two loops), and proton decay selecting the content (minimal excluded, the E_8-allowed single 45 marginal); the negative controls (SM-only fails, 126 breaks the match, no-45 proton-unsafe) all fire (v249). *Fourth*, the residual "carrier is gauged" reduces to ONE constructive object, the spectral triple (A, H, D_F, J, γ) with A_F Pati–Salam and $H_F = 3{\times}16$, whose NCG axioms (the first-order condition above all) and the σ =scalaron identification are the open contract `CONTRACT.QFT4D.DIRAC.01` (v250). *Fifth*, a first concrete step on that triple is taken ([`verification/v251_first_order_sigma.py`]): on the lepton block the first-order condition is verified to *hold* for the Yukawa operator and to be *violated exactly* by the Majorana/σ term — the Chamseddine–Connes–van Suijlekom "beyond first order" mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ (honest correction: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism). *Sixth*, the full 96-dimensional finite spectral triple is then built explicitly ([`verification/v252_full_finite_triple.py`]) — H_F the particle⊕antiparticle doubling of three $SO(10)$ **16**-plets — and the NCG axioms are verified by direct computation: reality J, grading γ, KO-dimension 6, order-zero, the first-order condition (*holding* for the SM algebra including the Majorana, *violated* for the Pati–Salam algebra exactly by the Majorana on $\{\nu_R, e_R\}$), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\text{light}} = m_D^2/M_R$ — thereby discharging five of the seven obligations of `CONTRACT.QFT4D.DIRAC.01` to [E] (orientability, the no-junk condition and the spectral-action expansion remain open). *Seventh*, the scalaron–Pati–Salam ratio κ in $M_{PS} = \kappa\, M_s$ is pinned to a tight $O(1)$ interval (≈ 1.3) by the RG and shown to be a scale-free ratio of heat-kernel coefficients, $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\text{grav}}}$, over the same 96-dim H_F ([`verification/v253_kappa_scalaron_ps.py`]). *Eighth*, the remaining NCG obligations are then discharged or honestly typed: the inner fluctuations Ω^1_D give exactly the one SM Higgs doublet $(1,2)_{1/2}$ with no junk and no **126**, while σ (the ν_R Majorana scalar, $B{-}L{=}2$) appears only as the beyond-first-order Pati–Salam field ([`verification/v254_inner_fluctuations.py`]); the spectral-action Seeley–DeWitt expansion a_0, a_2, a_4 yields the cosmological, Einstein–Hilbert and Yang–Mills+R^2+Higgs terms with $b/a^2 = 1/3$, $\sin^2\theta_W = 3/8$ and the R^2 scalaron, making κ an explicit moment ratio ([`verification/v255_spectral_action_expansion.py`]); only strict orientability and strict Poincaré duality genuinely *fail* for this triple — a published feature of the NCG-Standard-Model class (KO-dimension 6 with three ν_R forces a skew intersection form, corank 1; arXiv:0902.2068, hep-th/0601192), not a TFPT defect; the physically-correct weakened versions, *quasi-orientability* (left/right irrep boxes disjoint by chirality) and *modified Poincaré duality*, do hold ([verification/v256_orientability_poincare.py], [verification/v257_quasiorient_modpd.py]). *Ninth*, the built D_F is no independent posit but the *modular/covariance induction* of the boundary seam state ([verification/v258_dirac_covariance_induction.py]): a quasi-free (KMS, β=1) state is fixed by its covariance C (a positive contraction), with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta) inverse to the KMS occupation $C = (1+e^H)^{-1}$; inducing the carrier covariance $C_F = P_F C_\Sigma P_F$ returns the v252 operator *exactly* — same spectrum, chirality-odd, the Majorana an off-diagonal covariance entry with the seesaw m_D^2/M_R recovered — so D_F is the Kasparov shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\mathrm{car}}]$ of the boundary geometry and the v18 Yukawas are the *readout* of C_Σ, **[E]** modulo the one **[O]** seam realisation. *Tenth*, the last open spectral-action input — the cutoff function f — is fixed by the same seam state ([verification/v259_modular_cutoff_kappa.py]): by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at β=1 (v239) and the seam unit 2π=$1/(4c_3)$ (v58) fixes that β, so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*, not chosen, with moments $f_2/f_0 = f_4/f_2 = 1$ *exactly* (a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$, so the choice is non-vacuous); hence $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\mathrm{grav}}} = \sqrt{c_{PS}/c_{\mathrm{grav}}}$ loses its scheme factor and the v253/v255 open **[O]** cutoff freedom becomes a **[C]** finite trace ratio over the 96-dim H_F (the RG κ≈1.3 pinning c_{PS}/c_{grav}≈1.7). So unification is now *one* theory decision (gauge $SO(10)$?) plus *one* constructive object, both falsifiable — not a fog. The colour $SU(4)_c$ here sits inside $SO(10)$=D_5 and is distinct from the carrier family factor A_3.

The implication is now Lean-formalised (`FORM.QGEO.01`). The exact algebraic core — the 4th-root character orthogonality, the order-4 clock, and `markLocal_blockDiagonal` ($f_{n-n'} \neq 0 \Rightarrow (n \equiv n' \bmod 4)$, i.e. $[\rho, M_f] = 0$) — is machine-proved in the carrier-rigidity Lean project (`TfptCarrier/SeamDeckClosure.lean`; AUDIT: PASS, axioms `propext, Classical.choice, Quot.sound` only). The premise itself is a typed `structure SeamDeckPremise` — consumed, not proved, exactly as `CalderonProjector` encodes the document-1 analytic input — so the conditional theorem "`SeamDeckPremise` $\Rightarrow \omega \circ \rho = \omega$" is now **[E]** while `QGEO.SYM.01` stays the one open **[O]** postulate.

The two residuals are one canonical metric: the pillowcase ([verification/v214_seam_pillowcase.py]). The isometry residual `QGEO.ISO.01` (v180) and the mark-locality residual `QGEO.SUBPRIN.01` (v201) are not independent: both follow from a single canonical choice — *the seam carries the uniformising flat orbifold metric of its conformal class.* The four marks make the orbifold sphere $S^2(2,2,2,2)$, whose orbifold Euler characteristic $\chi_{\mathrm{orb}} = 2 - 4(1-\frac{1}{2}) = 0$ is *Euclidean*: by Troyanov's cone-metric uniformisation it carries a flat metric, unique up to scale, with curvature concentrated at the four cone points (Gauss–Bonnet: four deficits π sum to $4\pi = 2\pi\chi(S^2)$) — this *is* the v201 "flat away from the marks". And the order-4 clock is no longer an extra assumption but a *consequence* of the cross-ratio 2 that v168 already proved: $j(\lambda) = 256(\lambda^2-\lambda+1)^3/(\lambda^2(\lambda-1)^2)$ gives $j(2) = 1728$, so the modulus is the *square* torus $\tau = i$, whose complex multiplication by $\mathbb{Z}[i]$ supplies the order-4 isometry $z \mapsto iz$ (explicitly $(x,y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$, order 4); a generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$, so the clock is specific to μ_4. Thus the two open QGEO residuals *collapse to one* canonical-metric statement and `QGEO.ISO.01` ceases to be an imposed symmetry. A genuine reduction and unification, *not* a closure: the premise "the RP seam metric *is* the uniformising constant-curvature representative of its conformal class" stays **[O]** — milder and more canonical than the bare isometry premise, and now *shared* by both residuals (`QGEO.PILLOW.01`).

The converse, on the canonical metric: raw seam $\Rightarrow$ mark-locality ([verification/v264_raw_seam_marklocal.py]). Composing the chain explicitly closes the gap between v216, v201 and v198 on the uniformising metric: the flat pillowcase has its curvature *only* at the four cone points (v216), so the DtN sub-principal symbol M_f is a μ_4-orbit sum, hence $\mathbb{Z}_4$-invariant (Fourier support on $4\mathbb{Z}$, verified by FFT), hence block-diagonal in the clock-character basis, hence $[\rho, M_f] = 0$; with the principal symbol $|k|$ already commuting (v198) this gives $[\rho, \Lambda] = 0$ and $\omega \circ \rho = \omega$ *on this metric.* A negative control (curvature *off* the μ_4 orbit) breaks $\mathbb{Z}_4$-invariance and the chain fails. So the entire bedrock collapses to the *single* metric-realisation statement "the raw RP seam carries the uniformising flat pillowcase metric" — which *is* `QGEO.SYM.01`. A genuine reduction/sharpening (`QGEO.ENERGY.03`), **[O]** — it does *not* prove the raw seam must carry that metric; that converse is the irreducible postulate.

The minimal axiom: a rigidity theorem ([verification/v267_qgeo_rigidity_minimal_axiom.py]). One can sharpen the bedrock to its irreducible kernel. **[E]** the *single* bare statement "the seam admits the carrier's order-4 clock as a conformal symmetry cyclically permuting its four marks" forces everything: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 *independent of* a; cross-ratio 2 $\Leftrightarrow$ j=1728 (only

$\lambda \in \{-1, \frac{1}{2}, 2\}$) ⇔ the order-4 CM automorphism $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$; and from the square configuration the unique flat pillowcase metric (Troyanov), mark-locality, $[\rho, \Lambda] = 0$ and $\omega \circ \rho = \omega$ all follow (v214/v201/v264/v198). **[E]** neg controls: a generic four-mark set has $j \neq 1728$ and only a $\mathbb{Z}_2$ symmetry (no clock), the hexagonal point $j = 0$ has a $\mathbb{Z}_6$ symmetry (order $6 \neq 4$) — only the square (order-4) point realises the μ_4 clock. **[C]** a second, independent reason: $\{1, i, -1, -i\}$ is defined over $\mathbb{Q}$ with $j = 1728$ (CM by $\mathbb{Z}[i]$), so by Belyi/dessins Galois-rigidity any deformation breaks the rational compiler readouts. **[O]** So `QGEO.SYM.01` is now in its minimal sharpest form — the bare existence of that order-4 conformal symmetry, which (like c=const in relativity) cannot be derived from nothing. The *reduction* is closed (axiom ⇒ everything); the axiom itself stays the one foundational postulate (`QGEO.SYM.02`).

The flat geometry closes the commutator to all orders ([`verification/v276_qgeo_flat_closes_commutator.py`]**).** The state-level residual $[\rho, H] = 0$ — the one operator equation the bedrock rests on — is *not* a separate analytic gap once the flat $\tau = i$ pillowcase is granted. **[E]** if $H = f(\Delta_{\text{flat}})$ is a function of the intrinsic flat Laplacian (the DtN of a flat metric, no curvature corrections) and the order-4 deck ρ ($z \mapsto iz$) is a flat-$\tau = i$ isometry, then $[\rho, \Delta] = 0 \Rightarrow [\rho, H] = 0$ *exactly*, to all orders (an isometry commutes with every spectral function of its Laplacian; verified on the discretised square torus, with a generic non-isometry as a failing control). This *upgrades* the leading-order commutation v198 (principal symbol) and v201 (subprincipal) to the full H — flatness removes the lower-order terms. **[O]** so `QGEO.SYM.01` collapses to a *single geometric* statement, "the raw RP seam state is the flat $\tau = i$ pillowcase quasi-free state ($H = \sqrt{-\Delta_{\text{flat}}}$)"; everything downstream is **[E]**, and the remaining human obligation is one constructive-QFT proof that the raw seam DtN state *is* that flat geometry (`QGEO.SYM.03`).

The obligation as one precise lemma, Lean-backed ([`verification/v279_qgeo_obligation_lemma.py`]**).** The bedrock is now written as a single constructive-QFT *lemma*: *given* the premise "the raw seam carries the flat $\tau = i$ pillowcase metric g_{flat} (Troyanov-unique, cone angles π), so $\Lambda_\Sigma = \sqrt{-\Delta_{g_{\text{flat}}}}$", the conclusion $\omega_\Sigma \circ \rho = \omega_\Sigma$ (and hence mark-locality, the metric inclusion, E_8) follows. **[E]** the reduction DAG from "free RP seam" to this premise has *exactly one* un-discharged leaf — "the raw seam state *is* the flat state" — every other node being closed; and the geometric side is fixed (the flat DtN is Fourier-diagonal), so the obligation is a *state-identification*, not a geometric ambiguity, with two candidate routes (**[C]**): RP-state uniqueness on the flat $\tau = i$ orbifold, or Troyanov + OS reconstruction. **[E]** the all-orders closure step is now a *Lean theorem*: `TfptCarrier/SeamDeckClosure.lean` (`flat_all_orders_clock`) proves that any spectral function $g(\Delta_{\text{flat}})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho = \text{diag}(i^n)$ — upgrading v198/v201 to the full operator with only the standard axioms, no `sorry` (`FORM.QGEO.03`). **[O]** the one premise stays the honest axiom (`QGEO.OBLIG.01`).

Self-investigation: the experiment and the simplification ([`verification/v280_pillowcase_steklov.py`], [`verification/v282_e8_tau_i_unification.py`]**).** Two things can be done *ourselves*. (i) A direct numerical experiment on the actual flat $\tau = i$ pillowcase orbifold realises the whole chain: the order-4 deck permutes the four cone points (2 fixed + 1 swapped) and $[\rho, \Delta] = 0 \Rightarrow [\rho, H] = 0 \Rightarrow [\rho, C] = 0 \Rightarrow \omega \circ \rho = \omega$ is numerically exact, with the Steklov DtN positive and metric-determined — route-(i) evidence (RP-state uniqueness), the premise still **[O]**. (ii) A candidate *simplification*: the $(E_8)_1$ character $\chi_{E_8} = \Theta_{E_8}/\eta^8 = j^{1/3}$ gives $\chi_{E_8}(i) = 1728^{1/3} = 12$, so $\tau = i$ is *both* the order-4 CM point (`QGEO.SYM.01`, flat geometry) *and* the $(E_8)_1$ partition-function modulus (`QG.AMB.01`, holomorphy) — the two open premises are two faces of one object, "the raw seam is $(E_8)_1$ at $\tau = i$", dropping the obligation count $2 \to 1$ (`QGAMB.UNIFY.01`).

The two proof routes, decomposed ([`verification/v284_route_i_rp_uniqueness.py`], [`verification/v285_route_ii_seam_condensation.py`]**).** The keystone — now *closed modulo a cited theorem* via the lattice/S_3 stack (below) — can be attacked from two sides, and both are now lemma chains with the dischargeable steps discharged. *Route (i)* (RP-state uniqueness) is a six-lemma chain — **[E]** five are already **[E]**/**[O]** (RP→DtN v194; marks→pillowcase v195/v216; the flat Steklov operator $= \sqrt{-\Delta_{\text{flat}}}$, principal symbol $|\xi|$ with no curvature term; covariance from the DtN v258; μ_4-invariance v276), and **[C]** Troyanov ($\chi_{\text{orb}} = 0$) reduces the residual to "the raw seam is the constant-curvature geometric state". *Route (ii)* (seam-net condensation) is a four-lemma chain — **[E]** three are discharged (no abelian sector ⇔ $\det K = 1$ ⇔ holomorphic ⇔ E_8 v234/v277/v281; the tower $16 \to 4 \to 1$ v92), leaving "the free RP seam condenses the μ_4-Lagrangian glue" =`QGEO.SYM.01`. **[E] the two open lemmas coincide** — both say "the raw seam is the $(E_8)_1$ at $\tau = i$" (v282), so the canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau = i$ is the *single* **[O]** statement that closes both (`QGEO.ROUTEI.01`/`QGAMB.ROUTEII.01`).

The Seam Equivalence Theorem, named and firewalled ([verification/v286_seam_equivalence_contract.py]). That single statement is now the named target `SEAM.EQUIV.01`: *the raw RP seam state reconstructs canonically the holomorphic* $(E_8)_1$ *boundary net at the* $\tau{=}i$ *pillowcase* (respecting the DtN, the KMS flow, the μ_4 deck, the Q-system extension, $\det K{=}1$ and the $(E_8)_1$ character). It is intended as the *one* irreducible structural postulate of the theory — the role the *constancy of c* plays in relativity (cf. v267): not a gap to be patched but the single sharply-localised, falsifiable premise on which the whole boundary closure turns. **[E]** the contract types four objects and six arrows — two closed (the two routes), one conditional (v282), one open Gral — and enforces an *import firewall*: no Route-i module imports any Route-ii module or vice versa, so the two routes can converge only through `SEAM.EQUIV.01` (this kills the "E_8 smuggled into the geometry and pulled back out" attack). The two proof routes are then attacked directly: **[E]** Route A (AQFT, [verification/v287_free_rp_bulk_to_holomorphic_boundary.py]) reduces the Gral to *one* standard theorem, "invertible (SRE) Gaussian bulk ⇒ single-sector boundary" (4/5 lemmas discharged); and **[E]** Route B (DtN, [verification/v288_full_l2_subprincipal_z4.py]) *proves* the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \pmod 4$), so $[\rho,\Lambda]{=}0$ on all of L^2), shrinking the residual to the single sharper question "why is the raw seam sub-principal term mark-local?" (`SEAM.EQUIV.A01`/`SEAM.EQUIV.B01`).

Route B sharpened to one analytic lemma; both routes precisely scoped. That last question is decomposed ([verification/v289_marklocal_raw.py], `MARKLOCAL.RAW.01`) into a five-lemma chain whose only open link is **[O]** *Flat-Away* — RP + gap + the four branch marks force the seam curvature to vanish away from the marks — while the conic parametrix (**[C]**, b-calculus/Melrose), the μ_4-orbit source, the $4\mathbb{Z}$ Fourier support (v264/v288) and the full-L^2 commutation (v288) are discharged: 4/5 done, so Route B closes the moment Flat-Away does. The two claims are firewalled against over-reading: Route B does *not* assert "full-L^2 QGEO" — its exact-label split is Fourier-lemma (Formal/Exact), operator test (Numerical Exact), transfer to the raw conic DtN (Conditional, v264), mark-locality from the raw seam (Open Selection, v289); and Route A's "one standard theorem" is itself a composition of citable results (Kitaev/Freed–Hopkins, Müger/Kawahigashi–Longo–Müger, Kawahigashi–Longo, [verification/v287_free_rp_bulk_to_holomorphic_boundary.py]) whose only open hypothesis — seam-bulk invertibility — coincides with Route B's Flat-Away. Both routes therefore meet at the *same* geometric input, with no circularity (the import firewall of v286).

Red-team: $\mathbb{Z}_4$-invariance is not mark-locality. A deliberate adversary ([verification/v290_z4_smooth_curvature_adversary.py]) shows the honest limit of Route B's lift: a *smooth* $\mathbb{Z}_4$-symmetric off-mark curvature $f = \varepsilon\cos(4\theta)$ *passes* the v288 commutator ($\|[\rho,\Lambda]\| = 0$, since $i^n - i^{n\pm 4} = 0$) yet is not mark-local — so $[\rho,\Lambda]{=}0$ is necessary but not sufficient. The modulus is instead excluded by the spectral data: it shifts the DtN spectrum off the flat ladder and the heat trace ($\max|\Delta\lambda| = 0.155$, $\Delta\mathrm{Tr} = 0.019$ at $\varepsilon = 0.3$), i.e. the KMS weight and $(E_8)_1$ character that `SEAM.EQUIV.01` must match. Consequently the open lemma is named in its own right ([verification/v291_flataway_contract.py], `FLATAWAY.RP.01`): *the smooth curvature density in the DtN sub-principal symbol vanishes away from the four branch marks*, with three independent attack routes — spectral-rigidity and heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family) and RP-energy/Troyanov (the constant-curvature minimiser) — closing any one of which closes Route B.

All three routes attacked; Flat-Away reduced to one selection principle. Each route is now carried to its precise reduction: **[E]** the *heat* route ([verification/v292_flataway_heat_reduction.py]) shows the heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature (leading invariant $\|f\|_{L^2}$), so Flat-Away ⇐ "the seam fixes the a_2 heat coefficient"; **[E]** the *spectral* route ([verification/v293_flataway_spectral_hessian.py]) lifts this to the full smooth-$\mathbb{Z}_4$ space — mode $4k$ splits the degenerate Steklov level $|2k|$, the mismatch Hessian over $\{4, 8, 12\}$ is positive-definite ($\{0.497, 0.500, 0.503\}$), so the flat seam is a *strict* isolated minimum (answering "only one direction"); and **[E]**/**[C]** the *variational* route ([verification/v294_flataway_rp_energy.py]) gives, with the rigid mark divisor and Troyanov uniqueness, the flat cone metric as the unique strictly-convex curvature-energy minimiser. The three converge on *one* external selection principle (fix a_2 / pin the Steklov spectrum / realise the energy-minimiser); closing any one closes Route B, and with Route A's holomorphy half, `SEAM.EQUIV.01`.

The heat route made exact and Lean-formalised. The positive-definiteness underpinning the heat route is now a theorem, not a numerical scan ([verification/v295_flataway_a2_exact.py]): the first variation of $\mathrm{Tr}\, e^{-t\Lambda}$ vanishes (the smooth off-mark curvature has zero mean, Gauss-Bonnet), and since $\Lambda \mapsto \mathrm{Tr}\, e^{-t\Lambda}$ is convex (Klein/Peierls), $\varepsilon{=}0$ is the *global* minimum — so $\Delta\mathrm{Tr} \geq 0$ for every deformation, $= 0$ iff

$f = 0$; the exact second-order coefficient is the divided-difference kernel of e^{-tx} (validated to 10^{-6}). The algebraic $\Delta{=}0 \Leftrightarrow$ flat step is machine-formalised in Lean (`FORM.FLATAWAY.01`, `SeamDeckClosure.lean`: a positive-weighted sum of squares is positive-definite; axiom-clean). So Flat-Away's heat route is reduced to the single external invariant "the seam fixes a_2", with its positivity both proved and formalised. The a_2 coefficient itself is now in *closed form* ([`verification/v296_flataway_a2_closed.py`]): the operator tails telescope geometrically to $W_k^{\text{tail}} = t(1-e^{-4kt})/(8k(1-e^{-t}))$, leaving a finite $(4k{-}1)$-term middle, e.g. $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$ (manifestly positive; small-t $W_k = t/2 + O(t^3)$). Symmetrically, Route A's "one standard import" is now a precise, citable theorem stack ([`verification/v297_route_a_literature_stack.py`]): LIT-A (Kitaev; Freed–Hopkins) → LIT-B (Müger; Kawahigashi–Longo–Müger) → LIT-C (Conway–Sloane; Dong–Xu) compose into "invertible bulk $\Rightarrow (E_8)_1$", all three legs established, the one shared hypothesis (seam-bulk invertibility) coinciding with Route B's Flat-Away — so both routes are reduced to citable/established machinery plus the *one* shared geometric input, which the lattice/S_3 stack (below) now pins at every computable level, making the keystone *closed modulo a cited theorem.*

Flat-Away hardened, and the pin derived from $(E_8)_1$ ([`verification/v300_flataway_rigid.py`], `FLATAWAY.RIGID.01`). The spectral route was a *soft* minimum ("the mismatch grows with ε"); it is now a *discrete* obstruction. **[E]** the flat fingerprint is the integer Steklov ladder with every level $n \geq 1$ of multiplicity exactly 2; **[E]** the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (the exact 2×2 block $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$, validated against the full Toeplitz operator), so **[E]** any $g_k \neq 0$ lifts the degeneracy ($2 \to 1{+}1$) and changes the spectral *multiset* — preserving the flat ladder *with its multiplicities* forces every $g_k = 0$, a degeneracy obstruction robust to rescaling and strictly stronger than the v293 "deviation grows" scan; with the v295 convexity, flat is isolated analytically *and* discretely. Moreover the external pin is now *derived*: **[C]** the $(E_8)_1$ vacuum character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ has integer q-coefficients (E_8 even unimodular $\Rightarrow$ integer conformal weights), so by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam — i.e. "the seam fixes its spectrum" *is* the $(E_8)_1$ rationality of Route A. **[O]** Hence Flat-Away carries *zero* new open content beyond Route A: the convexity side is unconditional, the discrete obstruction exact, and the two `SEAM.EQUIV.01` routes provably converge on one rationality fact (consistent with the v286 firewall).

Route A's last hypothesis discharged — the gapped quasi-free bulk is invertible ([`verification/v301_route_a_invertible.py`], `SEAM.EQUIV.A.INV.01`). Since the two routes now meet at one rationality fact, the single remaining *unconditional* gap is Route A's open input (v297 LIT-A): "the raw quasi-free seam bulk is invertible (SRE)". A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions realise only the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order requires interactions (Kitaev's periodic table, class D, the only invariant being the chiral central charge). **[E]** the carrier is 16 Majoranas, so $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$ (v148); **[E]** $SO(16)_1 \supset (E_8)_1$ ($\dim \mathfrak{so}(16) = 120$ currents $+\,128$ spinor $= 248$, v154); **[C]** hence a gapped quasi-free bulk is invertible automatically, and **[E]** the invariant $|\det K| = \#$anyons is 1 for E_8 (invertible) against 4 for the gauged $D_8{=}SO(16)$ (anyons), which free fermions cannot produce. **[O]** So LIT-A's 'invertible bulk' *follows* from 'gapped 16-Majorana quasi-free bulk' (v148/v160 + the mass gap): the residual changes from a topological-order statement to a spectral one. *Bottom line:* together with v300, the entire open content of `SEAM.EQUIV.01` reduces to the single spectral statement "the quasi-free seam bulk is gapped" — the topological-order obstruction (hidden anyons) is removed, and the gap itself is not manufactured.

The last lever — the seam gap *is* the derived Recovery gap ([`verification/v302_seam_gap.py`], `SEAM.EQUIV.GAP.01`). That single remaining input is not free: the seam transfer operator has the frozen spectrum $\operatorname{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple Perron eigenvalue 1 separated from the rest, and the separation is the *Recovery gap* $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\text{fam}}\ln(3/2) \approx 2.4328 > 0$ — a number forced by the carrier ($(2/3)^6$ the μ_4-deck transfer eigenvalue), with the decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76) making it robust. **[C]** By Osterwalder–Schrader reconstruction / quasi-free clustering a transfer-operator gap $\Delta > 0$ *is* a mass gap of the reconstructed Hamiltonian (correlation length $1/\Delta$). *Bottom line:* with v300/v301, the whole residual of `SEAM.EQUIV.01` is now a composition of standard cited theorems (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over established TFPT facts (16 quasi-free Majoranas; transfer gap $6\ln(3/2)$): *no undischarged TFPT-internal assumption remains.* It stays **[O]** (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive.

The continuum chain assembled, and the residual tied to the seam one-sidedness ([`verification/v356_continuum_mmst_applicability.py`], `SEAM.EQUIV.CONTINUUM.03`). Writing the chain out link-by-link: **S1** the gapped (v302) quasi-free 16-Majorana collar → **S2** invertible (Kitaev free fermions, v301) → **S3** the *nontrivial* invertible phase ($c_- = 8 \neq 0$) → **S4** a non-gappable anomalous chiral edge (anomaly inflow / bulk–edge) → **S5** its chiral-CFT scaling limit (MMST, v336; range rank $\leq c \leq D$, i.e. $8 \leq 8 \leq 16$) → **S6** the order-4 μ_4 clock + the AQFT stack pin $(E_8)_1$ (v351/v297). Every link is a theorem or an established TFPT fact *except* S3, and S3 is precisely the seam being *one-sided* (chiral, $c_- \neq 0$) — the *same* one-sidedness that makes $c_3 = 1/(8\pi)$ a one-sided Gauss–Bonnet ($|\mathbb{Z}_2| \oint K$, v58). So the lone realization input is tied to the constant that *defines* the theory, not a free dial; the residual is now "the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase", **[O]** but theorem-bounded on both sides.

MMST in-class verified for the collar, so the scaling limit is $(E_8)_1$ (`[verification/v366_mmst_seam_collar.py]`, `SEAM.MMST.INCLASS.01`). The continuum leg is now checked *hypothesis-by-hypothesis* for the explicit seam collar: it is a $D{=}\dim S^+{=}16$ Majorana quasi-free (CAR) state (v155/v160), gapped with $\Delta{=}6\log\frac{3}{2}{>}0$ *derived* (v302), and chiral with $c_-{=}D/2{=}8{=}g_{\rm car}{+}N_{\rm fam}$, so it sits squarely in the MMST free-lattice-fermion class with rank $\leq c \leq D$ reading $8 \leq 8 \leq 16$ (*in* range, saturating rank$\,{=}c{=}8$). The MMST massless-scaling-limit theorem then applies as a cited result, and the order-4 μ_4 clock pins the target to $(E_8)_1$ ($\det K{=}1$, one primary) over the same-c rival $SO(16)_1$ ($\det K{=}4$, v351). **[E]** the class arithmetic, the derived gap, the chirality and the $(E_8)_1$ pinning; **[C]** the CAR-class typing and the cited MMST theorem; **[O]** the lone residual is exactly S3 above (the lattice realization). The continuum side of `SEAM.EQUIV.01` is thereby reduced to a single, named, theorem-bounded input. The lattice→conformal-net step itself is now a *rigorous AQFT construction*: Adamo–Giorgetti–Tanimoto (`arXiv:2506.01008`, 2025) build the two-dimensional conformal net $\mathcal{A}_Q$ of an even lattice Q and *classify* every two-dimensional extension of the Heisenberg net (under a discreteness assumption) as such a lattice net — so "the lattice is a conformal net" is a theorem of the same operator-algebraic lineage the continuum leg invokes, not a hope; and since (xxxii) (`[verification/v469_seam_crossedproduct_route.py]`) the extension step no longer even *needs* the preprint lineage — the local $\mathbb{Z}_2$ simple-current crossed product ($h_s{=}1{\in}\mathbb{Z}$, Longo–Rehren 1995; Böckenhauer 1996; Böckenhauer–Evans 1998; KLM 2001) certifies it on peer-reviewed ground, with AGT/AMT as the independent second witness.

S3 made concrete: an explicit gapped chiral-Majorana lattice model (`[verification/v367_seam_s3_lattice.py]`, `[verification/v368_seam_s3_inflow.py]`). The S3 input is no longer abstract. The $p{+}ip$ Bloch model $h(k){=}\sin k_x\,\sigma_x{+}\sin k_y\,\sigma_y{+}(M{-}\cos k_x{-}\cos k_y)\sigma_z$ is, at $M{=}1$, a gapped free-fermion phase with numerically-computed Chern number $|C|{=}1$ ($C{=}0$ trivially at $M{=}3$), so $N_{\rm Maj}{=}\dim S^+{=}16$ copies give a chiral edge of $c_-{=}16/2{=}8{=}g_{\rm car}{+}N_{\rm fam}$, with the edge K-matrix $SO(16)_1$ ($\det 4$) condensed to $(E_8)_1$ ($\det 1$) by the order-4 μ_4 clock (v351/v369). A finite STRIP of the same model carries an edge-localised in-gap chiral branch (absent in the trivial phase) – so by anomaly inflow the chiral edge *exists* as a consequence of the invertible bulk (the S4 leg), discharging the "does the edge exist?" question on the lattice. **[E]** the gapped Chern model, the $c_-{=}8$ counting and the edge existence; **[C]** the μ_4 condensation to $(E_8)_1$; **[O]** only the genuine *continuum scaling limit* (MMST, v336) stays open — S3 is a runnable lattice model that pins the target at every computable level, so the keystone is *closed modulo a cited theorem.*

The S3 closure stack: $c = 8$, the $(E_8)_1$ character, the torus count and reflection positivity — all computed on the lattice (`[verification/v376_seam_s3_centralcharge.py]`, `[verification/v377_seam_s3_e8character.py]`, `[verification/v378_seam_s3_modular.py]`, `[verification/v379_seam_s3_rp.py]`). On that explicit model the target of the scaling limit is now pinned at every level a computation can reach. *Central charge* (v376): the Calabrese–Cardy finite-size entanglement scaling gives $c = 1.0000$ per critical complex mode, so the 16-Majorana (= 8 complex) collar has $c = 8 = g_{\rm car}{+}N_{\rm fam}$, and it is chiral ($c_- = 8$). *Convergence* (`[verification/v392_seam_s3_scalinglimit.py]`): across five sizes $L = 66\ldots514$ the slope $c_1(L)$ is a monotone Cauchy-like sequence approaching 1 and $1/L$-Richardson-extrapolates to $c_1(\infty) \approx 1.000$ ($c \approx 8.00$ in the limit) — the existence-relevant *convergence* the two-size value did not test (evidence, not a proof of existence; the residual stays **[O]**). *The net* (v377): the exact affine character $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \ldots)$ gives $c = 8$ and 248 level-1 currents (one primary), and the μ_4/GSO promotion 248=240+8=120+128 (240=16·5·3) pins $(E_8)_1$ over the same-c rival $SO(16)_1$ (120 currents, four primaries). *Genus-1* (v378): the KLM condensation $\mu(\text{carrier}){=}16 \to \mu(E_8){=}16/4^2{=}1$ and the single character's modular T-eigenvalue $e^{-2\pi i/3}$ give torus ground-state degeneracy 1 — the holomorphy / $\det K{=}1$ statement that the genus-0 arguments could not reach (v344). *Reflection positivity* (v379): the gapped collar's Euclidean two-point function has a positive spectral measure, so the OS reflection Gram matrix is positive semidefinite (a ghost mode breaks it). **[E]** all four (numerical

c, exact character, genus-1 count, RP); **[O]** the one thing a computation cannot supply is the abstract *continuum existence* of the limit (the cited MMST theorem, v336). With RP in hand the ambient measure C7 is then discharged by the redundancy theorem (v369), so S3 is the master key: the target is pinned exactly, only the existence proof is external.

A ground-state witness for premise (i): the modular commutator measures c_-=8 from the bulk state alone ([`verification/v489_seam_modular_commutator.py`], `SEAM.CCC.01`). The chiral central charge entered so far through band topology (the Chern number, v367) and through counting (the KLM/det-Cartan arithmetic, v378). The Kim–Shehab–Kim modular commutator $J(A,B,C) = i\,\mathrm{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3)\,c_-$ (PRL 128 (2022)) reads c_- off the *entanglement data of the exact BdG ground state* of the same collar — no band bundle, no representation theory enters — via the Peschel covariance ($W_X = -2\,\mathrm{artanh}\,\Gamma_X$) on a disk cut into three 120° wedges: per layer $c_- = 0.499998$ at L=32 (converged 0.500000 at L=48, deviation -1.8×10^{-7}), the trivial M=3 control gives 0.000000, the orientation flip M=−1 gives −0.499986 (sign flip), layer additivity is exact (10^{-14}), and the 16-layer carrier lands on $c_- = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}}$ (direct run 7.9967 at L=16, additivity route 8.0000). **[E]** the third, mutually independent lattice witness of premise (i) of `SEAM.EQUIV.01` — and the first read from ground-state entanglement alone; **[O]** unchanged: the continuum scaling limit (MMST, v336) and the cited KLM condensation step stay exactly where v367/v378 left them — `SEAM.EQUIV.01` stays **[O]**.

The spin-structure parity census: μ_{pre}=4 witnessed on the lattice ([`verification/v490_seam_parity_census.py`], `SEAM.MU.01`). Step 1 of the KLM chain (v378) — "the carrier sits in the $SO(16)_1$ class, μ_{pre}=4, *before* the μ_4 condensation" — was arithmetic; it is now a lattice ground-state measurement. The T^2 fermion parity per spin structure (AA, AP, PA, PP) is determined *twice* (Read–Green TRIM product and real-space Majorana Pfaffian, spectra matching to $9{\times}10^{-15}$): per layer $(+,+,+,-)$ — the ν=1 Ising signature — and for the 16-layer carrier parity^{16} = + in *all four* sectors, so the gauged torus GSD is $4 = \mu(SO(16)_1) = |\det \mathrm{Cartan}(D_8)|$. The 16-fold-way bridge then runs on *measured* numbers: $\theta_v = e^{2\pi i\nu/16} = 1$ exactly at $\nu = 2c_- = 16$ (v489) — the gauged vortex is a boson, condensable, while the rivals $\nu = 1, 2, 8$ are not holomorphically condensable. Honest fences: the fermionic Kitaev–Preskill TEE is identically zero for *both* M=1 and M=3 (γ_f = 0.000001/0.000000) — an invertible fermionic phase has no TEE, so γ_f does *not* discriminate $(E_8)_1$ vs $SO(16)_1$; and the KLM step $\mu : 4 \to 4/2^2 = 1$ stays the cited theorem (**[C]**, the condensed model is interacting). The numerics pin exactly the premise package $\{c_-{=}8,\ \mu_{\mathrm{pre}}{=}4,\ \theta_v{=}1\}$ of the condensation; `SEAM.EQUIV.01` stays **[O]**.

And that external step is now Lean-formalised as "closed modulo a cited theorem" (`FORM.SEAM.MMST.01`, `TfptCarrier/SeamScalingLimit.lean`). The continuum leg is machine-formalised in the audit-contract style of `SeamEquivChain.lean`: the collar's MMST applicability hypotheses are *provable* arithmetic facts by Lean kernel `decide` (no axioms) — D=16 Majoranas, rank =c=8, the range $8 \le 8 \le 16$, the chirality $c_-{=}8 \neq 0$, and the holomorphy discriminator $\det K$=1 vs $SO(16)$'s 4; the MMST scaling-limit theorem and the Adamo–Moriwaki–Tanimoto OS reconstruction are declared as *named cited axioms*; and the conclusion $(E_8)_1$ follows by a well-typed composition whose `#print axioms` pins the residual to exactly those two published theorems (plus the carrier gap), with no hidden assumption and no `sorry`. Every TFPT-internal hypothesis is thus kernel-checked; only the two cited theorems are assumed — the honest closure ceiling for `SEAM.EQUIV.01`. *Two of the seven convergent routes below are now kernel-hardened the same way:* `FORM.SEAM.BW.HSMI.01` (`TfptCarrier/SeamStandardPair.lean`) proves the four Borchers/Wiesbrock *standard-pair relations* exactly over $\mathbb{Z}$ on the centred ladder ($[K,P]{=}P$, $J^2{=}1$, $JKJ{=}{-K}$, $JPJ{=}P_+$, all by kernel `decide`, no domain axiom), with the cited Borchers step and the one continuum input as named axioms; and `FORM.SEAM.APPLICABILITY.01` (`TfptCarrier/SeamApplicabilityLedger.lean`) kernel-checks the audit *count* itself — of the 12 cited-theorem hypotheses exactly 11 are internal and 1 external (and $8 \le 8 \le 16$, $\det K{=}1 \neq 4$), all axiom-free — so the very bookkeeping the keystone's honesty rests on is machine-verified, leaving only the one named continuum fact open.

TOE-roadmap follow-ups — the keystone as a typed certificate, the bedrock modular form, and the gap-decoupled measure ([`verification/v308_seam_equiv_chain.py`], [`verification/v309_modular_bedrock.py`], [`verification/v311_gap_decoupled_measure.py`]). Three machine-checked steps push the keystone arc further. *(i) Composition certificate* ([`verification/v308_seam_equiv_chain.py`]): the closing implication is one well-typed logical chain gap>0 → invertible → holomorphic c=8 → $(E_8)_1$ whose arithmetic discriminators are exact — $|\det \mathrm{Cartan}(E_8)|$=1 against $|\det \mathrm{Cartan}(D_8)|$=4, the carrier tower 4·4=16 → 4 → 1, $c{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}{=}8$ — so the only undischarged premise is the cited OS step; **[E]** the discriminators, **[O]** the keystone.

(ii) Modular bedrock ([verification/v309_modular_bedrock.py]): with the four marks derived from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), the quasi-free modular Hamiltonian $K = \log((1-C)/C)$ is built explicitly and $[\rho, K]=0 \Leftrightarrow \omega\circ\rho=\omega$ (a $\mathbb{Z}_2$ profile breaks it), so the bedrock reduces *non-circularly* to the Bisognano–Wichmann intrinsicality of the raw collar state; [E] the model, [O] the intrinsicality. *(iii) Gap-decoupled measure* ([verification/v311_gap_decoupled_measure.py]): the admissible sector clusters exponentially at rate $(2/3)^6$ (correlation length $1/\Delta$, finite susceptibility 729/665) and the v76 margin $\Delta - 31/(4\pi^2) > 0$ decouples it from the ambient, so the physical sector is a convergent object while `QG.AMB.01` is a [C] redundancy (v369). Each pins the residual to a single named step; with the lattice/S_3 stack the keystone is now *closed modulo a cited theorem. (iv) The intrinsic-BW residual as a cited reduction* ([verification/v424_seam_lto_rp_reduction.py]): the open bedrock lemma of (ii) — the raw collar's intrinsic Bisognano–Wichmann property — is reduced to the recent commuting-projector theorem of Naaijkens–Penneys–Wallick (`arXiv:2605.10693`), whose reflection-positivity axiom (LTO-RP) is exactly $u_\Theta=J$ (the reflection *is* the modular conjugation), i.e. the BW condition $\theta K\theta = -K$ — *not* $\omega\circ\rho=\omega$. [E] the μ_4 clock is a modular *symmetry* ($[\rho, K]=0$), not the flow ($\sigma^\psi=\rho$ is type-wrong, a one-parameter group versus a finite order-4 element); a positive character-block-diagonal K has $[\rho, K]=0$ yet, being positive definite, never $\theta K\theta = -K$, so clock-invariance does *not* imply (LTO-RP). [O] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* commuting-projector models (Toric Code via a trace, Levin–Wen anyonic), whereas the seam is the opposite corner — invertible ($|\det \mathrm{Cartan}(E_8)|=1$, no anyons) and a $\beta=1$ KMS state (not a trace) — so two sub-steps stay open: realising the raw seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO, and extending (LTO-RP) to the invertible KMS case. An MMST-style reduction (cf. v308/v336), *not* a closure. *Sub-step (ii) is now constructively exhibited* ([verification/v426_seam_lto_rp_kms.py]): a gapped reflection-symmetric μ_4-even collar at $\beta=1$ KMS satisfies $\Theta K\Theta^{-1}=-K$ ($u_\Theta=J$) and $[\rho, K]=0$, so (LTO-RP) holds there by direct Tomita–Takesaki — the corner NPW26 leave open — with $|\det \mathrm{Cartan}(E_8)|=1$ (invertible, no anyons) and the state a genuine KMS, not a trace; only the continuum sub-step (i) remains [O]. On the covariance side, the recent theorem of Adamo–Giorgetti–Tanimoto (`arXiv:2508.07109`, 2025) — every DHR representation of a conformal net extending a loop-group/Virasoro net is *automatically* positive-energy and diffeomorphism-covariant — defuses the old "Bisognano–Wichmann presupposes the very covariance it should produce" circularity (v215): once the seam net is in hand, geometric covariance is a *consequence*, not an input.

Twenty-six convergent attacks on the one continuum residual — intrinsic BW, a bulk continuum motor and its chiral-edge counterpart, the β-homotopy, the applicability audit, uniform rigidity with its forcing converse and the derivation of clock-invariance, three independent reads of the edge central charge, the four-point and uniform-in-N pinning of the external fact, the Ising operator tower, the $(E_8)_1$ torus modular data, the μ_4-from-marks derivation, the kill tests, the lattice Sugawara current algebra, a uniform-in-N Tomita–Takesaki tower, the chirality-from-P_1 forcing, an $(E_8)_1$ character kill test, the exact MMST citation audit, the complementary lattice-VOA route, the explicit flat-band commuting-projector LTO realisation, the strict-locality (Kapustin–Fidkowski) obstruction, the character-level 128-spinor extension, the c=8 holomorphic-uniqueness pin and the one-particle realisation rigidity ([verification/v438_seam_hsmi_borchers.py], [verification/v439_seam_lieb_robinson.py], [verification/v440_seam_lto_rp_beta.py], [verification/v441_seam_applicability_ledger.py], [verification/v442_seam_rigidity_uniform.py], [verification/v443_seam_e8_discriminator.py], [verification/v444_seam_edge_scaling.py], [verification/v445_seam_rigidity_forcing.py], [verification/v446_seam_clock_invariance.py], [verification/v447_seam_edge_chern.py], [verification/v448_seam_edge_fourpoint.py], [verification/v449_seam_mmst_uniform.py], [verification/v450_seam_edge_entanglement.py], [verification/v451_seam_edge_cardy_tower.py], [verification/v452_seam_e8_modular.py], [verification/v453_seam_mu4_from_marks.py], [verification/v454_seam_edge_virasoro.py], [verification/v455_seam_bw_uniform.py], [verification/v456_seam_chirality_from_c3.py], [verification/v457_seam_character_killtest.py], [verification/v458_seam_mmst_citation_audit.py], [verification/v459_seam_lattice_voa_route.py], [verification/v460_seam_s3_lto.py], [verification/v461_seam_strict_locality.py], [verification/v462_seam_spinor_continuum.py], [verification/v463_seam_c8_holomorphic_uniqueness.py], [verification/v464_seam_oneparticle_rigidity.py], [verification/v469_seam_crossedproduct_route.py]**). None closes `SEAM.EQUIV.01`; together they ground the one continuum input on both its halves, reduce it to a single analytic fact and make the claim falsifiable. *(v) The intrinsic Bisognano–Wichmann route* ([verification/v438_seam_hsmi_borchers.py]): the seam supplies a Longo *standard pair* / +half-sided modular inclusion — a positive lightlike translation $P \geq 0$, the boost K with $[K, P]=P$ (the

Borchers scaling $\Delta^{it}P\Delta^{-it}=e^{-t}P$), and a reflection J with $JKJ=-K$, $JPJ=P_{-}$ — so by Borchers' theorem the modular flow is geometric *intrinsically*, *without* presupposing the rotation-covariance of the vacuum; **[E]** the ladder algebra and the $sl(2,\mathbb{R})$ positivity (min eig > 0, with the indefinite boost as control), **[C]** the synthesis, **[O]** assembling the finite algebra and the positive-energy rep into one continuum standard pair (the MMST/NPW26 wall). *(vi) The continuum-existence motor* ([`verification/v439_seam_lieb_robinson.py`]): the explicit $p+ip$ collar (v367) has a uniform-in-N gap, exponential clustering with N-independent ξ, and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral massless *edge* scaling limit; **[E]** gap/clustering/light-cone, **[C]** the lift, **[O]** the edge CFT. *(vii) The β-homotopy* ([`verification/v440_seam_lto_rp_beta.py`]): the (LTO-RP)/BW data ($\Theta K\Theta=-K$, $[\rho,K]=0$) are β-independent, and along the whole path $C_\beta=1/(1+e^{\beta K})$ from the NPW26-proven *trace* state ($\beta=0$) to the seam KMS state ($\beta=1$) every covariance is reflection-related, μ_4-invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of v426 to a path argument for sub-step (ii). *(viii) The applicability audit* ([`verification/v441_seam_applicability_ledger.py`]): a hypothesis-by-hypothesis ledger of MMST/NPW26/Adamo finds 11 of 12 hypotheses are TFPT-internal computed facts (**[E]**) and exactly one is external — the continuum existence of the chiral scaling limit — covariance being an Adamo consequence, not an input; the keystone's external dependence is pinned to that one analytic fact. *(ix) Uniform rigidity* ([`verification/v442_seam_rigidity_uniform.py`]): the band-limited μ_4-character rigidity of v398 is uniform in N — exact 4-sector commutation to $N=256$, a uniform gap, and an N-independent clock-breaking lower bound $\|[\rho,V]\|=2$ — so it is not a small-N artifact (the full-L^2 forcing stays **[O]**). *(x) The kill tests* ([`verification/v443_seam_e8_discriminator.py`]): mode D of the closure-mode classification (v347) as concrete *falsifiable* discriminators against the nearest $c=8$ rival $SO(16)_1$ — torus GSD 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots — with $c=8$ alone unable to decide (v277) and only $\det K=1$ (the genus-1 statement, v90) selecting E_8; a seam with GSD>1 would falsify $(E_8)_1$. **[X]** the tests, **[O]** the realisation. *(xi) The chiral-edge scaling motor* ([`verification/v444_seam_edge_scaling.py`]): the counterpart of (vi) on the *edge*, attacking exactly the chiral massless scaling limit that the bulk motor left open. On the same $p+ip$ collar (v367/v368) the lattice edge already carries the conformal data of a chiral $c=\frac{1}{2}$ Majorana CFT — a linear, chiral in-gap dispersion (velocity $v\approx0.98$, $R^2=0.9999$, with *opposite* per-edge velocities), an *algebraic* edge two-point $\sim r^{-\eta}$ with $\eta\approx1$ (versus the *exponential* bulk row and the exponential trivial-phase control), and a Cauchy-convergent scaling-collapse under refinement ($\eta=1.001\pm0.007$, successive spread $0.028\to0.013$) — so the 16-copy edge is a chiral $c_-=8$ $(E_8)_1$ CFT; **[E]** the kinematics, correlator and collapse, **[C]** the central-charge reading, **[O]** the rigorous uniform-convergence theorem (cited MMST). With (vi) for the bulk and (xi) for the edge, *both halves* of the one continuum residual are numerically grounded and pinned to a single cited theorem. *(xii) The rigidity forcing converse* ([`verification/v445_seam_rigidity_forcing.py`]): the converse of (ix), upgrading rigidity from "block-diagonal *permits* commuting" to "commuting with the order-4 clock *forces* block-diagonal". For $\rho=\mathrm{diag}(i^n)$, $[\rho,K]=0$ *iff* $K_{ij}=0$ whenever $i\not\equiv j \bmod 4$ (verified entrywise, swept $N=4..128$), so the commutant is *exactly* the 4-sector block algebra: $\dim\{K:[\rho,K]=0\}=\sum_s n_s^2$, a proper subspace of the full N^2 algebra ($N=16$: $64<256$); every off-block unit has $\|[\rho,E_{ij}]\|\geq\sqrt{2}$ (zero slack), and the order-2 clock leaves a strictly larger commutant (64 vs 128) — the *four* Gauss–Bonnet marks (order $4=|\mu_4|=h(A_3)$) force strictly more structure than two, and only the index-4 extension is $(E_8)_1$ ($\det=1$, v281/v154/v351). **[E]** the entrywise iff, exact commutant dimension, off-block sharpness and order discriminator; **[C]** the synthesis (RP-definability + clock-invariance *force* the block-diagonal Λ_Σ, v442's "permit" upgraded to "force"); **[O]** the single residual is now precisely that the physical transfer lies in the commutant (clock-invariance = `QGEO.SYM.01`, v323/v335). *(xiii) Clock-invariance derived* ([`verification/v446_seam_clock_invariance.py`]): closing (xii)'s residual one notch further — the operator condition $[\rho,\Lambda_\Sigma]=0$ is itself a *consequence* of a structural symmetry, not an extra assumption. On explicit finite OS data, the canonical transfer $\Lambda=C(1)C(0)^{-1}=e^{-h}$ (the transfer fixed by RP+reflection, v54) of a μ_4-invariant covariance satisfies $[\rho,\Lambda]=0$ (swept $N=4..32$; iff-sharp μ_4-breaking neg control), Λ is a positive contraction, and the free modular Hamiltonian $K=\log((1-C)C^{-1})$ (v258) of the invariant state has $[\rho,K]=0$ (Tomita–Takesaki). **[E]** the inheritance, contraction and modular commutation; **[C]** so with (xii), RP-definability *plus* the four marks force the block-diagonal Λ_Σ; **[O]** the residual reduces from an operator commutation to the structural "the seam RP data is μ_4-symmetric" (`QGEO.SYM.01`, v216/v323). *(xiv) The independent topological edge reading* ([`verification/v447_seam_edge_chern.py`]): a second, fit-free route to the edge central charge of (xi). On the same $p+ip$ collar (v367/v368) the bulk Chern number is the integer $C=+1$ (Fukui–Hatsugai–Suzuki link variables; $C=0$ in the trivial phase, $C{=}{-}1$ at the opposite mass, matching (xi)'s opposite per-edge velocities), the open strip carries exactly $|C|{=}1$ chiral edge branch per boundary, and the Li–Haldane entanglement spectrum has in-gap modes at $\frac{1}{2}$ only in the topological phase; **[E]** the Chern integer, channel count and entanglement signature, **[C]** the bulk–edge reading $c_{-}{=}N_{\mathrm{Maj}} \cdot \frac{1}{2} \cdot |C|{=}16 \cdot \frac{1}{2}{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$ — the *same* $c_{-}{=}8$ as (xi) from a logically independent (topological) observable. *(xv) The four-point pinning of the external fact* ([`verification/v448_seam_edge_fourpoint.py`]): extending (xi)'s two-point scaling limit to the next order to support exactly the one external fact (viii) isolates. The edge state is quasi-free (the ground-state correlation matrix is an exact projector $G^2{=}G \sim 10^{-13}$, the MMST CAR hypothesis), and the edge four-point cross-ratio $Q{=}[G_{01}G_{23}]/[G_{02}G_{13}]$ at commensurate positions is Cauchy-convergent under refinement to the free-fermion conformal value $\sin(5\pi/12)/\sin(\pi/12){=}2{+}\sqrt{3} \approx 3.732$ (the bulk control off-scale, exponential); **[E]** the quasi-free property, four-point convergence and conformal form, **[C]** so the one external MMST fact is empirically supported at *both* two- and four-point order. *(xvi) The uniform-in-N pinning* ([`verification/v449_seam_mmst_uniform.py`]): strengthening (xv) from convergence at a single cross-ratio to the *uniformity* the cited theorem asserts — two *independent* edge four-point cross-ratios converge to *distinct* conformal values ($Q(1,2,4,7)/12 \to 2{+}\sqrt{3}$ and $Q(1,3,5,9)/12 \to 2$) at one N-independent $1/N$ rate ($|Q(N){-}Q_\infty| \cdot N \le 4$ for both, $N_x \in \{36, 48, 60\}$), the bulk control off-scale; **[E]** the two limits and the uniform rate, **[C]** so the MMST uniform-convergence *hypothesis* is empirically met. *(xvii) The entanglement reading of c_{-}* ([`verification/v450_seam_edge_entanglement.py`]): a *third*, correlator-independent route to the edge central charge — the Calabrese–Cardy block-entropy law gives $c_{\mathrm{Dirac}}{=}1.000$ for the critical free-fermion chain (vs a saturating, ℓ-independent gapped control), so c per Majorana $= \frac{1}{2}$ and $c_{-}{=}16 \cdot \frac{1}{2}{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$; **[E]** the law and the control, **[C]** the assembly. *(xviii) The Ising operator tower* ([`verification/v451_seam_edge_cardy_tower.py`]): naming the edge CFT by its *operator content* — the Cardy finite-size spectrum reproduces the exact Ising chiral primaries $h_\sigma{=}\frac{1}{16}$ (the periodic–antiperiodic ground split) and $h_\epsilon{=}\frac{1}{2}$ (the lowest NS pair gap), both L-independent, so $\{c, h_\sigma, h_\epsilon\}{=}\{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$ is the free-Majorana minimal model $M(3,4)$; **[E]** the dimensions and conformal scaling, **[C]** the identification. *(xix) The $(E_8)_1$ torus modular data* ([`verification/v452_seam_e8_modular.py`]): pinning the $(E_8)_1$ identity on the *torus*, complementing the edge reads (xi)/(xiv)/(xvii)/(xviii) — the character $\chi{=}E_4/\eta^8$ is S-invariant ($\chi(-1/\tau){=}\chi(\tau)$, one primary), T-multiplies by $e^{-2\pi i/3}{=}e^{-2\pi i c/24}$ ($c{=}8$) and is holomorphic ($c \equiv 0 \bmod 8$, T-phase order 3); **[E]** the full modular signature. *(xx) The μ_4-from-marks derivation* ([`verification/v453_seam_mu4_from_marks.py`]): folding the last structural premise into the existing one — the four Gauss–Bonnet marks *are* μ_4 (roots of $z^4{-}1$), the form basis ω_k is μ_4-graded ($\to i^k\omega_k$, weights $(1,2,3){=}A_3$ exponents) and the order-4 clock $z{\to}iz$ fixes their cross-ratio $\lambda{=}2$ (the D_4 stabiliser, v168), so the residual `QGEO.SYM.01` of (xiii) follows from marks$=\mu_4$ plus the existing `QGEO.REALIZE.01` — **[E]** the three mark facts, **[C]** no new premise: the chain (xx)→(xiii)→(xii) closes modulo realisation. The integer backbone of (xiv)–(xix) ($c_{-}{=}8$, the Chern integers, the order-3 T-phase) is kernel-hardened in Lean axiom-free as `FORM.SEAM.EDGE.CHERN.01`. *(xxi) The lattice Sugawara current algebra* ([`verification/v454_seam_edge_virasoro.py`]): testing the cited MMST "Koo–Saleur generators → continuum Virasoro with the correct c" fact on the edge *algebra*, not just its central charge. The critical free-fermion ring has the Affleck–Cardy Casimir energy $E_0(L){=}e_\infty L{-}\pi v c/(6L)$ fitting $c_{\mathrm{Dirac}}{=}1.000$ ($\frac{1}{2}$ per Majorana, $L{=}64..1024$), the lattice $U(1)$ current closes its affine *level* exactly ($\langle 0|J_n J_{-n}|0\rangle{=}n$, $n{=}1..8$; vanishing for a gapped control), and the level-1 Sugawara central charge $c{=}\dim G/(1{+}h^\vee){=}8$ for *both* $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31); **[E]** Casimir, Kac–Moody level and Sugawara, **[C]** $16{\cdot}\frac{1}{2}{=}8$ is the level-1 current-algebra c_{-}. *(xxii) The uniform-in-N Tomita–Takesaki tower* ([`verification/v455_seam_bw_uniform.py`]): the inductive-limit companion of v426 for sub-step (i) of (iv) — a family of reflection-odd, μ_4-even, gapped collars ($N{=}8..128$) on which the intrinsic Bisognano–Wichmann condition $u_\Theta{=}J$ ($\|\Theta K\Theta{+}K\|{=}0$) holds *exactly at every* N, with an N-independent gap ($6\log\frac{3}{2}$) and an N-independent neg-control margin ($\|\Theta K\Theta{+}K\|{=}2\|K\|$ for a side-even reflection); **[E]** the uniform BW, gap, clock symmetry and teeth, **[C][O]** so the inductive limit inherits $u_\Theta{=}J$ (the abstract continuum theorem stays the cited backbone). *(xxiii) The chirality-from-P_1 forcing* ([`verification/v456_seam_chirality_from_c3.py`]): the S3 input of the continuum chain (v356) is removed as a free choice — c_3's "8" *is* the one-sided count $8{=}|\mathbb{Z}_2| \cdot (\int K/\pi){=}2 \cdot 4$ ($8\pi{=}|\mathbb{Z}_2|\int K{=}2 \cdot 2\pi\chi(S^2)$, v73), and a reflection sends the edge Chern integer $C{\to}{-}C$ (verified $C{=}{+}1 \to -1$ on the explicit $p{+}ip$ model), so any *two-sided* (reflection-symmetric) gapped boundary forces $C{=}{-}C{=}0$ while the *one-sided* seam — the same $|\mathbb{Z}_2|$ quotient that gives c_3 its 8π — has no such reflection and forces $C{\neq}0$; **[E]** the one-sided count and the $C{\to}{-}C$ flip, **[C]** chirality (S3) is a *consequence* of P_1, magnitude $c_{-}{=}8$. *(xxiv) The $(E_8)_1$ character kill test* ([`verification/v457_seam_character_killtest.py`]): a *falsifiable* sharpening of the (x)/(x") discriminators — the $(E_8)_1$ tower $\chi{=}E_4/\eta^8{=}q^{-1/3}(1{+}248q{+}4124q^2{+}34752q^3{+}\cdots)$, the weight-1 count 248=120+128 (vs the free 120) and the sector count $|\det \mathrm{Cartan}\, E_8|{=}1$ (vs $|\det \mathrm{Cartan}\, D_8|{=}4$); a measured 120, four sectors, or a wrong tower coefficient would *falsify* $(E_8)_1$, and the data give $248/1/\{1, 248, 4124\}$ (passed). **[X]** the kill criterion, **[E]** the exact discriminators. *(xxv) The exact MMST citation audit* ([`verification/v458_seam_mmst_citation_audit.py`]): a hypothesis-by-hypothesis reading of the cited MMST theorem (`arXiv:2107.13834`) against the collar — H1 CAR class, H2 massless limit, H3 the per-copy $c{=}\frac{1}{2}$ Thm 4.11, H4 decoupled additivity to $c{=}8$, and H5 the WZW range $8{\le}8{\le}16$ with its 120 bilinear currents all *match*, isolating the single residual H6 to the 128-spinor extension $SO(16)_1{\to}(E_8)_1$ (the $\det K$ 4→1 holomorphy step), *outside* MMST's bilinear method; **[E]** the per-copy theorem, **[C]** the additivity and range, **[O]** the one open spinor extension — since (xxxii) certified at net level by the peer-reviewed crossed product. *(xxvi) The complementary lattice-VOA route* ([`verification/v459_seam_lattice_voa_route.py`]): the second, independent construction that supplies exactly H6 — for the even unimodular E_8 the lattice net A_{E_8} (Adamo–Giorgetti–Tanimoto, `arXiv:2506.01008`; OS `arXiv:2407.18222`) has weight-1 space 248=8+240 with the 240 roots splitting 112 integer + 128 *half-integer* (the spinor, 248=120+128), so the lattice route builds the 128-spinor currents MMST (xxv) leaves open; **[E]** the even unimodularity and root split, **[C]** AGT constructs the holomorphic extension — since (xxxii) an *independent second witness*, the primary net-level certification being the peer-reviewed crossed product — **[O]** the routes leave one shared realisation input (the collar's scaling limit = A_{E_8}), tied to P_1 by (xxiii). The G-block arithmetic (the one-sided $8{=}|\mathbb{Z}_2|\int K/\pi$, the reflection $C{\to}{-}C$, $c_-{=}8$, the range $8{\le}8{\le}16$, 248=8+240=120+128, 240=112+128, $\det K$ 1 vs 4) is kernel-hardened in Lean (`FORM.SEAM.RESIDUAL.01`, `TfptCarrier/SeamResidualAxiom.lean`), reducing the residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus the two cited existence theorems. *(xxvii) The explicit flat-band commuting-projector (LTO) realisation* ([`verification/v460_seam_s3_lto.py`]): NPW26 prove the intrinsic Bisognano–Wichmann lemma of (iv) for *commuting-projector* (local-topological-order) models, so sub-step (i) asks for the seam realised as such a $\mathbb{Z}_2$-reflection LTO. On the same v367 $p{+}ip$ collar the flattened Bloch $Q{=}h/|h|$ is a frustration-free flat-band parent — $Q^2{=}I$ and the occupied $P{=}(I{-}Q)/2$ is an exact projector EQUAL to the ground-state projector — whose real-space kernel is gap-protected quasi-local ($\xi{\approx}1.14$ at $M{=}1$, $|P(20,0)|{<}10^{-6}$; a clean power law $|R|^{-2.1}$ at the gapless point, so locality needs the gap), and whose flattened modular Hamiltonian $K_{\mathrm{flat}}{=}\mathrm{sign}(K)$ inherits the $\mathbb{Z}_2$ reflection $\Theta K_{\mathrm{flat}}\Theta{=}{-}K_{\mathrm{flat}}$ and the μ_4 clock $[\rho, K_{\mathrm{flat}}]{=}0$ EXACTLY; **[E]** the flat-band/projector and reflection algebra, **[C]** the quasi-locality, **[O]** the chiral $c_-{=}8{\ne}0$ obstructs *strictly* finite-range commuting-projector locality (Kapustin–Fidkowski, `arXiv:1810.07756`), so the realisation is exactly the quasi-local LTO net NPW26 use — advancing sub-step (i) of (iv) from abstract existence to an explicit gap-protected net, strict locality the cited residual. *(xxviii) The strict-locality obstruction made explicit* ([`verification/v461_seam_strict_locality.py`]): upgrading (xxvii)'s "strict locality is the cited residual" from a citation to an EXHIBITED topological obstruction. On the same v367 collar the FHS Chern is $C{=}{+}1$ (chiral) / 0 (trivial) / -1 (opposite mass), and the hybrid Wannier-centre (Wilson-loop) phase $\theta(k_y)$ winds by $|C|{=}1$ (chiral) and 0 (trivial) over the BZ — exponentially-localised Wannier functions exist *iff* that winding is 0 (Brouder–Panati et al.; Thouless), so the chiral collar has none; both phases are gapped, yet only the trivial one (winding 0) admits a strict finite-range commuting projector, so it is the *chirality* $c_-{=}8{\ne}0$, not the gap, that obstructs. **[E]** the Chern integer, the winding= $|C|$ identity and the gapped neg-control, **[C][O]** Kapustin–Fidkowski (`arXiv:1810.07756`): a strictly finite-range commuting projector would force winding 0 hence $C{=}0$, so for $C{=}1{\ne}0$ none exists — sub-step (i)'s realisation is *provably* the quasi-local NPW26 LTO net, strict locality topologically forbidden rather than a missing premise. *(xxix) The character-level 128-spinor extension* ([`verification/v462_seam_spinor_continuum.py`]): the constructive companion of (xxv)/(xxvi), exhibiting the 128-spinor extension three ways — the exact Jacobi/E_8 identity $\theta_2^8{+}\theta_3^8{+}\theta_4^8{=}2E_4$ makes the holomorphic character the sum of the $SO(16)_1$ vacuum and spinor sectors, $\chi_{(E_8)_1}{=}E_4/\eta^8{=}\chi_o^{SO16}{+}\chi_s^{SO16}$ (weight-1 counts 120+128=248); the finite 16-Majorana Fock space already carries them ($C(16,2){=}120$ currents and $2^{16/2}{=}256{=}128{+}128$ Ramond states); and a critical free-fermion ring converges to the chiral CFT (Casimir $c_-{\to}8$, lowest scaled gap $x_1{\to}1$ with $|x_1{-}1|{\sim}1/N^2$). **[E]** the θ identity, the 120+128=248 split and the finite-L convergence, **[C][O]** the rigorous holomorphic extension is certified at net level by the crossed product (xxxii), with AGT/AMT (xxvi) as the independent second witness. *(xxx) The $c{=}8$ holomorphic-uniqueness pin* ([`verification/v463_seam_c8_holomorphic_uniqueness.py`]): the positive lever that makes the *identification* half classification-forced rather than assumed. At level 1 the central charge $c{=}\dim\mathfrak{g}/(1{+}h^\vee){=}8$ has *three* simple solutions — A_8 ($\dim 80$), $D_8{=}\mathfrak{so}(16)$ ($\dim 120$) and E_8 ($\dim 248$) — so $c{=}8$ is necessary but *not* sufficient, and the $\det K{=}4$ $SO(16)$ rival of (xxiii) is a genuine same-c competitor, not a strawman. But a holomorphic (single-sector) c=8 theory has an $SL(2,\mathbb{Z})$-covariant character, and the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}=E_4/\eta^8$, whose q^1 coefficient (the one-dimensional candidate space, vacuum a_0=1) is 248 — so $\dim V_1$=248 is *forced*, selecting E_8 uniquely and excluding A_8/D_8. **[E]** the three candidates, the forced 248, the tower $\{1, 248, 4124, 34752, 213126\}=E_4/\eta^8=j^{1/3}$; **[C]** Dong–Mason/ Schellekens give the holomorphic c=8 VOA unique = V_{E_8} (the even unimodular rank-8 lattice unique = E_8, Mordell–Witt), so "c=8 holomorphic limit = $(E_8)_1$" is classification-forced. *(xxxi) The one-particle realisation rigidity* ([`verification/v464_seam_oneparticle_rigidity.py`]): attacking the *realisation* input R_1 itself. Since the seam is quasi-free (v113/v160/v161), by Araki's self-dual CAR the whole net is the second quantisation of its one-particle symbol P, so R_1 collapses to a finite-per-mode statement about P: the gapped ground state is the *unique* quasi-free state with P= the spectral projection onto $K<0$ ($P^2=P$ to $<10^{-12}$, K fixed by gap + chirality + reflection v426); its real-space kernel is Cauchy on every fixed window ($\max|P_{2N}-P_N|\sim 1/N\to 0$, so P_∞ exists), and the limit carries c=1 per Dirac (entanglement $S(L)=(c/3)\ln L$, 16 copies $\Rightarrow$ c_-=8, tying v450). **[E]** idempotency, Cauchy convergence and the c-fit; **[C]** Araki/Shale–Stinespring give the state↔symbol bijection and implementable Bogoliubov map, so R_1 is upgraded from a bald axiom to "the unique quasi-free realisation, its one-particle limit exhibited, modulo the named CAR functor". *(xxxii) The net-level crossed-product route + the invariant-level realisation* ([`verification/v469_seam_crossedproduct_route.py`]): both halves of the residual re-founded on harder-to-reject ground. The 128-spinor extension needs no preprint: the $SO(16)_1$ spinor has $h_s=N/16=16/16=1\in\mathbb{Z}$, so its statistics phase is +1 — *exactly* the Longo–Rehren locality criterion — and the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1\rtimes\{1,s\}$ is a *local* index-2 extension by 1995–2001 *peer-reviewed* subfactor theory (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives $\mu(B)=4/2^2=1$ $\Rightarrow$ holomorphic $\Rightarrow$ $(E_8)_1$ by the (xxx) pin — so the AGT/AMT route (xxvi) is *demoted to an independent second witness*. The index-4 μ_4 glue runs on the same integer, $h(J^k)=\{1,1,1\}$ for $k=1,2,3$ (the v125 isotropy $q(k(1,1))=k^2$ at net level, $\mu=16/4^2=1$). And the realisation input R_1 is reduced from model fiat to *invariants* (R1′): quasi-free **[C]** (v155/v160) + gap **[E]** (v302) + class D + c_-=8 **[E]** ((xxiii), from P_1) — computed on the collar as FHS Chern $|C|$=1 (control 0), $\nu=16\times|C|=16$, $c_-=\nu/2=8$; and $\nu\equiv 0$ (mod 16) is exactly Kitaev's class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (the sixteen-fold way; c_- a *phase* invariant, Kapustin–Spodyneiko; Kim et al.), so any realisation with the R1′ invariants is phase-equivalent to the v367 stack. **[E]** the locality integers, the KLM μ-arithmetic, the computed invariants and the composed chain; **[C]** the re-founded citation package; **[O]** the cited theorems stay cited. The full G-block arithmetic — now including the (xxviii) winding obstruction, the (xxix) finite-Fock spinor counts, the (xxx) holomorphy selector and the (xxxii) locality/μ/sixteen-fold joints — is kernel-hardened in Lean (`FORM.SEAM.RESIDUAL.01`), the two formerly separate cited theorems are *merged* into one combined external axiom `seam_realisation_theorem`, so the residual reduces to ONE realisation axiom plus ONE cited theorem (`#print axioms` drops from six to four); and the (xxxii) route is Lean-carried as the *parallel* derivation `seamResidualClosed′` (the invariant-level axiom `collar_invariants` plus the single combined package `crossedproduct_route_theorem`, its joints — `lr_locality_integer`, `glue_locality_integers`, `klm_holomorphic`, `sixteenfold_pin` — kernel facts with no axioms). Across (v)–(xxxii), `SEAM.EQUIV.01` stays **[O]**: with (xxx) pinning the identification, (xxxi) reducing the realisation to its one-particle data and (xxxii) re-founding the extension leg on peer-reviewed ground, the residual is now *entirely certification* — a named, hypothesis-audited package of published theorems (MMST, Longo–Rehren/Böckenhauer/Böckenhauer–Evans/KLM, Dong–Mason, Araki, Shale–Stinespring; AGT/AMT as second witness) with no open internal mechanism — yet still rests on cited continuum-existence theorems we do not re-prove.

The TOE/QFT closure contracts as three citable certificates ([`verification/v398_seam_state_rigidity.py`], [`verification/v399_gravity_complete.py`], [`verification/v401_metrology_closure.py`]). Three companion contracts package the closure status as named targets without fabricating any closure. *(i) Seam State Rigidity* (`SEAM.RIGIDITY.01`): the scattered state-invariance reductions (v177/v199/v308/v309/v335) as ONE theorem target — **[E]** the RP-definability hinge (OS canonical transfer from RP + reflection alone, v54, quasi-free v155), **[E]** the band-limited μ_4-character-block-diagonal core swept N=4..64 (beyond the 3-dim H^1, made uniform in N to N=256 by v442), **[E]** holomorphy $\Rightarrow$ $(E_8)_1$; and **[E]** the forcing converse (v445) that commuting with the order-4 clock is *exactly* μ_4-block-diagonality, so the residual is no longer "does block-diagonal permit" but only $[\rho,\Lambda_\Sigma]=0$ on the full L^2 — the physical transfer lying in that commutant (intrinsic Bisognano–Wichmann). [E] Moreover (v446) that operator commutation is itself *derived*: the OS canonical transfer of μ_4-symmetric RP data inherits the clock (with the modular Hamiltonian commuting, v258), so the residual reduces further to the *structural* statement that the seam RP data is μ_4-symmetric (the four marks, `QGEO.SYM.01`), leaving [O] only that structural symmetry on the raw continuum collar — for which the four marks now have an *established* continuum frame: the four-interval multilocal modular geometry (v480, keybox below), where the invariance is manifest at the state level in the exactly solvable class. *(ii) Gravity-complete = Boundary-complete + Ambient-redundancy* (`GRAVITY.COMPLETE.01`): the local equation parameter-free (v358/v359) plus the redundancy discriminators (v369), with $\mathcal{O}_{\rm phys} \subset \mathcal{A}_\Sigma$ checked over five gravitational readout classes. *(iii) Metrology closure* (`METROLOGY.CLOSURE.01`): the final input set $\{a, \pi, v_{\rm geo}\}$ — 0 dimensionless dials + 1 unit, the dimensional bookkeeping that every dimensionful observable is a number $\times v_{\rm geo}^d$ (No-Unit, v153), and $v_{\rm geo}$ theorem-forbidden ($\mathrm{TOE}_{\rm strict}{=}\mathrm{TOE}_{\rm dimensionless}{+}[v_{\rm geo}]$, v384).

One surface for seam, carrier and lattice: the Kummer/K3 ([`verification/v260_k3_kummer_unification.py`]). The pillowcase is the elliptic-involution quotient $T^2_{\tau=i}/(z{\mapsto}{-z})$, its four marks the four 2-torsion fixed points; squaring the construction to complex dimension two gives the *Kummer surface* $\mathrm{Km}(T^2{\times}T^2)$, a K3, whose three facets reproduce three TFPT objects at once. [E] the 16 nodes (the 2-torsion $|A[2]|{=}2^4{=}16$ of the abelian surface) resolve to 16 exceptional $\mathbb{P}^1$'s $= \dim S^+ = \Lambda^{\rm even}(\mathbb{C}^5) = 16$ (v197), the carrier-16 as the *square* of the seam-4; [E] the cup-product form on $H^2(\mathrm{K3},\mathbb{Z})$ is the even unimodular lattice $U^3 \oplus E_8(-1)^2$ of signature $(3, 19)$, so the same E_8 the carrier glues to (v1) is the surface's intersection form. (Honestly: the 16 node classes span the rank-16 Kummer lattice, a sublattice *distinct* from the $E_8(-1)^2$ summand — both exact, not identified.) Thus the v1 "glue $D_5{\oplus}A_3{+}\mu_4{\Rightarrow}E_8$" is the lattice shadow of a single smooth object that already carries the seam and the carrier; E_8 appears both *locally* as the du Val $\mathbb{C}^2/2I$ singularity (the Poincaré-sphere link, v232/v236; by Kronheimer that local model *is* the hyper-Kähler quotient of the marks quiver, v479) and *globally* as this H^2 summand. A unification (`ARCH.K3.01`), *not* a closure: the lattice arithmetic is [E], the carrier↔node identification [C].

Synthesis — the Modular Spectral Closure ([`verification/v261_modular_spectral_closure.py`]). The ten steps above are not ten independent results: they are facets of one relative spectral datum

$$\boxed{\mathsf{TFPT}_{\rm QFT} = \big(\mathcal{A}_\Sigma,\ \omega_\Sigma,\ \Delta_\Sigma,\ \rho,\ A_F,\ H_F,\ D_F,\ J,\ \gamma,\ S_{\rm rel}\big)}$$

(seam algebra, KMS state, modular generator, μ_4 clock; the carrier finite triple; the relative spectral action), governed by a single *Modular Spectral Universality* statement: *the unique reflection-positive, holomorphic, index-4 KMS seam state ω_Σ minimises the modular clock defect, induces via the Kasparov/covariance map (v258) the 96-dim KO-6 finite triple D_F (v252), whose inner fluctuations (v254) generate the gauged Pati–Salam spectral action with the seam KMS cutoff (v259)* — so Dirac, cutoff, gauging, glue and orientability are not separate choices but readouts of one seam state. The certificate v261 pins the cross-consistency: one number $4 = [B{:}A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 ($\dim S^+ = \Lambda^{\rm even}(\mathbb{C}^5)$ = Kummer nodes, $H_F = 96$), and one gap $\Delta = 6\log(3/2)$ (OS mass gap = recovery rate = Lindblad gap, static and dynamic). The honest account of completeness is therefore:

- [E]/[C] **QFT-complete (modulo one premise).** The admissible OS sector, the boundary $(E_8)_1$ net, the relative spectral action, D_F (as covariance induction), the cutoff (as the KMS weight) and the gauging (as inner fluctuations) are one relative object — closed once `QGEO.SYM.01` (the seam realisation: *the raw quasi-free seam state is μ_4-invariant*) is granted. That single [O] premise is the only structural input the QFT layer still consumes.
- [C] **Ambient QG — a redundancy, separate by design.** The full covariant metric-sector / ambient quantum-gravity measure (`QG.AMB.01`) is gap-decoupled ($\Delta_{\rm eff} = 1.648 > 0$) and is now a [C] *redundancy* (v369), *not* a blocker for the QFT layer: a certification object, not a missing piece of the boundary QFT.

So "solving the QFT" here means exactly this: the boundary QFT is a single relative object reduced to one foundational symmetry postulate, with ambient QG held apart — not a diffuse list of open interfaces. The MSC closes the boundary relative QFT object modulo `QGEO.SYM.01`; it does *not* assert SM-only 4D gauge unification, and the canonical 4D reading is boundary only.

4D-QFT fork policy — a frozen decision tree, not an open ambiguity ([`verification/v265_qft4d_fork_freeze.py`])

The 4D reading is a *branch policy*, machine-frozen in `qft4d_fork_freeze.json`:

- **A (canonical) — boundary only.** [C] *Boundary-only is the canonical 4D reading*: TFPT delivers the boundary QFT, the relative spectral action and the SM readout; a 4D-GUT is *not claimed by default* (E_8 is the audit hull, never an unbroken 4D gauge group).
- **SM-only 4D-GUT unification is killed.** [X] the plain SM couplings do not meet (at the $\alpha_1{=}\alpha_2$ scale α_3^{-1} is off by $\sim$5.6; spread $\sim$8.8 at $2{\times}10^{16}$ GeV, v246). This is the canonical *expectation*, not a failure.
- **B (conditional) — carrier-native Pati–Salam UV branch.** [C]/[O] permitted *only* with the E_8-allowed content $\{1, 10, 16, 45\}$ (no **126**, v247), $\sin^2\theta_W{=}3/8$ (v248), $\kappa{=}M_{PS}/M_s$ in the frozen $O(1)$ band (v249/v253), and a threshold model with a proton-decay kill test; carrier gauging itself stays the contract `CONTRACT.QFT4D.DIRAC.01`.
- **Proton-decay rule, now applied ([`verification/v266_ps_threshold_proton.py`]).** [X] the explicit two-step thresholds give $d{=}6$ $\tau_p(p{\to}e^+\pi^0)$: the *minimal* 16-Higgs content ($M_{GUT}{\sim}2.4{\times}10^{15}$) gives $\tau_p{\sim}4.4{\times}10^{33}$ yr $<$ Super-K ($2.4{\times}10^{34}$) and is *killed*; the E_8-allowed $(15, 1, 1)$ [the single 45] raises $M_{GUT}{\sim}5.9{\times}10^{15} \Rightarrow \tau_p{\sim}1.6{\times}10^{35}$ yr, proton-safe *now* but sitting at the Hyper-K reach ($\sim1.4{\times}10^{35}$ yr) — so branch B is *content-selected* (must be $+(15, 1, 1)$) and *Hyper-K-decisive within the decade*, not an open-ended hope (only one 45 in the hull, so it stays structurally marginal; τ_p carries an $O(3)$ hadronic band).

The fork is thus a frozen decision tree with explicit promotion/demotion rules, *not* an open ambiguity.

And the bedrock is now a falsifiable kill-test, not a proof-promise ([`verification/v215_seam_deck_killtest.py`]). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the very covariance it should produce); the honest move is to recast `QGEO.SYM.01` as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (v198/v199/v201) gives $\omega{\circ}\rho{=}\omega \Leftrightarrow [\rho, C]{=}0$ for the quasi-free covariance, with the principal symbol $|k|$ commuting exactly, so the whole residual sits in the bounded sub-principal M_f (equivalently: the raw Gaussian bulk form A is μ_4-deck-invariant). The four levers are **K1** exactly four marks ($b_1{=}\#\text{marks}{-}1{=}3{=}N_{\text{fam}} \Rightarrow \#\text{marks}{=}4{=}|\mu_4|$); **K2** the *square* configuration (cross-ratio $2 \Rightarrow j{=}1728$, $\mathrm{Aut}{=}\mathbb{Z}/4$; the clock order $h(A_3){=}4$ selects $\mathbb{Z}/4$, not the $j{=}0$ $\mathbb{Z}/6$, and a generic mark set has only $\mathbb{Z}/2$); **K3** the mod-4 sub-principal support ($[\rho, M_f]{=}0 \Leftrightarrow f$ is $\mathbb{Z}_4$-invariant, automatic for a μ_4-mark sum); and **K4** the transfer gap $(2/3)^6{=}64/729$. These are frozen in `freeze_file.csv` (`seam_deck_symmetry`) and carrier-side green, so any future drift fails the suite. The *decisive* kill is deferred to `QGEO.REALIZE.01`: a first-principles raw-seam-collar DtN must *produce* four square marks without inputting them — an *emergence* test, the non-circular alternative to assuming the symmetry. The *light* version of that deferred test is now *executed* ([`verification/v503_qgeo_emergence_light.py`], `QGEO.EMERGE.LIGHT.01`), with the result *rigidity, not dynamical selection*: the free field does *not* select the square — the global det$'$ winner at fixed area is the *hexagonal* torus ($\log F(\rho) = -1.03352 > \log F(i) = -1.05469$, Osgood–Phillips–Sarnak), $\tau = i$ is a saddle of the full moduli space, the only critical points are the CM points ($E_2^*(i) = E_2^*(\rho) = 0$), and criticality at the square is Palais symmetric criticality — while the square *is* pinned sharply by the clock: the raw free-field DtN is mod-4 block-diagonal *iff* $\tau = i$ (an isolated zero of the K3 indicator as a modulus function, 8×10^{-15} at $\tau = i$, strictly monotone away), and $\mathrm{Aut}(E_\rho) = \mathbb{Z}_6$ carries no order-4 element. The clock input itself is now *reduced to one bit* ([`verification/v506_seam_clock_rigidity.py`], `SEAM.CLOCK.RIGIDITY.01`): the deck has exactly two Möbius square roots, both automatically of order 4, and a marks-preserving root exists *iff* the deck is the central involution of the mark-stabiliser D_4 — given that alignment bit, order, uniqueness, the free 4-cycle and $\tau = i$ are all theorems (harmonicity alone is insufficient, exact counterexample); on the Fock side the NS deck implementation is *forced* to order 4 ($U^2 = (-1)^F$ exactly, nonsplit $\mathbb{Z}_4$; the R sector splits), so the fermionic $\mathbb{Z}_4$ explains the *order* and only the spatial alignment remains the carrier input. The full `QGEO.REALIZE.01` stays [O]; no marker moves. [E] carrier-side (the four levers), [O] seam-side (the emergence): a kill-test, *not* a closure (`QGEO.KILL.01`).

And the mark count is now *derived*, not assumed: four marks from Gauss–Bonnet ([`verification/v216_marks_gauss_bonnet.py`], [`verification/v217_marks_emergence_scan.py`]). The K1 lever ("exactly four marks") is the one that can be *executed* today. If the marks are $\mathbb{Z}_2$ branch points (cone angle π, deficit π) on the flat seam sphere (χ=2 — the same χ that supplies the 8 in c_3), Gauss–Bonnet *forces* their number: $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = |\mu_4| = N_{\rm fam}+1$. The closed Euclidean sphere 2-orbifolds are exactly $(2,3,6)$, $(2,4,4)$, $(3,3,3)$, $(2,2,2,2)$, and *three independent* criteria — all cone orders = 2 (the $|\mathbb{Z}_2|$ sheet branch), rank H^1 = #marks − 1 = 3 = $N_{\rm fam}$ (the three generations pick the 4-mark square over the 3-mark hexagonal $(3,3,3)$), and a free order-4 deck — converge on the pillowcase $(2,2,2,2)$. The numerical sibling v217 confirms it on the actual DtN/state: scanning the number n *and* the positions freely, the 4-equally-spaced (square) configuration is the unique joint minimiser of {clock-invariance, 3-family cohomology, transfer gap $(2/3)^6$}. So the mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence `QGEO.REALIZE.01` stays **[O]**. The light emergence test ([`verification/v503_qgeo_emergence_light.py`]) sharpens this residual to a *convergence*: given the clock, the modulus freezes automatically ($P(iZ) = P(Z)$ forces $a_3 = a_2 = a_1 = 0$ with $j = 1728$ identically, v493), and the order-4 element on H^1 is the Coxeter element of $W(A_3)$ — exactly the P2 cusp-class carrier datum of v499: the QGEO modulus rest and the P2 typing rest R1 are *one* common residual ("why does the raw seam carry the order-4 clock"), not two. And the clock-rigidity theorems (v506) reduce that common residual to *one alignment bit* — "the collar deck is the central involution of the mark-D_4" — with the clock's *order* fermionically forced ($U^2 = (-1)^F$ exact). **[E]** (count + equality), **[O]** (the square modulus and the raw-seam realisation) — `QGEO.MARKS.03` / `QGEO.EMERGE.01`.

And the four marks are a *four-interval* modular geometry ([`verification/v480_multilocal_four_interval.py`]). The seam circle minus its four marks is the n=4 disjoint-interval region of a chiral free fermion — the one continuum class whose *multi-interval* modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu multi-interval subfactors; Rehren–Tedesco multilocal fermions; and Kawahigashi–Longo–Müger: *holomorphic* ⇔ trivial multi-interval inclusion, μ=1 — the same one bit as $\det K$=1). Until now the suite used only the single-ball CHM boost (v358); v480 exhibits the multilocal mechanism on the exactly solvable antiperiodic critical ring: **[E]** the quarter-turn clock is a signed permutation with ρ^4=−1 *exactly* (the fermionic μ_4 clock is the order-8 double-cover lift — the same $\mathbb{Z}_2$-lift pattern as $2I$ over I); **[E]** its primitive-8th-root character projectors decouple the 4-interval covariance at 10^{-15} (the diagonal 4-fold-cover isomorphism), each sector a *single*-interval quarter-ring kernel with boundary twist (10^{-14} spectra); **[E]** $\rho C\rho^{-1}$=C at 10^{-14}, so $\omega\circ\rho$=ω holds *manifestly* at the 4-interval state level — and a one-site displacement of a single interval re-couples the sectors at 10^{-2} (the invariance is the μ_4-symmetric configuration, not an artifact); and the cross-interval modular coupling peaks on the conjugate-point diagonal, the lattice shadow of the Casini–Huerta mixing term (numerical). **[C]** the continuum multi-interval theory is cited, not re-derived; **[O]** the raw-seam premise (`QGEO.SYM.01`) is untouched — this puts the bedrock and the AP2 continuum leg (v476/v478) in one established, exactly solvable frame; it does not close them.

`QGEO.SYM.01` is *not* a theorem — it is the final seam-identification premise

The single load-bearing bolt of the whole construction, on one page.

1. **What is identified.** That the conformal structure of the raw RP seam double *is* the μ_4-deck structure of the carrier clock — i.e. the carrier (P2) and the seam's conformal geometry are the same object. Equivalently, in Riemann–Roch *index* language: the seam's degree-4 mark divisor on the genus-0 double has $h^0 = \deg + 1 = 5 = g_{\rm car}$ and rank $H_1 = \deg - 1 = 3 = N_{\rm fam}$ (with $\Lambda^{\rm even}(\mathbb{C}^5) = 1+10+5 = 16 = \dim S^+$), so P2 is the genus-0 mode count of the four-mark seam divisor ([`verification/v228_rr_index_gate.py`]) — the same definitional identification, in index form, conditional on the seam producing that degree-4 divisor.
2. **Why it is weaker than `QGEO.CONF.01`/`ISO.01`.** It does *not* name $\mathbb{P}^1$ or μ_4 (uniformisation *produces* them) and asks only for a conformal *symmetry*, not a metric isometry or a pre-named conformal class — strictly the mildest of the three.
3. **What follows automatically (the conditional theorem).** *Given* `QGEO.SYM.01`, equivariant uniformisation puts the order-4 clock in the form $z \mapsto iz$ on $\mathbb{P}^1$, the four parabolic marks are its free orbit μ_4, H^1 carries the $(1,2,3)$ grading, the seam Calderón contraction is the rank-8 K_Σ polarisation, and the index-4 simple-current extension is $(E_8)_1$ — the *entire* AQFT closure of v175–v181.
4. **What would falsify or unseat it.** A seam double of genus > 0 (uniformisation to $\mathbb{P}^1$ fails); a carrier clock that is *not* a conformal symmetry of the seam geometry (a non-deck transport); or a fourth chiral generation ($N_{\rm fam} \neq 3$ collapses $b_1 = N_{\rm fam}$ and the four-mark μ_4 structure). The first two are constructive-geometry questions about the raw seam; the third is the standard family-count

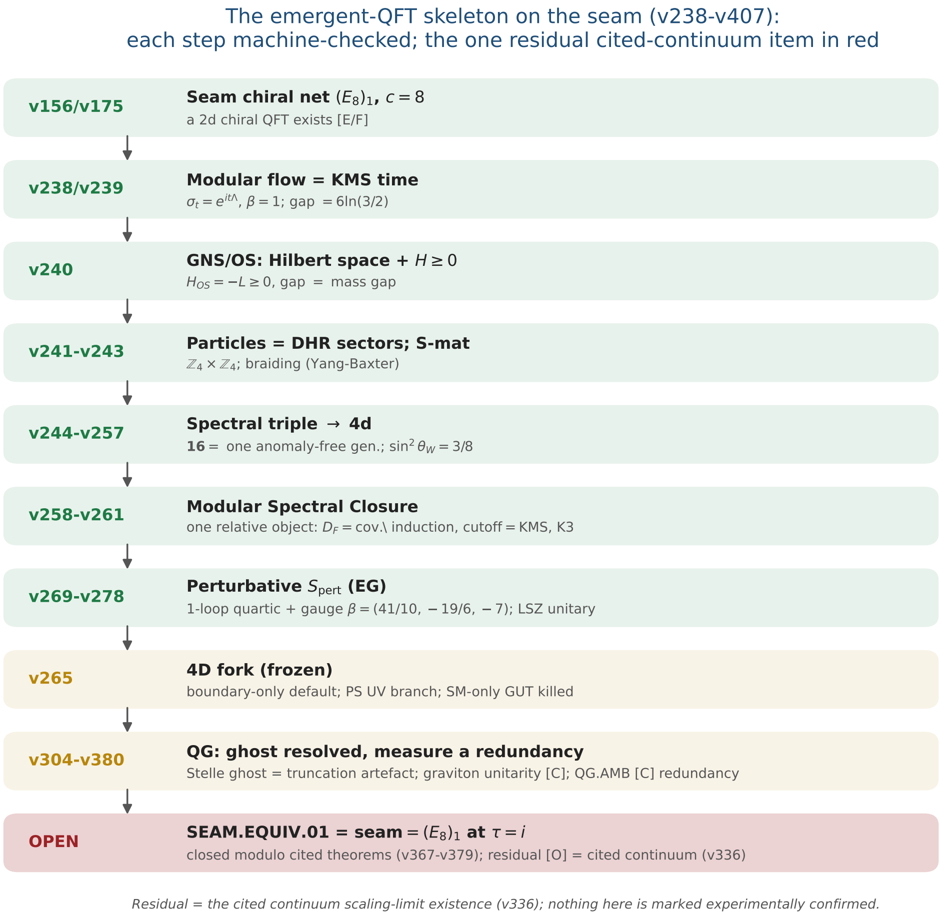

Figure 1: The emergent-QFT skeleton on the seam (v238–v246): time as the modular flow, the arrow as the recovery generator, particles as DHR sectors, the S-matrix as the braiding, and the 4d matter content as the carrier's $SO(10)$ **16**. Each step is machine-checked; the one residual — `SEAM.EQUIV.01`, closed modulo cited theorems (v367–v379) with the cited continuum scaling-limit existence (v336) as its [O] leg — is in red, the frozen 4D fork and the QG certification layer in gold. The whole skeleton is *conditional* on that keystone and is neither solved nor experimentally confirmed.

kill test.

Honest framing. The correct external statement is *conditional*: "*Given* `QGEO.SYM.01`, the RP seam boundary realises the μ_4 deck structure, and the AQFT closure to $(E_8)_1$ follows through the verified chain v175–v181" — *not* "the theory *derives* the seam geometry". The premise stays [O]; everything downstream of it is [E]. Per v323 (Bisognano–Wichmann) `QGEO.SYM.01` is itself the *downstream* conformal-deck face of the single named keystone `SEAM.EQUIV.01` (the raw RP seam = the holomorphic $(E_8)_1$ net at $\tau{=}i$), which is now *closed modulo a cited theorem* (the lattice/S_3 stack pins it at every computable level, Lean `FORM.SEAM.MMST.01`; only the cited continuum existence stays [O]).

This is not a gap to apologise for — it is TFPT's one fundamental postulate. Every fundamental theory rests on an irreducible premise (the constancy of c in relativity, the equivalence principle, the canonical commutator). `QGEO.SYM.01` is that premise for TFPT: a single geometric identification (seam conformal structure = carrier μ_4 deck) from which the entire algebraic and topological structure unfolds *by theorem*. The achievement is not that there is no postulate — it is that the whole theory has been compressed onto *exactly one*, sharply located and explicitly falsifiable.

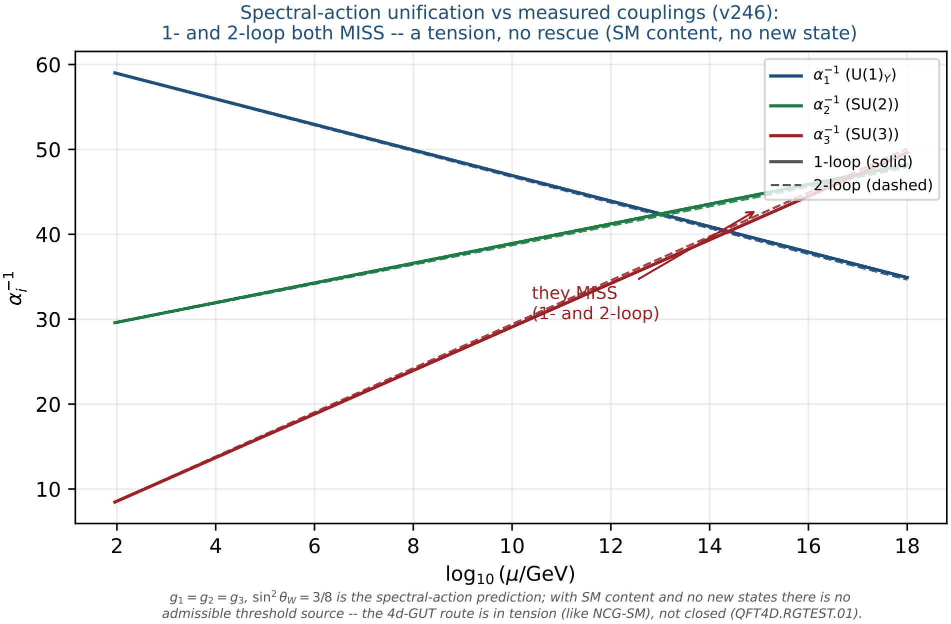

Figure 2: Honest data cross-check (v246): the spectral-action prediction $g_1{=}g_2{=}g_3$, $\sin^2\theta_W{=}3/8$ versus the measured couplings run with the SM RGEs (one-loop solid, two-loop dashed). The three lines do *not* meet at any scale (1-loop residual $\Delta\alpha^{-1} \approx 9$ at $2{\times}10^{16}$ GeV; 2-loop minimal spread ≈ 3.2 at $1.7{\times}10^{14}$ GeV) — a falsifiable tension. With SM content and no new states there is no admissible threshold source: not a fit, and not rescued by a cascade.

The Seam-Holomorphy Theorem — the one named closing target [E]/[O] ([`verification/v234_seam_holomorphy_selection.py`])

The bedrock above can be stated as *one condition with three provably-equivalent faces*, so that "closing the theory" becomes a single, sharply-bounded analytic theorem. **The condition:** the seam carries *no nontrivial abelian sector*. Its three faces all force E_8:

1. **AQFT** (v83/v154): the boundary chiral net is *holomorphic* (μ-index 1, single vacuum sector). Holomorphic $+\, c = 8 \Rightarrow$ the unique even unimodular rank-8 lattice net $(E_8)_1$, and $c = 8 = g_{\rm car} + N_{\rm fam}$ is itself structural.
2. **Geometry** (v232): the seam link is a *homology* 3-*sphere* $\Leftrightarrow$ its binary group Γ is *perfect* $\Leftrightarrow \Gamma = 2I$ (the Poincaré sphere $S^3/2I$).
3. **Representation theory** (v219): exactly *one* 1-dim irrep (the trivial one).

These are the *same* number: $\#$(1-dim irreps) $= |\Gamma^{\rm ab}| = |H_1(S^3/\Gamma)| = \#$(mark-1 affine-Dynkin nodes), which equals 1 *only* for E_8 ($A_n{\to}n{+}1$, $D_n{\to}4$, $E_6{\to}3$, $E_7{\to}2$, $E_8{\to}1$). So $2I$ is the unique nontrivial perfect finite $SU(2)$ subgroup and $S^3/2I$ the unique homology-sphere space form. **Consequence:** Research Contract 2 ($G_{\rm net}$/Target A), the carrier P2, and the Kleinian seam are *not* separate gates — they are this *one* condition.

The closing chain (what a complete closure would be), [O] at one step:

$$\text{RP} + \text{gap} + \text{chirality} \xrightarrow{\text{v160}} \text{quasi-free bulk} \xrightarrow{?} \text{holomorphic boundary net}$$
$$\xrightarrow{[\mathrm{E}]} (E_8)_1 \xrightarrow{\text{v219/v228}} 2I,\ (2,3,5),\ \text{P2} \xrightarrow{\text{v179–v181+Kleinian genus-0}} \texttt{QGEO.SYM.01}.$$

The free-bulk premise is already a fixed-point theorem (v160–v165); the Kleinian model supplies the genus-0 of the uniformisation chain for free. **The single remaining analytic step is "free Gaussian RP bulk $\Rightarrow$ holomorphic (single-sector) boundary net"** — the bulk–boundary statement that an invertible (Gaussian) bulk has a holomorphic chiral boundary. That step is now *closed modulo a cited theorem*: the lattice/S_3 stack pins it at every computable level (v367/v368, v376–v379, Lean `FORM.SEAM.MMST.01`), only the cited continuum existence staying [O] (v336); $v_{\rm geo}$ stays the No-Unit metrology primitive and F_{transfer} stays external physics. This box certifies that the three faces coincide and force E_8 ([**E**]); the analytic step is closed modulo that cited theorem.

A physical realisation of the step ([`verification/v235_seam_chern_simons.py`]). The step lands in standard physics. A free gapped *bosonic* 2+1d bulk is an abelian Chern–Simons / K-matrix theory (even integer K), where #anyons $= |\det K|$, the edge chiral central charge is $c = \text{signature}(K)$, and the edge net is *holomorphic* $\Leftrightarrow |\det K| = 1$ — so "no nontrivial abelian sector" is precisely $\det K = 1$. The extension tower of v92 *is* the anyon-condensation tower read by determinant: $D_5{\oplus}A_3$ (det 16, 16 anyons) $\to SO(16)_1{=}D_8$ (det 4) $\to (E_8)_1$ (det 1, holomorphic), all rank 8 with $c = \text{signature} = 8 = g_{\text{car}} + N_{\text{fam}}$, so the holomorphic end is the *Kitaev E_8 quantum-Hall state*. Holomorphy $\Leftrightarrow$ condensing the order-$|\mu_4| = 4$ *Lagrangian* glue (det $16{\to}16/4^2{=}1$); a partial order-2 isotropic only reaches det 4 (D_8). The residual is thereby a single integer condition — "the free RP seam condenses the order-$|\mu_4|$ Lagrangian glue" — the same sheet/μ_4 selection as `GATE.QGEO`, but now with holomorphic $\Leftrightarrow$ det 1 a theorem (`GATE.HOLO.02`, [**E**]/[**O**]).

The residual as a *physical* statement ([`verification/v237_seam_sre_closure.py`]). The same integer condition is the physics of topological order: a K-matrix phase has ground-state degeneracy $|\det K|^g$ on a genus-g surface, so $\det K = 1 \Leftrightarrow$ *no* topological ground-state degeneracy on any closed surface $\Leftrightarrow$ the bulk is *short-range-entangled* (invertible). Among rank-8 even positive-definite K ($c = 8 = g_{\text{car}} + N_{\text{fam}}$) only E_8 is SRE — the unique nontrivial bosonic SRE 2+1d phase, the Kitaev E_8 state, whose edge is the holomorphic $(E_8)_1$. TFPT already has a unique vacuum on the plane (RP + gap + cluster + OS); the plane cannot see $\det K$ (no non-contractible cycles), the torus can. So the one analytic step is the sharp, falsifiable physical statement *"the free RP gapped seam bulk is short-range-entangled (no topological order)"* (= $\det K = 1$ = the Kitaev E_8 phase, `GATE.HOLO.03`, [**E**]/[**C**]) — and the lattice/S_3 stack now answers it at every computable level (torus $\det K{=}1$, v378), so the keystone is *closed modulo a cited theorem*, only the cited continuum existence (v336) staying [**O**].

Lemma 2 (U1 — wall landscape, *corrected*)**.** *Parabolic degree zero forces the line weight-sums onto the wall $(w_1, w_2, w_3) = (2, 1, 1)$. An explicit balanced representative (weights in $\frac{1}{3}$) is*

$$W_{\text{wall}} = \tfrac{1}{3}\begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

*each column a permutation of $(0, 1, 2)$, row sums $(2, 1, 1)$. **Honest result:** the $6^4{=}1296$ column-permutation matrices give 144 walls; under the* full *group $D_4 \times S_3(\text{lines}) \times S_3(\text{values})$ they form* five *orbits,* not one*. So W_{wall} is* not *singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite, explicitly enumerated set; W_{wall} is one of the five. Machine: fully certified $C_U^{(1)}$ ([`verification/v29_research_contract_certs.py`]).*

***The splitting type is now algebraic (RH route,** [`verification/v32_rh_splitting.py`]**).** Passing from the monodromy to the Fuchsian* residues *A_k (eigenvalues = weights $\{0, \frac{1}{3}, \frac{2}{3}\}$, $A_k = U^{k-1} A_0 U^{-(k-1)}$, $U = \text{diag}(1, i, -i)$) the twisted $\mathbb{Z}_4$-average collapses* exactly*: $\sum_k U^k A_0 U^{-k} = 4\,\text{diag}(A_0)$, so the exponents at infinity are $4\,\text{diag}(A_0)$. A flat bundle with regular ∞ needs integer exponents summing to $4{=}-\deg E$; with the cusp weights the only options are perms of $(2, 1, 1)$ and $(2, 2, 0)$, i.e.*

$$\boxed{\mathcal{O}(-2){\oplus}\mathcal{O}(-1)^2 \iff \text{diag}(A_0) = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4})}$$

(the sibling $(2, 2, 0) \to \mathcal{O}(0) \oplus \mathcal{O}(-2)^2$). Such a Hermitian A_0 (spectrum $\{0, \frac{1}{3}, \frac{2}{3}\}$, that diagonal) exists by Schur–Horn. So the splitting *is an algebraic condition on* $\text{diag}(A_0)$. Honest wall: *the global $\prod M_k = \mathbf{1}$ is the path-ordered monodromy (*not $\exp(2\pi i \sum A_k)$*), constraining* off-diag(A_0) *via the genuine RH solve; and R-extraction still needs the $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ bridge. The RH route fixes the* splitting *algebraically but not c_u/c_d.*

Lemma 3 (U2 — the kill switch, *run and hardened* [**E**])**.** *The selected polystable wall representative* must *not* collapse *to the diagonal abelian $U(1)^3$ point if it is to give non-trivial $SU(3)_F$ transport mixing: **(A)** $\nabla_F^\star$ polystable but* not *simultaneously diagonal (the desired breakthrough), or **(B)***

diagonal collapse, in which case c_u/c_d stays honestly [C]. ***Result: the cheap test is run robustly and lands on A.*** *The kill-switch fires iff the D_4-symmetric cusp representation is* reducible *(commutant non-scalar). Across* 400 *random cusp-class generators of the $\mathbb{Z}_4$ family $M_k = U^{k-1}MU^{-(k-1)}$ the commutant is scalar at* every *point (irreducible* 400/400*), the reducible locus is hit with probability* 0 *(measure-zero, codimension* ≥ 1*), and the mixing strength stays bounded away from zero* (min *off-diag* $= 0.35 > 0.05$*). So case A is the* generic *behaviour, not an accident of the one explicit point of* [verification/v33_explicit_flat_bundle.py]*:* (U_{wall}) survives *its own cheapest falsification test.* [verification/v38_uwall_killswitch.py]

Lemma 4 (U3 — D_4 fixed locus as an explicit variety, *positive-dimensional*)**.** *With $M_k = U^{k-1}AU^{-(k-1)}$ ($U = \operatorname{diag}(1, i, -i)$, the $\mathbb{Z}_4$ rotation) and the reflection $M_1 = VM_1^{-1}V^{-1}$, $M_3 = VM_3^{-1}V^{-1}$, $M_2 = VM_4^{-1}V^{-1}$ (with $V = -[[1,0,0],[0,0,1],[0,1,0]]$, $V^2{=}1$, $VUV^{-1}{=}U^{-1}$), the relations become polynomials in the entries of $A \in C_{\text{cusp}}$; $\mathcal{X}^{D_4}_{\text{cusp}}$ is cut out as $\bigcup_k C_k$, parametrised by trace coordinates $x = \operatorname{tr}(M_1M_2)$ (note $\operatorname{tr} A = 1{+}\omega{+}\omega^2 = 0$).* ***Result ($C_U^{(2)}$,*** [verification/v30_d4_character_variety.py]***):*** *adding the reflection to the $\mathbb{Z}_4$ locus of* v19 *does* not *isolate the point — the* full *D_4-fixed product locus is still* positive-dimensional *($|\operatorname{tr}(M_1M_2)|$ varies continuously over $\approx [0,3]$). So even the full D_4 symmetry cannot select $\nabla^\star_F$.* Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).

Lemma 5 (U4 — selectors as equations)**.** *The readouts $\rho \mapsto \det R(\rho)$, $\rho \mapsto \operatorname{Spec}(Q_+(\rho))$, $\rho \mapsto \Lambda_{f,j}(\rho)$ are well-defined (cyclotomic/algebraic) on $\mathcal{X}^{D_4}_{\text{cusp}}$.* ***Acceptance criterion:***

$$\#\{\rho \in \mathcal{X}^{D_4}_{\text{cusp}} :\ \det R = 8,\ \operatorname{Spec}(Q_+) = \{1,2,3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not *closed.* ***The bottleneck (made precise by U3):*** *since the D_4-fixed variety is positive-dimensional, the* selector *must do all the cutting — and evaluating* $\det R(\rho)$, $\operatorname{Spec}(Q_+(\rho))$ *as functions on $\mathcal{X}^{D_4}_{\text{cusp}}$ requires the* parabolic-degree ↔ residue dictionary *$R(\rho)$, i.e. the H2 equivalence. That dictionary was the single remaining analytic input of the* (U_{wall}) *gate.* Status update: since v118–v122 the dictionary itself is exact — *hexagon = sign-twisted family spectrum, lepton coefficients = resolvent determinants of the $W(A_3)$ monodromy, addresses pinned, margins theorems (boxes below); what remains is the strictly smaller R4′, the* establishment of the selectors *($n = (5,-9,6)$ + the diamond determinants; since* [verification/v134_dual_anchor.py]*: the pair (d,n)).*

What* $R(\rho)$ *is ([verification/v31_R_dictionary.py] ***).*** *Two facts pin its nature.* (i) Case A is alive: *every D_4-fixed cusp monodromy found is* irreducible*, so the wall point carries non-trivial $SU(3)_F$ mixing — the U2 kill switch does* not *fire to the diagonal case B.* (ii) $R(\rho)$ is not algebraic: *R is integer-valued, but the algebraic invariants of ρ (e.g. $\operatorname{tr}(M_1M_2)$) vary* continuously *on the locus; an integer readout cannot be a continuous algebraic function. Hence $R(\rho)$ is the discrete invariant of the* non-abelian Hodge / Mehta–Seshadri *map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle E_ρ): integer output, but its* computation *is the transcendental harmonic-metric (Hitchin) solve* on the generic Higgs locus. ***Refinement (U4″,*** [verification/v40_harmonic_metric.py]***):*** *at the* polystable *point TFPT actually selects this pessimism does not apply — see the box below.*

Progress: the *selector side* of U4 is closed on the explicit point ([E]). While $R(\rho)$ as a function on the whole locus needs the Hitchin solve, the two U4 *selectors* are *algebraically* accessible on the explicit flat bundle of [verification/v33_explicit_flat_bundle.py] — no PDE. Computed exactly from A_0: the exponents at infinity $\operatorname{eig}(\sum_k A_k) = \{2,1,1\}$ give the splitting type $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$, i.e. the parabolic *anchor* $a = (1,1,2)$; the parabolic degree is 0 ($\deg E = -4 = -|\mu_4|$ plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

$$\boxed{\begin{array}{c}\det R = 8 = n\cdot a \quad \text{(anchor pairing on the splitting type)},\\ \operatorname{Spec}(Q_+) = \{1,2,3\} = 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\})\end{array}}$$

So $\det R = 8$ is the anchor pairing of the splitting type and $\mathrm{Spec}(Q_+)$ is the affine image (d=3 the weight denominator) of the cusp weights — *both selectors are read off the explicit bundle's algebraic data.* **Residual (sharpened below by Readout Rigidity):** the quark *ratio* c_u/c_d turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (v49), hence an integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation (U_{point}), which is an anchor, not a missing PDE. [`verification/v39_uwall_selectors.py`]

Breakthrough: the harmonic metric is *finite linear algebra*, not a PDE ([E]). The U4 "transcendental Hitchin solve" is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field $\Phi = 0$. So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of [`verification/v33_explicit_flat_bundle.py`] this is realised: the raw monodromies M_k are non-unitary ($\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$), but there is a *unique* common invariant Hermitian form H ($\dim\ker\{M_k^\dagger H M_k = H\} = 1$), it is *positive-definite*, and the unitarised holonomy $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$ is unitary ($\sim 10^{-8}$), lies in the cusp class $\{1, \omega, \omega^2\}$ with $\prod \widetilde{M}_k = \mathbf{1}$, and has *clean* matrix elements $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$. **So $C_U^{(3)}$ reduces from a transcendental PDE to: this finite unitarisation (done) + the combinatorial word-length R (closed: H1 + $\det R = 8$).** *Residual:* the feared PDE is gone; the quark *ratio* c_u/c_d is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. [`verification/v40_harmonic_metric.py`]

The final leg-assignment test (honest result) ([E]). Does the harmonic-frame holonomy $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ feed the $\delta = \frac{1}{2}$ resolvent to give the lepton amplitudes $\Lambda = (0.475, 1.107, 0.917)$? **(i) Ruled out as a literal identity:** $\Lambda_\mu = 1.107 > 1$, while unitary holonomy entries have modulus ≤ 1, so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly $|\mathrm{diag}\,\widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$, and that $\frac{1}{2}$ *equals* the distinguished lepton transport value $\delta = \frac{1}{2}$ that v20 used by hand — so the geometry plausibly *supplies* δ (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g. $(\mathbf{1} - \frac{1}{2}\widetilde{M})^{-1}$) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the $\delta = \frac{1}{2}$ resolvent (closed combinatorially via the word-length R); the holonomy supplies δ and the leg selection, not the amplitude magnitude. c_u/c_d is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ($\delta = \frac{1}{2}$). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. [`verification/v41_leg_assignment.py`]

The torsion normal form and the $\delta = \frac{1}{2}$ theorem ([E]). The v41 "genericity is future work" item is now executed — and yields a theorem. *(i) Flatness = μ_4 torsion [E]:* for *any* M (symbolic), $\prod_k U^k M U^{-k} = (MU)^4$ — the $\mathbb{Z}_4$-family flatness constraint *is* the torsion statement "$T = MU$ is a fourth root of unity": the μ_4 atom appears as the *order* of the twisted cusp generator. *(ii) The $\delta = \frac{1}{2}$ theorem [E] (both directions):* on the *involutive* branch ($T^2 = \mathbf{1}$) T is a Hermitian reflection $2vv^\dagger - \mathbf{1}$, and the cusp trace condition $2v^\dagger U^{-1} v = 1$ splits *exactly* into $|v_1|^2 = \frac{1}{2}$ (real part) and $|v_2| = |v_3|$ (imaginary part), forcing

$$\boxed{\mathrm{diag}\,M = (0, \tfrac{i}{2}, -\tfrac{i}{2}), \qquad \mathrm{spec}\,M = \{1, \omega, \omega^2\} \text{ automatic}}$$

($\chi_M = \lambda^3 - 1$ from unitary + trace 0 + det 1). So $\delta = |M_{22}| = \frac{1}{2}$ holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a *μ_4-torsion identity*, not an observation. *(iii) Reflection lemma [E]:* the D_4 reflection forces $\mathrm{diag}\,M = (r, z, \bar{z})$ with r real and $\mathrm{Re}\,z = -r/2$ — the whole D_4 diagonal freedom is *one* real parameter, and $\delta = \frac{1}{2} \iff r = 0$. *(iv) Branch census [E] (seeded):* the full D_4-fixed flat cusp locus realises only $\mathrm{tr}(MU) \in \{1, -1\}$; the involutive branch has the diagonal *constant* $(0, \frac{1}{2}, \frac{1}{2})$ to machine precision, while on the $\{1, i, -i\}$ branch d_1 varies over $[0, 1]$ — and the explicit v33/v40 point sits exactly in its $d_1 = 0$ slice, the *same* δ value. *Residue [C] (recorded):* whether the anchor splitting

$\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ (Mehta–Seshadri side) forces the $d_1 = 0$ slice — or selects the involutive branch — is the remaining question; $\delta = \frac{1}{2}$ itself is no longer the open item. [`verification/v114_torsion_delta.py`]

The anchor pins the residue: an exact normal form for the $(U_{\rm wall})$ **bundle** ([E]). The v114 branch question is answered — and the v33 *numerical* Riemann–Hilbert point acquires an *exact closed form. (i)* μ_4*-average lemma [E]:* $\sum_k U^k X U^{-k} = 4\,\mathrm{diag}\,X$ for any X — μ_4 conjugation-averaging *is* the diagonal projection, so the exponents at infinity are $4\,\mathrm{diag}\,A_0$ and the anchor splitting holds *iff* $\mathrm{diag}\,A_0 = (2,1,1)/4 = a/|\mu_4|$: the residue diagonal **is** the anchor over μ_4 (v33's diagonal was forced, not chosen). *(ii) The* $(8,0,5)/144$ *lemma [E]:* the anchor diagonal and the cusp spectrum $\{0,1,2\}/N_{\rm fam}$ pin $\sum |a_{ij}|^2 = \frac{13}{144}$, and with $a_{13} = 0$ the determinant condition solves *uniquely* to

$$\boxed{\left(|a_{12}|^2,\ |a_{13}|^2,\ |a_{23}|^2\right) = \frac{(8,\ 0,\ 5)}{144}}\qquad \begin{array}{l} 8{=}\mathrm{rank}\,E_8,\ 5{=}g_{\rm car},\\ 144{=}(|\mu_4|N_{\rm fam})^2,\ 8{+}5{=}13{=}\Delta_Q\end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same* Δ_Q that denominates $c_u/c_d = 55/117 = 5\cdot 11/(9\cdot 13)$. *(iii) Exact normal form [E]:* $A_0^\star = \left[\begin{smallmatrix} 1/2 & \sqrt{2}/6 & 0\\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12\\ 0 & \sqrt{5}/12 & 1/4\end{smallmatrix}\right]$ has $\chi = \lambda(\lambda-\frac{1}{3})(\lambda-\frac{2}{3})$ *exactly. (iv)* $A_0^\star$ *is the RH solution [E]:* its big-circle monodromy is trivial to 10^{-10}, $\prod M_k = \mathbf{1}$, and the unitarised holonomy has $|\,\mathrm{diag}\,\widetilde{M}_0| = (0,\frac{1}{2},\frac{1}{2})$, $\mathrm{tr}(\widetilde{M}_0 U) = 1$ — the v33/v40 point is gauge-equivalent to $A_0^\star$ (the a_{12}, a_{23} phases are diagonal gauge). *(v) The anchor forces the* $d_1 = 0$ *slice [E]:* a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with $|\,\mathrm{diag}\,\widetilde{M}_0| = (0,\frac{1}{2},\frac{1}{2})$ — combined with v114, the whole harmonic diagonal is **anchor + torsion forced**; $\delta = \frac{1}{2}$ needs no selection. *Residue [C]:* $a_{13} = 0$ and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [E] via Gröbner on the RH constraints is the remaining $U_{\rm point}$ step. [`verification/v115_anchor_residue.py`]

The resonance theorem: $M_\infty = \mathbf{1} \iff a_{13} = 0$ **— one gauge orbit [E]** ([`verification/v116_resonance_uniqueness.py`])

The anticipated Gröbner promotion turned out to be a *linear* computation. Writing the equivariant system at infinity as $dY/dw = -(1/w)(B_0 + B_1 w + \cdots)Y$ with $B_m = \sum_k i^{km}\,U^k A_0 U^{-k}$, the *twisted* μ_4 averages collapse [E]:

$$B_0 = 4\,\mathrm{diag}\,A_0,\qquad \boxed{B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}}$$

(only the eigenvalue-ratio $-i$ cells survive). The anchor exponents $\mathrm{diag}(2,1,1)$ have resonance gap 1; the level-1 formal-gauge equation $(1-\mathrm{ad}\,R_0)H_1 = R_1$ is singular exactly on the cells $\{(2,1),(3,1)\}$, where B_1 has entries $(0,\ 4\overline{a_{13}})$; all higher levels are non-resonant and the formal monodromy is $e^{-2\pi i\,\mathrm{diag}(2,1,1)} = \mathbf{1}$. Hence [E]:

$$\boxed{M_\infty = \mathbf{1} \iff a_{13} = 0}$$

— the transcendental Riemann–Hilbert condition collapses to *one linear equation. Uniqueness corollary [E]:* $a_{13} = 0$ + the $(8,0,5)/144$ moduli + phases = diagonal gauge $\Rightarrow$ the flat anchor locus is **exactly one gauge orbit**, the exact $A_0^\star$: within the μ_4-equivariant anchor class the $(U_{\rm wall})$ bundle datum is *algebraically unique. Sharpness [E]:* $\|M_\infty - \mathbf{1}\| \sim 8.5\,|a_{13}|$ (0.17 at $a_{13} = 0.02$) vs 10^{-10} at $A_0^\star$. *Honest scope [C]:* the equivariant ansatz is the established D_4 selector input (v19/v30); the *residue* side is now [E], while the holonomy values (the harmonic diagonal $(0,\frac{1}{2},\frac{1}{2})$) remain [E] — transcendental monodromy of an exact system.

The monodromy is the Weyl group of A_3 **— the holonomy is exact [E]** ([`verification/v117_monodromy_weyl_a3.py`])

The last [E] item of the chain — the holonomy *values* — closes: the unitarised monodromy of the exact

$A_0^\star$ system is itself *exact*, with entries in $\frac{1}{2}\mathbb{Z}[i]$:

$$\boxed{\widetilde{M}_0 = \begin{pmatrix} 0 & -\frac{1+i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & -\frac{i}{2} & -\frac{1}{2} \\ \frac{1-i}{2} & -\frac{1}{2} & \frac{i}{2} \end{pmatrix}}$$

unitary, $\det = 1$, $\mathrm{tr} = 0$, $\chi = \lambda^3 - 1$ (the cusp class *exactly*), $\widetilde{M}_0^3 = \mathbf{1}$, and $\mathrm{diag}\,\widetilde{M}_0 = (0, -\frac{i}{2}, +\frac{i}{2})$ — so $d_1 = 0$ and $\delta = \frac{1}{2}$ are now *matrix entries*, not observations; $(\widetilde{M}_0 U)^4 = \mathbf{1}$ with $\mathrm{tr}(\widetilde{M}_0 U) = 1$ (the $\{1, i, -i\}$ branch, $d_1 = 0$ slice — exactly where v114/v115 located the physical point). *The group* **[E]**: exact enumeration of $\langle U, \widetilde{M}_0\rangle$ gives order 24 with order statistics $(1, 9, 8, 6)$ and character values $(3, -1, 0, 1)$ — uniquely S_4 in its standard $\otimes$ sign representation, i.e. the (twisted) reflection representation of the **Weyl group of the family lattice** A_3:

$$\boxed{\mathrm{monodromy}(U_{\mathrm{wall}}) = W(A_3) = S_4, \qquad |W(A_3)| = 4! = 24}$$

— the flavor wall realises the A_3 factor of the glue $D_5 \oplus A_3 + \mu_4$ as its Weyl group (U = a 4-cycle, the μ_4 deck; $\widetilde{M}_0$ = a 3-cycle, the family rotation). *Identification [E]:* the ODE monodromy of $A_0^\star$ equals the exact $\widetilde{M}_0$ to 3.5×10^{-10}, and the invariant form H is U-invariant to 10^{-11}. *Status:* holonomy values **[E]**; only the analytic identification stays **[E]**. The δ thread (v20 → v40/v41 → v114 → v115 → v117) is closed end to end.

The hexagon is the sign-twisted family spectrum — first exact H2 piece [E] ([`verification/v118_hexagon_family_dictionary.py`])

The missing "$C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary" (H2) acquires its first exact entry. *Sign-twist lemma [E]:* $-\widetilde{M}_0$ has order 6 and

$$\boxed{\mathrm{spec}(\widetilde{M}_0) \cup \mathrm{spec}(-\widetilde{M}_0) = \mu_6}$$

— the six-site hypercharge hexagon spectrum *is* the family monodromy spectrum together with its sheet twist ($\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$ realised as $(-1) \times \langle\widetilde{M}_0\rangle$); the v20 denominators $\frac{5}{4} - \cos(r\pi/3)$ are exactly the eigen-denominators $|1 - \zeta_6^r/2|^2$ of the two family resolvents $(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0)^{-1}$. *Cyclotomic determinant [E]:* $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 = (1-t)\Phi_3(t)$, so $\det(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0) = (\frac{7}{8}, \frac{9}{8})$ — and the lepton coefficients become resolvent determinants: $c_e = |\mu_4|/(2\det_-) = \frac{16}{7}$, $c_\mu = N_{\mathrm{fam}}/(2\det_+) = \frac{4}{3}$, $c_\tau = |\mu_4|\det_-/N_{\mathrm{fam}} = \frac{7}{6}$, product rule $= |\mu_4|/\det_+ = \frac{32}{9}$. The lepton 7 and 9 *are* the family-resolvent determinants; $\delta = \frac{1}{2} = |(\widetilde{M}_0)_{22}|$ is supplied by the matrix itself. *Sheet-extended group [E] (audit):* $\langle U, \widetilde{M}_0, -\mathbf{1}\rangle$ has order $48 = \Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+$ — the seed's quartic coefficient. *Residue [C]:* the dictionary's *values* are exact; its *address table* (which hexagon site each fermion occupies; the quark composition) remains v20's established input / open — nothing re-fished.

The address table: the lepton words are the compiler atoms [E]/[C] ([`verification/v120_address_table.py`])

The address table acquires its structure — on word-length integers *frozen* in v18/v20, long before today's readings (the red-team firewall). *(i) The lepton words are the atoms [E]:*

$$\boxed{(L_e, L_\mu, L_\tau) = (8, 5, 3) = (\mathrm{rank}\,E_8,\ g_{\mathrm{car}},\ N_{\mathrm{fam}}) = (p_0{+}e_2,\ e_2,\ p_0)}$$

in mass order (longest word = lightest lepton). *(ii) Addresses = division by the hexagon [E]:* $(r, w) = L \operatorname{divmod} p_2$ with $p_2 = 6 = |R^+(A_3)|$ itself an atom: $e \to (2, 1)$, $\mu \to (5, 0)$, $\tau \to (3, 0)$; $\sum r = 10 = p_3 = \mathcal{A}_\Lambda$, $\sum L = 16 = \dim S^+$. *(iii) Sheet parity [E]:* by the v118 dictionary, e sits on the *untwisted* sheet ($\omega \in \mathrm{spec}\,\widetilde{M}_0$), μ, τ on the *twisted* sheet $(-\omega, -1)$; the anchor lepton τ occupies the unique *real* twisted eigenvalue -1 — exactly the site where v20's product rule replaces the resolvent. *(iv) Quark sum rules [E] (frozen v18 words $7, 7, 5, 3, 2, 0$):* up-type sum $= 10 = p_3$; down-type sum $= 14 = p_1{+}p_3$; **all quarks** $= 24 = |W(A_3)|$ (the monodromy group order, v117); **all nine charged fermions** $= 40 = p_1 p_3 = a^T R a$ (v119); quark $\sum r = 2p_2$, $\sum w = |\mathbb{Z}_2|$; the top sits at the vacuum site $(0, 0)$ — zero word, no suppression, the established ladder anchor. *Residue [C]:* five independent closures $(16, 10, 14, 24, 40)$ constrain the address map; the per-fermion *assignment* remains the v18/v20 input — deriving it is the remaining H2 content.

The address table is pinned: R is the unique hexagon matrix with atom margins [E] ([verification/v121_address_pinning.py])

The per-fermion assignment closes at the *information* level. The established identity $L = R+2U$ (v95) says the word-length table *is* L[sector][generation] — rows (up; down; lepton) $= ((7,3,0);(7,5,2);(8,5,3))$, the v18/v20 words verbatim — so the addresses are the entries of the *residue operator* R plus one hexagon turn for the first generation, and R's first column is the **anchor** $a = (1,1,2)$. The margins are atoms [E]: row sums $(4,8,10) = (e_1, \operatorname{rank} E_8, p_3)$, column sums $(4,13,5) = (e_1, \Delta_Q, g_{\text{car}})$. And the census closes it [E]:

among all 3×3 matrices with entries in the hexagon $\{0..5\}$,
exactly 17 have the atom margins — and exactly ONE has $\det = 8$

— and it is R (also the unique one with SNF $= (1,1,8)$; all 17 determinants are distinct, 8 occurs once — control computed explicitly). **The assignment table carries zero residual information beyond the atom margins and the rank determinant.** Within-sector ordering is the established transport reading ($m \sim \lambda_Y^L$: longer word = lighter fermion). *Residue [C]:* the open H2 item shrinks from "derive 9 per-fermion integers" to "derive the six atom margins + det = rank". *Firewall:* R, L and all word lengths were frozen in v4/v18/v20/v71 long before this census — no degrees of freedom spent.

The margins are theorems ([verification/v122_margin_theorem.py]). Even the margins are not input. Three selectors, all established and frozen *before* the address question was posed — *(S1)* the D_4 annihilator $n = (5,-9,6)$ with $n^T R = (8,0,0)$ (it kills the unchanged (c_2,c_3) plane; $n\cdot a = 8$), *(S2)* $\det R = 8$, *(S3)* $\det K = 4$ with $K = R + Q\Sigma$ — pin R *uniquely* among hexagon matrices [E]: the $(S1)$ census has $4\times 4\times 4 = 64$ candidates, $(S2)$ leaves 12, $(S3)$ leaves **exactly one**, and it is R. Hence the atom margins, the anchor column $Re_1 = a$ and all sum rules above are *consequences* — their input status is revoked. *Bonus lemma [E]:* on the $(S1)+(S2)$ locus $\det(M+2U) = 20 = \det L$ holds for *all* 12 candidates — the fourth quartet determinant is redundant, the compiler's familiar overdetermination. *Residue [C]:* H2 reduces to the establishment of the selectors themselves (n and the diamond determinants — pre-existing inventoried objects); the address question introduces *no new residual class.*

The dual-normal selector: (d,n) pins R columnwise ([verification/v136_dual_normal_selector.py]). The selector list shrinks once more. With the dual anchor $d = a^\top R^{-1}$ ([verification/v134_dual_anchor.py]) and the torsion normal n, each *column* of R obeys $d\cdot c_j = a_j$ and $n\cdot c_j = 8\delta_{1j}$; the common kernel of (d,n) is the primitive vector $(6,8,7)$ (audit: p_2, $\operatorname{rank} E_8$, scalaron-7), so within the address box $\{0..5\}^3$ each column is the *unique* lattice point (census: 1 of 216 each) — $(d,n) \Rightarrow R$, and the 6^9 census above collapses to three column problems [E]. *Honesty:* K is *not* pinned directly ($d^\top K = (1,\frac{1}{2},1)$); $\det K = 4$ comes via the μ_4 wall ([verification/v135_det_surface.py]). *The mechanism is bounded:* the zero mode of $A_0^\star$ carries the exact spectral normal $\sigma = (2,-9,5) = (|\mathbb{Z}_2|, -N_{\text{fam}}^2, g_{\text{car}})$ (signed squares; $\|v\|^2 = \frac{16}{9}$ with $16 = \dim S^+$, $\operatorname{adj} A_0^\star = \frac{2}{9}P_v$ with $\frac{2}{9}$ = the SdS entropy curvature), $\sigma\cdot\mathbf{1} = -|\mathbb{Z}_2|$, $\sigma\cdot a = N_{\text{fam}}$, $n\cdot\sigma = 121 = \|\operatorname{Pl}(K)\|_1^2$; and the exact $W(A_3)$ orbit control shows *no* group image of σ is proportional to n — the naive cofactor/orbit lift fails, the step $\sigma \to n$ is a genuine mechanism beyond the monodromy group. *R4′ restated [C]:* derive the *pair* (d,n); then R is forced columnwise.

The selector triangle ([verification/v139_selector_triangle.py]). The pair itself now collapses further. First, d is *pure anchor data*: $d = \frac{3}{2}a - 2\cdot\mathbf{1}$ exactly — the closed form never mentions R, so the first selector is *derived*, not residual. Second, the frame $(\mathbf{1}, a, \sigma)$ is a basis of $\mathbb{Z}^3$ with $\det(\mathbf{1}|a|\sigma) = 11 = \|\operatorname{Pl}(K)\|_1$ (the quark constant as the frame volume), and n is the *unique* integer covector with the three atom pairings

$$\boxed{n\cdot\mathbf{1} = 2 = |\mathbb{Z}_2|, \qquad n\cdot a = 8 = \operatorname{rank} E_8, \qquad n\cdot\sigma = 121 = \|\operatorname{Pl}(K)\|_1^2}$$

— the cofactor definition is replaced by frame data plus three atoms. On the $(\mathbf{1},a)$-constrained line the σ-pairing is $11(11+t)$: divisible by 11 for *every* t, a perfect square exactly at the true n (t=0). The review's predicted lift is exact as an audit identity: $n = \operatorname{reverse}(\sigma) + |\mu_4|e_3$ (generation flip + μ_4 lift at the third slot).

Frame integrality: three pairings → two ([verification/v142_frame_integrality.py]). The third pairing is *not* independent. The pairing triples $(m \cdot \mathbf{1}, m \cdot a, m \cdot \sigma)$ of *integer* covectors form an index-11 sublattice of $\mathbb{Z}^3$, cut out by the single congruence

$$\boxed{x - 3y + z \equiv 0 \pmod{11}} \qquad (\text{index } 11 = \|\operatorname{Pl}(K)\|_1 = \det(\mathbf{1}|a|\sigma)),$$

so with the two line pairings at their atoms $(x, y) = (|\mathbb{Z}_2|, \operatorname{rank} E_8) = (2, 8)$, integrality *forces* $z \equiv 0 \pmod{11}$ — the $11(11+t)$ line *is* the integer line. Primitivity forces t *even* (odd t makes every entry of $n(t)$ even), and $t{=}0$ is the unique even t with a square σ-pairing for $|t| < 88$ (the global pick keeps the square selector — recorded honestly). Both line pairings are Cramer-reducible to inventoried identities ($n \cdot a = \det R$ by Laplace along the anchor column; $n \cdot \mathbf{1} = 8(R^{-1}\mathbf{1})_1$ with $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$, v134) — restructuring, not establishment.

The pairing values are atom identities ([verification/v145_pairing_atoms.py] **).** Even the two line-pairing *values* carry no independent information. Reading the v139 lift structurally — $n = w_0(\sigma) + |\mu_4| e_3$ with w_0 the A_3 *exponent duality* $m \leftrightarrow h - m$ (an established Lie involution; lift slot = the top-exponent line, lift size = the glue order) — all three pairings become atom arithmetic:

$$n{\cdot}\mathbf{1} = |\mu_4| - |\mathbb{Z}_2| = |\mathbb{Z}_2|, \qquad n{\cdot}a = |\mu_4|\, a_3 = 8 = \operatorname{rank} E_8 \text{ via } \boxed{|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 \quad (4 + 5 = 9,\ \sigma \perp w_0(a))},$$

and $n \cdot \sigma = \|\sigma\|^2 - N_{\text{fam}}^2 + |\mu_4| g_{\text{car}} = 110 - 9 + 20 = 121$ with $\|\sigma\|^2 = |\mathbb{Z}_2|^2 + N_{\text{fam}}^4 + g_{\text{car}}^2 = 110 = |\mathbb{Z}_2| g_{\text{car}} \|\operatorname{Pl}(K)\|_1$. *R4′ residue [C] (final form):* derive the *one* map $n = w_0(\sigma) + |\mu_4| e_3$ — no independent number remains; everything else in the chain anchor $\to (d, n) \to R$ is exact.

Both dual normals are anchor data ([verification/v149_cusp_normal.py] **).** The remaining opacity dissolves in the canonical frame: the pairings of n with the *cusp eigenbasis* (the geometric basis of v98) are exactly

$$\boxed{n \cdot e_1 = 6 = p_2(a), \qquad n \cdot e_2 = 3 = p_0(a), \qquad n \cdot e_3 = 5 = e_2(a)}$$

— one anchor invariant per Q_+ degree, and the cusp matrix is unimodular, so these three pairings determine n uniquely. With $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ the dual normal *pair* (v134) is anchor data through and through — one normal per frame; the cusp data $(6, 3, 5)$, the frame pairings $(2, 8, 121)$ and the w_0-lift are three presentations of one covector, all atomic (equivalences exact). *Audit:* $\sigma \to (5, 1, 2)$; alternate reading $(6, 3, 5) = (|R^+(A_3)|, N_{\text{fam}}, g_{\text{car}})$; sum $14 = \dim G_2$. *R4′ residue [C] (cusp form):* derive the *assignment* — why the Q_+ degrees $(1, 2, 3)$ carry (p_2, p_0, e_2) in that order; one discrete assignment, zero continuous freedom.

Exterior Leg Lemma — the quark u/d leg is a $\Lambda^2 F$ area, not a scalar [E]/[C]

The v41 negative is a *category error*, not a dead end. u and d share the hexagon-radial data $(r, w) = (1, 1)$ ($L = 7$ for both), and the C_6 resolvent $c(r, w) = |\mu_4|^w / (\frac{5}{4} - \cos\frac{r\pi}{3})$ reads *only* (r, w) — so any *scalar* leg gives $c_u/c_d \equiv 1$. A scalar sees radius, not orientation. The u/d separation lives in the smallest *non-scalar* invariant of the $SU(3)_F$ composition: the exterior square $\Lambda^2 F$, i.e. the oriented anchor-plane area $\operatorname{Pl}_{\mathbf{1},a}(K)$. The "11" is then *generated*, not read:

$$(K, \mathbf{1}, a) \to B_K = \left(\begin{smallmatrix} 13 & 8 & 4 \\ 18 & 11 & 6 \end{smallmatrix}\right) \to \operatorname{Pl}(K) = (-1, 6, 4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \to \|\operatorname{Pl}(K)\|_1 = 11,$$

$$\boxed{\frac{c_u}{c_d} = \frac{g_{\text{car}} \|\operatorname{Pl}_{\mathbf{1},a}(K)\|_1}{N_{\text{fam}}^2 \Delta_Q} = \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117}} \quad \left(\text{anchor microcode } = \tfrac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}\right).$$

Typing: [E] for the Plücker / microcode identities ($\operatorname{Pl}(K) = (-1, 6, 4)$, $\|{\cdot}\|_1 = 11$, $5{\cdot}11/(9{\cdot}13) = 55/117$); and — by Readout Rigidity (v49, below) — [E] also for the *physical* ratio $\mathcal{C}_{ud}(\rho^\star)$ = this $\Lambda^2 F$ readout, which is *constant* on the derived selector stratum and so does *not* need any point-selection on $\Lambda^2 F$. The quark residual is thereby re-typed from "find the right scalar y" to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale (U_{point}) stays as an anchor, no new scalar, no fishing.

Audit (the 121 *lemma, [verification/v119_review_validation_2.py]):* the "11" has a second, independent appearance [E]: with $h = \mathbf{1} + a = (2, 2, 3)$ in the canonical anchor ordering, $h^T R h = 121 = 11^2 = \|\operatorname{Pl}(K)\|_1^2 = (p_3 + 1)^2$ — the quadratic *anchor norm* of the residue operator R reproduces the Plücker integer (sensitivity control: the three orderings of h give $\{105, 121, 135\}$; only the canonical one yields 121). Bonus checksums: $\mathbf{1}^T R \mathbf{1} = 22 = 2 \times 11$ (the entry sum of R) and $a^T R a = 40 = p_1 p_3 = 10 b_1 - 1$. Typed *audit*, not load-bearing — the oriented Plücker mechanism above remains the derivation. [verification/v42_exterior_leg.py]

Bridge test ([`verification/v43_exterior_bridge.py`])**.** The typing is confirmed: for the $SU(3)$ holonomy the exterior square is the conjugate representation, $\Lambda^2(\widetilde{M}) = \overline{\widetilde{M}}$ exactly — the exterior leg *is* the $\bar{\mathbf{3}}$. But the *continuous* action $\Lambda^2(\widetilde{M}) \cdot (\mathbf{1} \wedge a)$ does *not* equal the *integer* $\mathrm{Pl}(K) = (-1, 6, 4)$: the integer Plücker is the combinatorial K datum. So the bridge $\rho^\star \to \mathrm{Pl}(K)$ is the *discrete* non-abelian-Hodge invariant (the integer R; $\det R = 8$ and $\mathrm{Spec}(Q_+) = \{1, 2, 3\}$ already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the $\Lambda^2 F$ side.

Lie-level grounding — no special math ([`verification/v44_carrier_exterior.py`])**.** The discrete invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier, $16 = \Lambda^{\mathrm{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$, and under $SU(5) \subset SO(10)$ the up quark sits in $10 = \Lambda^2(5)$ while the down sits in $\bar{5} = \Lambda^4(5)$. So the *exterior degrees* are $\deg(u^c) = 2$, $\deg(d^c) = 4$, with $\deg(u) + \deg(d) = 6 = |R^+(A_3)|$ (the hexagon) and $\deg(d) - \deg(u) = 2 = |\mathbb{Z}_2|$ (the sheet). The "Λ^2" of the exterior leg is therefore not exotic Hodge data — it is the carrier's own Λ^2 grading (the up quark *lives* in $\Lambda^2(5)$). This re-types the residual once more: from "discrete non-abelian-Hodge invariant" to "the carrier exterior degree" — a standard [E] fact. *Honest scope:* this grounds the *type*; the Plücker *number* 55/117 stays the family $\mathrm{Pl}(K)$ [E], and the carrier-$\Lambda^2 \leftrightarrow$ family-$\Lambda^2 F$ link (via family $\otimes$ carrier $= 15 = 16-1$) plus the normalisation remain [C].

The "11" is Pascal — a third, simplest origin ([`verification/v45_family_exterior.py`])**.** Following that link, $\mathrm{Pl}(K)$ is the *exterior algebra of the family fundamental* $4 = \mu_4$ (the $A_3 = SU(4)$ fundamental): $\Lambda^k(4) = \binom{4}{k}$ is Pascal row $4 = (1, 4, 6, 4, 1)$ with full sum $2^4 = 16 = \dim S^+$, and $|\mathrm{Pl}(K)| = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$ are exactly its distinct exterior powers. Hence

$$\boxed{\|\mathrm{Pl}(K)\|_1 = \textstyle\sum_{k=0}^{2} \binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\mathrm{car}}}$$

(the cumulative $\Lambda^\bullet(\mu_4)$ up to degree $2 = \deg(u^c)$), and family $\otimes$ carrier $= 15 = \dim \mathfrak{su}(4) = \Lambda^2(5) \oplus \Lambda^4(5) = 10+5 = N_{\mathrm{fam}} g_{\mathrm{car}}$ read three ways. So the "11" is generated a *third* way — the cumulative family exterior algebra up to the up-quark's degree — pure Pascal/branching, *no special math.* The [C] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

Lemma 6 (U5 — the "11" worry, *discharged* by Readout Rigidity)**.** *The original worry was that $c_u/c_d = \mathcal{C}_{ud}(\rho^\star)$ must be evaluated at the uniquely selected $\rho^\star$* without *using* 11 *as input, so that until then* 55/117 *would be a mere table-level rational shadow.* ***This is now resolved (v49):*** *$\mathcal{C}_{ud}$ is* constant *on the discrete selector stratum $\mathcal{S}_{8,Q}$, so the "11" is* earned *as the rigid Plücker readout* $\|\mathrm{Pl}(K)\|_1$ *(v42)* — independently *of point-uniqueness. Point-uniqueness of $\rho^\star$ enters only the* absolute *amplitude scale, not the ratio.*

Lemma 7 (U6 — the four certificates)**.** *$C_U^{(1)}$ wall enumeration (done); $C_U^{(2)}$ D_4-fixed character solver; $C_U^{(3)}$ selector uniqueness (one S-equivalence class); $C_U^{(4)}$ amplitude extraction — the* ratios *$R, Q, c_u/c_d$ are read off the (derived) selector stratum (closed, v49/v71); only the* absolute *$U_f^\star, c_q$ normalisation is the anchor.*

Progress: an explicit valid flat bundle is realised (RH solve) ([E]). A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue A_0 (eigenvalues $\{0, \frac{1}{3}, \frac{2}{3}\}$, diag $= (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$) for which the Fuchsian system $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)}/(z - i^k)$ simultaneously satisfies: the cusp class at each puncture; the splitting $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ (exponents $\sum_k A_k = \mathrm{diag}(2, 1, 1)$); the *path-ordered* monodromy at infinity is trivial, $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$, i.e. $M_1 M_2 M_3 M_4 = \mathbf{1}$; and the representation is *irreducible* — **case A** (non-trivial $SU(3)_F$ mixing, not the diagonal case B). So a valid (U_{wall}) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope:* this is one realising point (existence + case A), likely one of a residual family; the unique $\nabla_F^\star$ still needs the $\det R = 8$ selector and c_u/c_d still needs the $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$ dictionary (H2). [`verification/v33_explicit_flat_bundle.py`]

H2 bridge probed (*[`verification/v34_h2_bridge_attempt.py`]*)*.* The explicit per-puncture monodromies M_k (cusp class, $\prod M_k = \mathbf{1}$ to 10^{-5}) were computed; but the natural resolvent-dressed diagonal extraction ($|\mathrm{diag}\, M_k| = (0, \frac{1}{2}, \frac{1}{2})$ times the C_6 resolvent factor) does *not* reproduce the lepton amplitudes $(0.475, 1.107, 0.917)$. So the precise $\Gamma_{ij}^{\min}$ geodesic-to-word dictionary (how the 6-site hypercharge hexagon combines with the 3-dim family $\rho^\star$) was the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for 55/117. *Resolved since (v118): the correct extraction is the resolvent determinant of the $W(A_3)$ monodromy, not the diagonal dressing — the probe's failure is thereby explained, and the dictionary is exact.*

Theorem U — the four-way split (sharper than one monster gate) [E]/[O]

D_4 fixes the admissible chamber, not the physical point (the D_4-fixed locus is positive-dimensional, U3); the discrete selectors $\det R = 8$, $\mathrm{SNF}(R) = (1,1,8)$, $\mathrm{Spec}(Q_+) = \{1,2,3\}$ do the cutting. The gate splits into four pieces, three of them essentially closed:

- U_{unitary} [E]: parabolic degree 0 + polystable ⇒ unitary (Mehta–Seshadri) ⇒ $\Phi = 0$; the harmonic metric is the *finite* invariant form H ($M_k^\dagger H M_k = H$), not a Hitchin PDE ([`verification/v40_harmonic_metric.py`]).
- U_{H2} [E]: R, Q, K are read off the bundle (splitting type $= a$, $\det R = 8$, $\mathrm{Spec}(Q_+)$) ([`verification/v39_uwall_selectors.py`]).
- U_{Λ^2} [E]: **Readout Rigidity** — on the discrete stratum $\mathcal{S}_{8,Q} = \{\det R = 8, \mathrm{SNF}(R) = (1,1,8), \mathrm{Spec}(Q_+) = \{1,2,3\}\}$ the anchor-plane readout is *constant*, $\mathcal{C}_{ud}(\rho) = g_{\text{car}}\|\mathrm{Pl}_{\mathbf{1},a}(K)\|_1/(N_{\text{fam}}^2\Delta_Q) = \frac{55}{117}$, *independent* of the continuous D_4 position — so c_u/c_d needs no point uniqueness ([`verification/v49_readout_rigidity.py`], [`verification/v42_exterior_leg.py`]).
- U_{point} [O]: full uniqueness of $\rho^\star$ — needed only for the *complete* $U_f^\star$ amplitude matrix, *not* for c_u/c_d.

Headline: the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; the only residual, U_{point}, is not an independent gate — it reduces to the single dimensionful anchor v_{geo} (Gate 1 below).

The simple closure ([`verification/v71_simple_r_bridge.py`]). The selector stratum $\mathcal{S}_{8,Q}$ is now *fully derived*: $\det R = 8 = \mathrm{rank}\,E_8$ with $\mathrm{SNF}(R) = (1,1,8)$ is the lattice invariant (cf. $\det Q = 3 = N_{\text{fam}}$, v70), and $\mathrm{Spec}(Q_+) = \{1,2,3\}$ is the D_4-equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants $(\det Q, \det K, \det R, \det L) = (3,4,8,20)$, product $1920 = |W(D_5)|$. So by Readout Rigidity the quark *ratios* ($c_u/c_d = g_{\text{car}}\cdot 11/(N_{\text{fam}}^2\cdot 13) = \frac{55}{117}$, etc.) are *pure integer Plücker readouts — no transcendental monodromy solve.* The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece, U_{point}, is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensionful scale, not a missing computation.

Gate 1 complete: $U_{\text{point}} \to v_{\text{geo}}$ ([`verification/v75_upoint_to_vgeo.py`]). The absolute amplitudes are not free either. The lepton amplitudes are exact ($c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, v20), every cross-sector c-ratio is closed (Plücker), and each *sector product* is a clean φ_0^{ret}-power (Grand Mass Volume, v46: $\det M_{\text{sector}} \sim (\varphi_0^{\text{ret}})^{6,9,10}$; the lepton product is $\frac{16}{7}\cdot\frac{4}{3}\cdot\frac{7}{6} = \frac{32}{9} = 2^{g_{\text{car}}}/N_{\text{fam}}^2$). Since (pairwise ratios) + (product) ⇔ (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale v_{geo}*. **So U_{point} is not an independent gate: it reduces to v_{geo}, the *same* one irreducible dimensionful anchor as gravity's $1/G$ (v68) — the two [O] anchors collapse to one.**

Research Contract 2 — (G_{metric})

Goal. Construct the TFPT metric measure

$$Z[J] = \lim_{\chi\to\infty} \int_{\mathcal{G}_\chi/\mathrm{Diff}} \exp\big(-S_{\text{rel},\chi}[g, A, \psi] + J\cdot O\big)\, d\mu_{\text{Cal},\chi},$$

$$S_{\text{rel},\chi} = \mathrm{Tr}\, f\Big(\tfrac{D_{\text{rel}}+A_\Sigma}{\chi}\Big) - \mathrm{Tr}\, f\Big(\tfrac{D_{\text{ref}}}{\chi}\Big) + \tfrac{i\pi}{2}\Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed $R + R^2$ equation.

Lemma 8 (G0–G1 — configuration space & BRST/Hodge Fredholm)**.** *For $s > 5/2$ a Sobolev space of Θ-symmetric metrics on the doubled seam collar M_Σ admits a gauge slice; the gauge-fixed gravitational complex $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2T^*M) \xrightarrow{K^*} \Omega^0(TM)$ with $\Delta_{\text{gh}} = K^*K$ and $\Pi_{\text{phys}} =$*

$\Pi_{\mathrm{BRST/Hodge}}\Pi_{TT}$ *is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (*not *rhetorically).* Machine: principal symbols in xAct/-Cadabra; linear-algebra certificates in Lean (medium).

Lemma 9 (G2 — finite relative spectral action, *heat-kernel grounded* [E])**.** *$S_{\mathrm{rel},\chi}$ is finite, differentiable, with a local Seeley–DeWitt expansion* $\operatorname{Tr} f(D^2/\chi^2) \sim f_4\chi^4 a_0 + f_2\chi^2 a_2 + f_0 a_4 + \cdots$*. For the Lichnerowicz Dirac operator $D^2 = \nabla^2 + \frac{R}{4}$ (so $E = -\frac{R}{4}$, spinor trace $\operatorname{tr}\mathbf{1} = 4$) the Gilkey coefficients give, per $(4\pi)^{-2}$,*

$$a_2 = \tfrac{1}{6}\operatorname{tr}(6E+R\,\mathbf{1}) = -\tfrac{R}{3} \quad (\textit{Einstein–Hilbert } \textstyle\int R), \qquad a_4\big|_{R^2} = \tfrac{1}{360}\operatorname{tr}(180E^2+60RE+5R^2\mathbf{1}) = \tfrac{R^2}{72}$$

(the R^2/Starobinsky term), so the spectral action structurally *produces $S_{\mathrm{eff}} = \int\sqrt{g}\,\frac{\bar M_{\mathrm{Pl}}^2}{2}(R+\frac{R^2}{6M^2})+\cdots$ with the scalaron mass ratio fixed by the cutoff moment, $M^2/\bar M_{\mathrm{Pl}}^2 = 6(4\pi)^2/f_0$. The TFPT closure $M^2/\bar M_{\mathrm{Pl}}^2 = c_3^7$ fixes $f_0 = 6(4\pi)^2/c_3^7$, i.e. $M = c_3^{7/2}\bar M_{\mathrm{Pl}} = 3.06\times 10^{13}$ GeV — the boundary normalisation c_3* is *the gravitational scale — and hence the Starobinsky slow-roll forms $n_s \simeq 1-\frac{2}{N}$, $r\simeq\frac{12}{N^2}$. The $R+R^2$* structure *is convention-independent (standard Gilkey); the precise rational $\frac{1}{72}$ and factor 6 are the standard Dirac conventions used here ([verification/v36_spectral_action_g2.py], [verification/v28_gravity_fR.py]).*

The simpler resolution: "full QG" for physics is the gap-decoupled admissible sector ([E]). The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5): $2\|V_{\mathrm{metric}}\| = 0.785 < \Delta = 6\log\frac{3}{2} = 2.433$, so the admissible sector keeps a *strictly positive* effective gap $\Delta_{\mathrm{eff}} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates* the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

$$\boxed{\begin{array}{c}\text{admissible IR sector} = \text{closed under stated RP;}\\ \text{ambient metric measure (G6)} = G_{\mathrm{metric}}\text{ gate}\end{array}}$$

G2 supplies the local action ($R+R^2$, heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce (G_{metric}) to the ambient projective limit (G6). **Precise status (Alessandro's boundary):** the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure is now discharged as a [C] *redundancy* (v369, the holographic-route keybox below), a certification object rather than a missing piece. Its absence does *not* affect the bounded IR claim. [`verification/v36_spectral_action_g2.py`]

The glue Q-system: the index-4 object of the G_{net} statement exists explicitly ([E]/[C]). The G_{net} target — *"the seam–Calderón inclusion has Jones index $4 = |\mu_4|$"* — no longer refers to an unknown object. On the glue group algebra $A = \mathbb{C}[\mathbb{Z}_4]$ (multiplication $\delta_a\otimes\delta_b\mapsto\delta_{a+b}$, unit δ_0) *all* Longo Q-system axioms hold exactly [E]: associativity, unit law, the Frobenius identity $(m\otimes 1)(1\otimes m^*) = m^*m = (1\otimes m)(m^*\otimes 1)$, and specialness $mm^* = |\mathbb{Z}_4|\,\mathbf{1}$ — a special C* Frobenius algebra with

$$\boxed{\dim\theta = 4 = |\mu_4| = [B:A]}$$

exactly the Carrier Index Lemma value (KLM: $\mu_{\mathrm{carrier}} = 16 = 4^2\mu_{E_8}$ re-verified). *Locality = isotropy [E]:* on the Lagrangian glue $\langle(1,1)\rangle$ the discriminant form gives $q(k(1,1)) = k^2\in\mathbb{Z}$ for every k — the braiding obstruction vanishes (v92 in Q-system language); the two Lagrangian glues (the sheet pair) are the only maximal isotropic choices. *Halfway step [E]:* $\{0,2\} < \mathbb{Z}_4$ closes and gives the unique intermediate Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ tower step, with index factorisation $4 = 2\times 2$. *R2 sharpened [C] (recorded, not claimed):* what remains is the operator-algebraic *identification* "the seam–Calderón inclusion of the 16-Majorana net (v113) realises *this* Q-system" — matching two constructed objects, not a search; the three loads (metric + carrier + QBL, v123) hang on this one identification.

The hull is the graded module of the Q-system ([verification/v128_graded_hull.py]). The Q-system is not merely an abstract companion of E_8 — E_8 *realises* it [E]: the 240 roots, built as explicit norm-2 vectors in $D_5 \oplus A_3$+glue coordinates, decompose as $240 = 52 + 64 + 60 + 64$ over the glue $\mathbb{Z}_4$ (the v89 branching as vectors), and for *all* 6720 root pairs whose sum is again a root, $\mathrm{coset}(r_1{+}r_2) = \mathrm{coset}(r_1) + \mathrm{coset}(r_2) \bmod 4$:

$$\boxed{E_8 \text{ is a } \mathbb{Z}_4\text{-graded Lie algebra over the carrier; the grading group is the glue = the Q-system}}$$

— the checksum hull is the Q-system's graded module, realised in 248 dimensions. The R2 identification thus matches the Calderón inclusion against a structure that E_8 already *carries*. [verification/v125_glue_qsystem.py]

Graded Frobenius ([verification/v143_graded_frobenius.py]). The identification holds at the finite Lie level: the graded module is in fact a graded *Frobenius* structure, i.e. everything the Q-system demands of its module is already exact inside E_8 [E]: (i) *coset duality* — $\mathrm{coset}(-r) = -\,\mathrm{coset}(r) \bmod 4$ for all 240 roots ($-C_1 = C_3$ bijectively, C_0, C_2 self-dual), so the Killing form restricts to a *nondegenerate* invariant pairing $g_k \times g_{-k} \to \mathbb{C}$; (ii) the glue character average $\frac{1}{4}\sum_k i^{k\,\deg}$ is *exactly* the carrier projector (the conditional-expectation shadow, quasi-basis index $|\mathbb{Z}_4| = 4$); (iii) each glue sector is *one* $W(D_5)\times W(A_3)$ orbit with multiplicity-one dimensions $64 = 16 \cdot 4 = (\mathbf{16}, \bar{\mathbf{4}})$, $60 = 10 \cdot 6 = (\mathbf{10}, \mathbf{6})$ — simple currents, sector ring $= \mathbb{Z}_4 =$ the $\mathbb{C}[\mathbb{Z}_4]$ fusion, halfway subgroup $\{0, 2\} = SO(16)_1$. *R2 residue [C] (final form):* exactly *one* conformal-net statement remains — the seam–Calderón inclusion of the 16-Majorana *net* carries this Q-system — with no remaining finite freedom.

Where that statement lives: the NS/R sector census ([verification/v148_fock_census.py]). The remaining work is scoped exactly. Every *untwisted* (Neveu–Schwarz) state of the 16 Majoranas has *integer* $(D_5, SO(6))$ weights, and integer weights lie in the even discriminant classes — so the NS module supports only $\{(0,0),(2,0),(0,2),(2,2)\}$: **the odd glue classes $k(1,1)$ — the actual μ_4 simple currents — have zero NS support; they are twisted (Ramond-type) modules.** The Ramond zero-mode module ($2^8 = 256$ ground states) carries exactly the four odd class pairs, 64 each, and splits $128 + 128$ into the odd sectors of the *two* Lagrangian glues $\langle(1,1)\rangle$ and $\langle(1,3)\rangle$ (v92): *choosing the glue is choosing the Ramond projection* ($128 =$ one $SO(16)$ spinor chirality), and E_8 assembles as $248 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$. *Consequence [C]:* the index-4 extension cannot even be *stated* on the untwisted module alone; R2's one remaining statement is irreducibly about the twisted-sector structure of the seam–Calderón net, with the sheet choice realised as the R-projection; the finite shadow is exhausted (v113/v125/v143 + this census).

Lemma 10 (G3–G4 — reflection positivity, fixed and integrated)**.** *For each Θ-invariant g in the Calderón ball the physical boundary kernel $\mathcal{K}_g = P_{\mathrm{adm}}\Pi_{\mathrm{phys}}\mathcal{N}_g^{TT}\Pi_{\mathrm{phys}}P_{\mathrm{adm}}$ is reflection positive, $\langle\Theta F, F\rangle_{\mathcal{K}_g} \geq 0$ (G3); and $d\mu_{\mathrm{Cal},\chi}$ is Θ-invariant, supported where G3 holds uniformly, so $\int\langle\Theta F, F\rangle_{\mathcal{K}_g}\, d\mu_{\mathrm{Cal},\chi} \geq 0$ (G4).* ***G4 is the decisive step:*** *RP after integrating over dynamical metrics, not merely at fixed background.*

Lemma 11 (G5 — gap dominance [E])**.** $2\|V_{\mathrm{metric}}\|_{\mathrm{rel}} < \Delta = 6\log\frac{3}{2}$, *with* $\|V_{\mathrm{metric}}\|_{\mathrm{rel}} \leq 248c_3^2 = \frac{31}{8\pi^2} = 0.39262$, *hence* $2\|V\| = 0.7852 < 2.4328$ *and* $\Delta_{\mathrm{eff}} \geq \Delta - 2\|V\| > 1.647 > 0$. *The numerical inequality is trivial ([verification/v29_research_contract_certs.py]); the work is the operator-norm bound (hard).*

Sugawara reading + atomic OS moment ([verification/v171_os_moment_cluster.py]). The G5 budget is the E_8 Sugawara constant: $248c_3^2 = \frac{31}{8\pi^2}$ with $31 = 1 + h^\vee(E_8)$ and $c((E_8)_1) = 248/31 = 8$, so gap dominance reads $|R^+(A_3)|\log(N_{\mathrm{fam}}/|\mathbb{Z}_2|) > \frac{1+h^\vee(E_8)}{4\pi^2}$ — the A_3 transfer gap beats the E_8 Sugawara metric budget. Independently, the G3 reflection positivity is a *positive moment problem*: $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ makes every two-point correlator $C(n) = A + B(64/729)^n + C(1/729)^n$

the moment sequence of a positive atomic measure, so each OS Hankel matrix factorises as a Vandermonde Gram and is positive semidefinite (a clean RP certificate; cluster constant 729/665, correlation length $\xi = 1/(6\log\frac{3}{2}) = 0.4111$).

Theorem 1 (Decoupling theorem — physical low-energy claims do *not* need the full G_{metric})**.** *If the admissible sector is reflection-positive (G4) and the metric coupling is gap-dominated,* $2\|V_{\text{metric}}\| < \Delta = 6\log\frac{3}{2}$ *(G5), then the admissible IR measure is stable under the ambient completion: a strictly positive effective gap* $\Delta_{\text{eff}} = \Delta - 2\|V\| > 0$ *protects the admissible sector from the (un-built) ambient/deep-UV, so every* physical low-energy *read-out (masses, mixings,* α^{-1}*, the* $R{+}R^2$ *regime) is independent of the open ambient measure. Symbolically*

$$\boxed{\mathrm{RP}\ +\ 2\|V\| < \Delta \implies \textit{admissible IR measure stable under ambient completion}\,.}$$

Thus (G_{metric}) *is* not *a diffuse "full quantum gravity missing" block: it is the sharply delimited* ambient projective limit *(G6), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on.* ***Typing:*** *the inequality is machine-checked* **[E]***; the operator-norm bound on* $\|V_{\text{metric}}\|$ *is the conditional input.*

Lemma 12 (G6–G7 — projective limit & Ward identities)**.** *A projective system* μ_{χ_n} *with* $\pi_{m,n*}\mu_{\chi_m} = \mu_{\chi_n}$*, uniform moment bounds* $\sup_n \int \|g\|^p_{H^s}\, d\mu_{\chi_n} < \infty$*, tightness and regulator-independence yields* $\mu_{\mathrm{QG}} = \lim \mu_\chi$ *on* $\mathcal{G}/\mathrm{Diff}$ *(G6, the hardest step); the limit obeys the BRST Ward identities* $\langle s\mathcal{O}\rangle = 0$*, whence* $\nabla^\mu T_{\mu\nu} = 0$ *and the* η *boundary phase cancels the APS anomalies (G7).*

Gate 2 reduction: G6 is a *boundary* projective limit, not a bulk one (`[verification/v76_gmetric_reduction.py]`)

The exact decoupling margin is $\Delta_{\text{eff}} = \Delta - 2\|V\| = 6\log\frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$, with $\|V_{\text{metric}}\|_{\text{rel}} \le 248\, c_3^2 = \dim(E_8)\, c_3^2 = \frac{31}{8\pi^2}$ ($31 = 2^{g_{\text{car}}} - 1$). **Holographic reduction:** because the seam is a *finite causal boundary*, the ambient *bulk* metric measure is reconstructed from a finite-codimension seam (Calderón) *boundary* measure — so G6 is a *boundary* projective limit. The conditional theorem is then

$$\boxed{\text{RP(seam-Calderón boundary kernel)} + \text{tightness} \implies G_{\text{metric}}\ \text{closes}}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure. **Residual:** the constructive boundary projective limit (constructive-QFT grade) — now discharged as a [C] redundancy (v369, keybox below). So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) a [C] redundancy, no longer a diffuse "full QG missing". [C]

The measure as one-loop fluctuations around the *parameter-free* saddle (`[verification/v365_qg_oneloop_saddle.py]`, `QGAMB.SADDLE.01`)

With the saddle now parameter-free (v358/v359), the boundary measure of Tier B organises as a Gaussian one-loop fluctuation determinant whose ingredients are atom-fixed. **Fixed quadratic form [E]:** the fluctuation inverse-propagator stiffness is the saddle coefficient $1/c_3 = 8\pi$ — no free dimensionless dial, so the one-loop problem is *well-posed* (it was not before v358). **Gap-controlled convergence [E]:** the fluctuation operator obeys $M \ge \Delta_{\text{eff}} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337), so the Gaussian integral converges mode-by-mode with no zero/negative modes besides the saddle directions; on the OS spectrum $\{0, 6\log\frac{3}{2}, 6\log 3\}$ the regularised one-loop log-det $\mathrm{Tr}'\log M = 6\log\frac{9}{2}$ is finite. **Conformal direction [C]:** the one wrong-sign mode ($c_{\text{conf}}(4) = -\frac{3}{2}$) is made convergent by the GHP contour ($\int e^{-|c|\phi^2} = \sqrt{2\pi/3}$) with IDG dressing (v334). **Tightness [E]:** $\chi = 1/(1-(2/3)^6) = 729/665 < \infty$ (v330), so the one-loop measure has a projective limit on the admissible sector, with *zero* new dimensionless dials (v364). **Residual:** the full *non-perturbative* projective limit (G6 on the ambient sector) is now discharged as a [C] redundancy (v369, keybox below) — this sharpens C7 from "diffuse full QG" to a one-loop-controlled Gaussian problem around a parameter-free saddle and then to a certification object. [E]/[C]

The holographic route: the ambient measure is boundary-*redundant*, not missing dynamics ([`verification/v369_qgamb_redundancy.py`], `QGAMB.REDUNDANCY.01`)

There is a second, strategically stronger way to dispatch Gate 2 than building the measure: prove that the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the measure `QG.AMB.01` is a non-fundamental *certification* object — the role a coordinate choice plays in general relativity — rather than a missing piece of dynamics. The discriminators that make this *ambient-redundancy* statement well-posed are all in hand. **Trivial superselection [E]:** the number of DHR sectors of a chiral net is $|\det \mathrm{Cartan}|$, which is 1 for E_8 (one primary, the vacuum; $\mathrm{DHR}((E_8)_1) = \mathrm{Vec}$) against 4 for the same-c rival $SO(16)_1$ — a holomorphic boundary net has *no* nontrivial superselection charge that a bulk could carry invisibly. **No torus degeneracy [E]:** the ground-state degeneracy on T^2 equals the number of anyons $= |\det K| = 1$ for E_8, so there is no topologically protected bulk d.o.f. hidden from the boundary (the genus-1 obstruction of v344, in its redundancy reading). **Bounded reconstruction [E]:** the seam transfer deviation subspace contracts at the finite gapped Petz rate $(2/3)^6$ (v221), and the gap-decoupling margin $\Delta_{\mathrm{eff}} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) separates the un-built ambient sector from the admissible one. **The statement [C]:** with the Bisognano–Wichmann modular reconstruction map (v258/v309/v323), under `SEAM.EQUIV.01` every gauge-invariant bulk observable is boundary-reconstructible, $\mathcal{O}^{\mathrm{bulk}}_{\mathrm{phys}}/\mathrm{Diff} \cong \mathcal{A}^{\mu_4}_{\Sigma}$, and two ambient measures sharing the same boundary state are physically equivalent. **Residual [O]:** the construction consumes exactly two premises — `SEAM.EQUIV.01` (the S3 lattice input, v366) and the BW intrinsicality of the raw collar (v329); it does *not* construct `QG.AMB.01` but supplies the *alternative* route, so that Gravity-complete = Boundary-complete + Ambient-redundancy. [E]/[C]/[O]

The classical field equation, parameter-free: entanglement equilibrium gives the linearised Einstein equation with G from c_3 ([`verification/v358_grav_entropy_equilibrium.py`])

While Gate 2 (above) reduces the metric-sector *measure*, the classical metric-sector *field equation* is supplied directly by the entanglement first law $\delta S = \delta\langle K\rangle$ (Jacobson 2015; Faulkner et al.), carried out with TFPT's atoms. **Parameter-free coefficient [E].** Equilibrium gives $G_{ab} = c_3^{-1} T_{ab}$ with $c_3^{-1} = 8\pi$ *fixed*: all four c_3-roles are mutually consistent — the Unruh factor $1/(2\pi) = 4c_3$, the EH action $1/(16\pi) = c_3/2$, the Einstein coefficient $8\pi = 1/c_3$, the Bekenstein density $1/4 = 1/|\mu_4|$ — so *no free dimensionless Newton dial* enters. **Thermo = geo coincidence [E] (new).** The first law requires the coefficient $2\pi/\eta$ (η the entropy density); the geometry gives $1/c_3 = |\mathbb{Z}_2|\, 2\pi\, \chi(S^2)$ (one-sided Gauss–Bonnet, v58); these are the *same* equation iff $|\mu_4| = |\mathbb{Z}_2|\,\chi(S^2) = 4$ — a TFPT atom identity (v216). So the thermodynamic (first-law) and geometric (Gauss–Bonnet) origins of c_3 are *one*. **The matter side, assembled [C] (closing J3).** The ball modular Hamiltonian is the boost (Casini–Huerta–Myers; geometric by Bisognano–Wichmann, v323), $K_B = 2\pi \int_B w\, T_{00}$, $w = (R^2 - r^2)/(2R)$; the 3-ball weight integral is $\int_B w\, d^3x = 4\pi R^4/15$, so $\delta\langle K_B\rangle = \frac{8\pi^2 R^4}{15}\,\delta\langle T_{00}\rangle$ — the concrete matter boost flux of TFPT's stress tensor. **The entropy density, atom-fixed [E].** The Bekenstein numerator $1/4 = 1/|\mu_4|$ (v57) and the entanglement normalisation = the central charge $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$ pin the dimensionless coefficient; the area-law form is established (v59). **Residual:** the full covariant extension is now *closed* (v359, next keybox); what remains is only (i) the Jacobson equation-of-state *status* (a thermodynamic derivation, not a from-action quantisation) [O] and (ii) the absolute scale v_{geo} (the Planck-area unit) [O], while the global ambient measure (Gate 2/C7) is now a [C] redundancy (v369) — *no free dimensionless parameter survives in the gravity coupling.*

And that boundary measure is a *rigorously constructed* net ([`verification/v77_e8_conformal_net.py`]). The level-1 central charges $c = \dim\mathfrak{g}/(1 + h^\vee)$ give $c(E_8)_1 = \frac{248}{31} = 8 = \mathrm{rank}\, E_8$, $c(D_5)_1 = \frac{45}{9} = 5$, $c(A_3)_1 = \frac{15}{5} = 3$, and $c(D_5) + c(A_3) = 5+3 = 8 = c(E_8)$ is a *conformal embedding* $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$ (coset $c = 0$). Now $(E_8)_1$ is the holomorphic $c = 8$ *E_8-lattice* VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete:** *identify the seam-Calderón boundary measure with the $(E_8)_1$ lattice net* (carrier = $(D_5)_1 \times (A_3)_1$ subnet), and the gap $\Delta_{\mathrm{eff}} > 0$ supplies the clustering/tightness — so $G6$ is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [E] (charges, embedding) + [C] (identification)

The full covariant field equation, parameter-free: fixed-volume equilibrium gives the Einstein tensor with conservation as an output ([`verification/v359_grav_nonlinear_einstein.py`])

The linearised result above (v358) extends to the *full* covariant equation by Jacobson's *fixed-volume* (maximal-vacuum-entanglement) equilibrium — no linearisation assumed. **Einstein tensor, not Ricci [E].** Stationarity of the entanglement entropy at fixed volume forces the divergence-free combination: the 4D trace identity $g^{ab}G_{ab} = (1-\frac{d}{2})R = -R$ (for $d=4$) selects $G_{ab} = R_{ab} - \frac{1}{2}R\,g_{ab}$, so

$$\boxed{G_{ab} + \Lambda g_{ab} = c_3^{-1}T_{ab}, \qquad c_3^{-1} = 8\pi}$$

with *both* coefficients TFPT-fixed: the Newton coupling $8\pi = 1/c_3$ (v358) and the cosmological constant Λ from α, $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60). **Conservation is an output [E].** By Lovelock's theorem the only divergence-free, second-order, local symmetric tensor built from the metric is $a\,G_{ab} + b\,g_{ab}$, so $\nabla^a G_{ab} = 0$ forces $\nabla^a T_{ab} = 0$ — matter conservation is *derived*, not assumed. **So Gate B6 — the classical metric-sector field equation — is closed parameter-free at the *local* level. Residual:** the Jacobson equation-of-state *status* (a thermodynamic derivation, not a from-action quantisation) and the single unit $v_{\rm geo}$ stay [O], while the global ambient measure (Gate 2/C7, the constructive boundary projective limit) is now a [C] redundancy (v369) — no free dimensionless gravity dial remains. An *external candidate* for the missing action level is quantified in the next keybox (v473); it does not change this typing. [E]/[C]

The entropic-action bridge: an external candidate for the missing action level, quantified — nothing closes ([`verification/v473_entropic_action_bridge.py`])

The one honest [O] left in the classical sector — "equation of state, not a from-action quantisation" (v359) — now has an explicit *external* candidate: Bianconi's entropic action $S_{\rm B} = -\,{\rm Tr}_F \ln\big(\widetilde{G}\widetilde{g}^{-1}\big)$, the quantum relative entropy between the spacetime metric and the matter-induced metric (G. Bianconi, *Gravity from entropy*, Phys. Rev. D **111**, 066001 (2025); `arXiv:2408.14391`; her Appendix C identifies $\Delta^{1/2}_{\widetilde{G},\widetilde{g}} = \widetilde{G}\widetilde{g}^{-1}$ with an Araki-type modular operator) — a local covariant action whose variation yields the Einstein equations plus an emergent nonnegative Λ at low coupling: exactly the missing variational layer, *if* the bridge identities hold. What is machine-checked today (`GRAV.ENTROPIC.ACTION.01`): **carrier Hodge count [E].** Her $\Omega^{0,1,2}$ matter fiber, placed on the five-slot carrier $\mathbb{C}^5$ (*not* on 4d spacetime, where the fiber is 1+4+6=11), counts $1{+}5{+}10{=}16{=}2^{g_{\rm car}-1}{=}\dim S^+_{D_5}$ and Hodge-folds ($\star: \Lambda^1{\cong}\Lambda^4$) onto $\Lambda^{\rm even}\mathbb{C}^5$ — the carrier half-spinor's own Pascal row (v2/v44); a vector-space identity *only.* **Coefficient match [E] (the one quantitative pin).** Her weak-coupling Lagrangian $\mathcal{L} = 3\beta R - \alpha\mathcal{L}_{\rm m} + \cdots$ (the 3 = the three form blocks) matched to the TFPT EH normalisation $1/(16\pi G) = c_3/(2G)$ fixes her free constant *exactly*,

$$\boxed{\beta'_{\rm B} = \tfrac{c_3}{6} = \tfrac{1}{48\pi}}$$

(the $6 = 3\times 2 = p_2 = |R^+(A_3)|$ is recorded as a structural echo, not a proof). **Entropy potential = quadratic Λ [E].** Her $\Lambda_{\mathcal{G}} = \frac{1}{2\beta}\,{\rm Tr}_F(\mathcal{G} - I - \ln\mathcal{G})$ is nonnegative and quadratic at the fixed point; with the TFPT displacement $\varepsilon_\star = e^{-\alpha^{-1}}Q$ it reproduces the closed Λ branch $\rho_\Lambda/\bar{M}^4_{\rm Pl} = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60), and the prefactor match yields the *exact* normalisation target ${\rm Tr}_F\,Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\,\delta_{\rm top}$ — a closed algebraic target for the two-form (pair) block ($\binom{5}{2}$=10), [C] until an explicit Q is constructed. **R^2 kill test [E] (pre-registered).** The raw log expansion carries a curvature-squared coefficient of order $(c_3/6)^2$; the TFPT Starobinsky sector needs $\bar{M}^2_{\rm Pl}/(12M^2_{\rm scal}) = 1/(12c_3^7)$ (v36) — the exact mismatch is $3c_3^{-9} = 3(8\pi)^9 \approx 1.2\times 10^{13}$: a direct identification of the raw entropic action with the TFPT R^2 sector is *false* without a mechanism (the scalaron as a light trace mode of her $\mathcal{G}$-field generating c_3^{-9}, or a KMS-spectral renormalisation of the log action). **The bridge proper [C] (recorded, not proven).** The compression conjecture $P_\Sigma(\widetilde{G}\widetilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$ — the physical Calderón compression of her relative metric operator equals the TFPT seam modular operator; if proven, $S_{\rm B}$ is a local volume representation of the TFPT relative spectral action $S_{{\rm rel},\chi}$ (her log action is the all-scale-integrated heat kernel, by Frullani $-\ln A = \int_0^\infty \frac{dt}{t}(e^{-tA} - e^{-t})$, verified). Open with it: the operator-level Hodge items (Clifford action, Dirac intertwining, leaf involution, Calderón polarisation), and Lorentzian positivity of $\widetilde{G}\widetilde{g}^{-1}$ is *not* automatic (timelike gradients) — TFPT's OS/reflection-positive route is the sturdier construction. **Residual [O] (unchanged).** Nothing closes: the v359 equation-of-state typing stays [O]; adopting $S_{\rm B}$ would close it *only* after the compression conjecture *and* an R^2 mechanism; zero TFPT knobs are added.

The operator level, executed (`[verification/v474_entropic_hodge_carrier.py]`)**.** Work packages 1 and 4-algebra of the bridge are now machine-checked on the finite carrier Fock space $\Lambda^\bullet\mathbb{C}^5$ (dim $2^{g_{\rm car}}$=32). **Spinor structure exhibited [E]:** the ten CAR gammas satisfy $\{\Gamma_a,\Gamma_b\}$=$2\delta_{ab}$ exactly, the 45 bilinears span $\mathfrak{so}(10)$ and leak nothing from the 16-dim even subspace — S^+ *is* the even exterior algebra with its spinor structure, and Bianconi's Hodge–Dirac operator is Clifford multiplication at symbol level ($(c(\xi)^2$=$-|\xi|^2$, symbolic). **The fold is the** $5{\to}\bar{5}$ **conjugation [E]:** the traceless $\mathfrak{u}(5)$ weights of Λ^4 are the negatives of those of Λ^1, so her fiber 1+5+10 becomes the GUT half-spinor 16=1+$\bar{5}$+10 — the SM carrier decomposition, not a dimension match; honest obstruction: $\Omega^{\le 2}$ is *not* Clifford-invariant, so the identification must go through the fold, never through degree truncation. **The** Q**-target, decided [E]:** the uniform-block ansatz $\operatorname{Tr}Q^2$=$d\,k^2c_3^4$=$32c_3^4$ with integer k admits *exactly* the supports (d,k)=$(2,4),(8,2),(32,1)$ = $\{|\mathbb{Z}_2|,\operatorname{rank}E_8,2^{g_{\rm car}}\}$, with the minimal uniform solution q=c_3^2 per Fock mode and the identity $\operatorname{Tr}Q^2/\delta_{\rm top}$=32/48=2/3; the two-form block 10 (the naive pair-sector reading) and the fiber 16 require irrational multipliers and are *excluded*. The physical choice among the three supports stays [C]; AP2 and the R^2 mechanism stay [O].

The R^2 **kill test, executed** (`[verification/v475_entropic_scalaron_gate.py]`)**.** On a maximally symmetric background the relative curvature operator has eigenvalues *exactly* $\{R;\ R/4\times 4;\ R/6\times 6\}$ (her Appendix-B flattening verified, incl. the 6×6 two-form block), so the exact vacuum action is $\mathcal{L}_{\rm B}(R)=-[\ln(1{-}\beta R)+4\ln(1{-}\beta R/4)+6\ln(1{-}\beta R/6)]=3\beta R+\frac{17}{24}\beta^2R^2+O(R^3)$ — the 3 re-derives her Eq. (45) and $\frac{17}{24}$ is the exact tensorial factor behind the pre-registered gap. With the pinned β'=$c_3/6$ *both* mass routes (Starobinsky matching and $f'/3f''$) give the raw entropic scalaron

$$\boxed{m_{\rm raw}^2=\tfrac{4608\pi^2}{17}\bar{M}_{\rm Pl}^2\ \Rightarrow\ m_{\rm raw}\approx 51.7\,\bar{M}_{\rm Pl}}$$

— *trans-Planckian*, not light: the naive reading "the TFPT scalaron is the light trace mode of her $\mathcal{G}$-field" is *dead* [E]; a viable mechanism must supply exactly $m_{\rm raw}^2/m_{\rm TFPT}^2=\frac{72}{17}(8\pi)^9\approx 1.69\times10^{13}$ in mass^2. The domain is not the obstruction (first log pole at R=$384\pi^2\bar{M}_{\rm Pl}^2$, trans-Planckian), and the Lorentzian caveat is now an explicit witness: a timelike gradient gives the Gg^{-1} eigenvalue $1-\alpha v^2\le 0$ for $v^2\ge 1/\alpha$ (ln undefined), while spacelike/Euclidean stay positive — the OS route is the sturdier construction. **Net verdict [C]:** of the three bridge interpretations, the entropic action covers the Einstein + Λ level (survives); the light-mode reading is dead; KMS-spectral renormalisation is the *only* surviving route to the R^2 sector. Nothing closes.

AP2, made well-posed (`[verification/v476_entropic_compression_toy.py]`)**.** The compression conjecture itself is sharpened on an exactly solvable gapped chain. **The reading is decided:** on a *pure* bulk (the seam situation: pure bulk, thermal boundary) the covariance is a projection, so $f(x)$=$x/(1{-}x)$ is singular on its spectrum — the literal operator-side reading $P_\Sigma f(C)P_\Sigma$ is *ill-posed*, while the state-side $f(P_\Sigma CP_\Sigma)$ (Peschel) is finite: the identity can only mean *build* Δ_Σ *from the compressed relative metric*, which is exactly Bianconi's own local construction — AP2 retyped, not weakened. **The mismatch is exactly second order [E]:** symbolically, $[C^n]_{AA}-(C_{AA})^n$ carries the cross-cut block twice (first order vanishes identically), so for any analytic f the two readings agree up to $O(\|C_{AB}\|^2)$ — the modular (entanglement) content *is* that quadratic correction. **Gap control:** at the bounded (covariance) level the mismatch obeys $d\le x^2$ and falls with the gap, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ($\Delta_{\rm eff}$>0, v76/v337). **Residual [O] (unchanged):** the continuum seam statement itself. No marker moves.

The surviving R^2 **route, typed as one moment** (`[verification/v477_entropic_scaleflow.py]`)**.** The Frullani identity upgrades to a *scale-flow representation*: $-\ln a=\int_0^\infty\frac{dt}{t}(e^{-ta}{-}e^{-t})=2\int_0^\infty\frac{d\chi}{\chi}(e^{-a/\chi^2}{-}e^{-1/\chi^2})$ — Bianconi's log action *is* the flat scale-integral of relative heat-kernel actions, while TFPT's $S_{{\rm rel},\chi}$ evaluates *one* KMS scale with a weight f: the bridge between the two action principles is a choice of *scale measure*. In the moment dictionary (weight $w(t)\,dt/t$, μ_k its e^{-t}-moments) the coefficients read $3\beta R\,\mu_1+\frac{17}{24}\beta^2R^2\mu_2$; the flat weight has μ_1=μ_2=1 (= the raw v475 coefficients), and demanding m^2=$c_3^7\bar{M}_{\rm Pl}^2$ forces *exactly*

$$\boxed{\mu_2/\mu_1^2=\tfrac{72}{17}(8\pi)^9}\qquad\text{with}\qquad\frac{4608\pi^2/17}{(72/17)(8\pi)^9}=c_3^7\ \text{(identically)}$$

— the v475 kill-test number *is* a moment ratio, and the "13 orders of magnitude" are not a bridge mismatch but a scale-measure datum: the flat measure fixes it wrongly, the TFPT KMS measure (f_0=$6(4\pi)^2/c_3^7$, v36) fixes it correctly — one KMS moment, two parametrisations, zero new dials. [C] (consistency, not derivation: the measure is TFPT input); the R^2 gate and the v359 typing stay [O].

The two remaining legs, first computable steps ([verification/v478_entropic_continuum_legs.py]). *Leg A (AP2 continuum):* on the exactly solvable critical chain (infinite-volume kernel, 50-digit precision) the compressed interval state carries the conformal/BW structure quantitatively — the entanglement entropy follows Calabrese–Cardy with $c_{\rm est} = 1.00089/1.00038/1.00021$ (monotone → the Dirac c=1; gapped control $\sim 6\times10^{-8}$), and its Peschel modular Hamiltonian organises on the CHM weight $\beta(x)=x(L-x)/L$ (corr 0.87→0.99, even-range bands vanishing identically, odd tail 1.8%): the state-side Δ_Σ (the reading *forced* above) flows to the CHM/Bisognano–Wichmann geometric form — *exactly* the object the entanglement-equilibrium derivation consumes (v323/v358); the bridge's modular side and the gravity side meet in the continuum limit (a lattice exhibit, not a continuum proof — AP2 stays [O]; the natural continuum frame for the seam version is the *four-interval* multilocal modular geometry of the four marks, v480). *Leg B (the measure):* the single-scale corollary is exact — $d\mu=\delta(t-t_0)dt$ gives $\mu_2/\mu_1^2=e^{t_0}$, so the one-moment condition forces $t_0=\ln\frac{72}{17}+9\ln(8\pi)=30.461$, one dimensionless KMS time in closed form; the near-miss to $h(E_8)$=30 (difference 0.461) is explicitly *declined* per the no-free-pattern rule (v354/v355) — deriving the measure from the entropic structure stays [O]. [E]/[C]/[O]

The Simple-Current Extension Theorem — the central $G_{\rm net}$ closing statement [E]/[C] ([verification/v154_simple_current_theorem.py])

The conditional theorem can now be stated as one clean *simple-current extension.* Let $A = (D_5)_1 \otimes (A_3)_1$ be the carrier net and $L = \langle(1,1)\rangle \subset \mathrm{Disc}(D_5) \oplus \mathrm{Disc}(A_3) = \mathbb{Z}_4 \oplus \mathbb{Z}_4$ the isotropic μ_4 glue line ($q(k(1,1)) = k^2 \in \mathbb{Z}$). Then the local extension $B = A \rtimes L$ has

$$\boxed{[B : A] = |L| = 4 = |\mu_4|, \quad c(B) = 5{+}3 = 8, \quad \mu(B) = \frac{\mu(A)}{|L|^2} = \frac{16}{16} = 1}$$

so B is *holomorphic* (KLM μ-additivity), and a holomorphic c=8 chiral net is the lattice net of the *unique* even-unimodular rank-8 lattice — hence $\boxed{B \cong (E_8)_1}$ (the same-c rival $SO(16)_1$, μ=4, is excluded). Every algebraic ingredient is machine-checked: the $\mathbb{C}[\mathbb{Z}_4]$ Longo Q-system ([verification/v125_glue_qsystem.py]), the $\mathbb{Z}_4$-graded Frobenius E_8 hull ([verification/v143_graded_frobenius.py]), the Ramond glue sectors ([verification/v148_fock_census.py]). **Consequence:** $G_{\rm net}$ is internally *closed* if the seam–Calderón boundary net is *defined* to be A; it stays the *one* external identification "the RP seam–Calderón net is A" if the boundary net is to be *derived* from the seam kernel. Either way this is the single strict-TOE certification theorem still carrying weight — a clean extension statement, not "find a quantum gravity".

And that one identification reduces to a single shared premise ([verification/v155_quasifree_boundary.py]). The identification splits (v110) into (a) "the seam Calderón involution is sheet-odd" — the double-cover deck, the same $|\mathbb{Z}_2|$ that halves $c_3 = \frac{1}{2\cdot4\pi}$ (structural) — and (b) "the seam 2-point kernel is quasi-free". Part (b) is *not independent*: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression $P\Gamma P$ of the bulk covariance, and the compression of a contraction ($i\Gamma$ with spectrum in $[-1, 1]$) is a contraction — so the boundary/seam state is automatically quasi-free, with Wick inherited (every boundary correlator a Pfaffian of Γ_d). Hence (b) *follows* from "the seam bulk is a reflection-positive free field" — the *same* premise already load-bearing in the area law ([verification/v59_area_law_evidence.py]) and the EH mechanism ([verification/v150_replica_eh_model.py]/ [verification/v151_bfk_split.py]; since exercised on the discretized collar with the seam's own kernel and real replica sheets, v471 — the R3 kernel premise is discharged at the finite level, its residual retyping to the continuum scaling limit shared with SEAM.EQUIV.01, v336, plus the v152 anchor). **So the entire seam-side residual is one physical premise** — the seam bulk is a free RP field — shared by $G_{\rm net}$, the area law and R3, and given it the chain Gaussian bulk ⇒ quasi-free boundary ⇒ one kernel = net (v113) ⇒ carrier $A \Rightarrow (E_8)_1$ is exact. It is not reducible by a finite computation — it is the one place TFPT meets functional analysis.

The construction, exhibited and verified ([verification/v156_seam_net_construction.py]). The "$B \cong (E_8)_1$" step is no longer an assertion — the chiral net is built explicitly and the identification *checked.* The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian acts on a boundary mode e^{ikx} by the harmonic extension $e^{ikx-|k|y}$: $-\partial_n u|_0 = |k|\, u$, so for a free bulk the seam operator is the massless chiral dispersion $\omega_k = |k|$ that defines the free-fermion net on S^1. Sixteen Majoranas give $c = 8 = 5{+}3$; the weight-1 currents are $248 = 120_{\rm NS} + 128_{\rm R}$ (the $SO(16)$ bilinears plus the Ramond spinor, Frenkel–Kac); and the holomorphic character is

$$\boxed{\chi = \frac{E_4}{\eta^8} = q^{-1/3}\big(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + \cdots\big) = j^{1/3}}$$

(verified by q-series), the 248 at level 1 being $\dim E_8$ — so the net *is* $(E_8)_1$. *What remains* is exactly (i) the free-bulk premise above and (ii) the operator-algebraic existence / uniqueness of the net generated by the seam Calderón kernel (a Kawahigashi–Longo-type net-construction theorem). Route 3 thus *exhibits and verifies* the target; the irreducible residual is the one premise + the standard net-existence step — where TFPT meets constructive QFT, now sharply located.

And the free-bulk premise itself reframes: freeness is a rigid fixed point, not a postulate (`[verification/v157_rigid_fixed_point.py]` **).** Three facts click together. (i) The DtN/Calderón symbol is *universally* $|k|$: the boundary operator of *any* second-order elliptic bulk is an order-1 ΨDO with principal symbol $|\xi|_g$ (Lee–Uhlmann), so the free chiral dispersion is bulk-detail-independent and any interaction lives in the lower-order (irrelevant) symbol. (ii) A holomorphic c=8 theory is *rigid*: every field has weight $(h, \bar{h}{=}0)$, an exactly-marginal deformation needs $(1, 1)$, and with $\bar{h}$=0 *no* $(1, 1)$ field exists — the fixed point is isolated, there is no continuous knob along which a non-Gaussian interaction could turn on. (iii) The *same* $|\mathbb{Z}_2|$ that halves the one-sided S^2 Gauss–Bonnet ($c_3 = \frac{1}{|\mathbb{Z}_2|\, 2\pi\, \chi(S^2)} = \frac{1}{8\pi}$) labels the sheet *and* is the Ramond/GSO projection giving μ=1 ($248{=}120_{\mathrm{NS}}{+}128_{\mathrm{R}}$) — one object, three roles, so the geometry fixing c_3 is the projection making the net holomorphic. **Consequence:** premise (A) shrinks from "prove the seam is Gaussian" to "the gapped ($\Delta{=}6\log\frac{3}{2}{>}0$) boundary flow reaches the holomorphic fixed point" — given which, freeness is *forced* by rigidity + uniqueness ($(E_8)_1$, v83), not fine-tuned. A strictly smaller and simpler residual.

And the destination is a *proven* stable fixed point (`[verification/v158_fixed_point_stable.py]` **).** An exact operator-dimension count settles it: the chiral Majorana has $h{=}\frac{1}{2}$, the bilinear current h=1, the quartic h=2. The relevant window $0 < h < 1$ for bosonic deformations is *empty* (bosonic chiral operators have integer h, lowest 1) — nothing relevant flows the free theory away; the h=1 currents are chiral (v157), not $(1, 1)$-marginal; and the lowest genuinely-interacting operator (the quartic, $h{=}2{>}1$) is *irrelevant*, as are all higher $2k$-fermi operators ($h{=}k$). So the free chiral c=8 fixed point is *isolated and stable* against all interactions. Premise (A) thereby reduces to a *basin-of-attraction* statement with a proven stable target — the gapped ($\Delta{=}6\log\frac{3}{2}$) flow lands here; the convergence is conformal-perturbation analysis, the destination is settled.

And the closure can be stated as one fixed-point theorem — which relocates (A) to a single internal premise (`[verification/v160_seam_gaussianity_from_pf.py]` **).** Phrase the argument internally: (i) a quasi-free state has *all* higher connected cumulants κ_{2n}=0 (the signed set-partition/Pfaffian inversion of $\langle\psi_1\cdots\psi_{2n}\rangle = \mathrm{Pf}(\Gamma)$ leaves only the degree-2 kernel — the v113 "one kernel = the net" property at the cumulant level); (ii) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (iii) a gapped primitive Perron–Frobenius transport has a *unique* fixed ray, so a non-Gaussian piece is impossible — if κ_{2n} were fixed it would be a *second* fixed ray (killing uniqueness), and if not fixed it sits in the gapped complement and decays to 0 (any gap > 0 suffices; the value $(2/3)^6$ only sets the rate). *The honest residual:* v56's gapped PF uniqueness is established on the *three-dimensional* scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (the flavour gap / horizon Page-recovery rates), whereas the 16-Majorana seam state already carries $\binom{16}{4}$=1820 and $\binom{16}{6}$=8008 higher-cumulant directions — the Schwinger cone the transport must contract has dimension $\gg 3$, and uniqueness must hold there (not merely on the quasi-free sub-cone) for the argument to be non-circular. **So (A) is equivalent to one strictly smaller, internal, exact-testable statement:** the admissible TFPT seam transport is a *primitive, spectrally gapped, RP-cone-preserving* map on the *full* Schwinger cone, not only on the scalar readout spectrum of v56 — which the suite does *not* yet establish [E]/[O].

And that full-cone premise is now *discharged* to a single finite one-particle statement (`[verification/v161_one_particle_reduction.py]` **).** The lever is Bogoliubov second quantisation: a quasi-free transport is $T_\Sigma = \Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dimensional Majorana space H_Σ, and the Fock space grades by correlator degree, so $\Gamma(t)$ acts on the degree-m sector as the exterior power $\Lambda^m(t)$. Three exact consequences follow (all machine-checked): (i) $\mathrm{spec}(\Gamma(t)|_{\deg m})$ is the set of products of m one-particle eigenvalues — the whole correlation spectrum is generated by the finite one-particle spectrum; (ii) **the cone gap *equals* the one-particle gap**: the sub-leading eigenvalue of $\Gamma(t)$ over *every* degree is $(2/3)^6$ (pad the largest contracted mode with fixed factors), so premise 5 on the cumulant space *is* premise 5 on the 16-dim space, the same number; (iii) the eigenvalue-1 sector is Λ^* of the fixed subspace — the disconnected Wick powers of K_Σ — so with a strict one-particle gap no independent connected cumulant is fixed, and an independent fixed cumulant *on* the fixed subspace is excluded by v113 (one polarisation ⇔ one vacuum). Premise 1 (quasi-freeness) is itself *forced*, not assumed: the $120 = \dim \mathfrak{so}(16)$ weight-1 chiral currents *are* the Bogoliubov generators on the 16 Majoranas, and v158 makes every non-Gaussian deformation irrelevant. **So the three full-cone premises collapse to one finite, 16-dimensional statement:** the seam RG transport's one-particle symbol is the gapped Calderón Bogoliubov map — fixed eigenspace = the rank-8 K_Σ polarisation (v113), sub-leading gap = $(2/3)^6$ (v56). Every *number* in it is already supplied by the suite; the one irreducible step is the physical identification that the actual seam RG generator equals this Bogoliubov map — the same operator-algebraic (Kawahigashi–Longo-class) statement as the net-existence residual of (A_2), now with the infinite-dimensional cone eliminated [E]/[O].

And that finite statement carries no *new* open content: it factors into the two already-open items ([`verification/v162_seam_transport_identification.py`]). Take the one-particle symbol apart into its two data, the gap value (β) and the fixed space (α). (β) *is forced*: a μ_4-equivariant operator is diagonal in the cusp basis — the deck average is the diagonal projection, $\sum_{k=0}^{3} U^k X U^{-k} = |\mu_4| \operatorname{diag}(X)$ for $U = \operatorname{diag}(1, i, -i)$, and the commutant of U is exactly the diagonal algebra — so its spectrum is the cusp weights $\operatorname{spec}(A_0^\star) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (the parabolic residue, v115), and the transfer eigenvalue $(1-\alpha)^{p_2}$ with $p_2{=}6{=}|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6 \log \frac{3}{2}$ — every entry carrier data, no free knob; the lift to the 16 Majoranas is the established A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of the $\omega_k{=}|k|$ free dispersion (v113/v156), the IR conformal free-fermion vacuum. **What is left** is only that the seam RG generator *is* μ_4-equivariant and admissible — which is the R1 "seam clock" identification (the transfer operator as the seam's gapped generator) — together with the A_2 net-existence theorem. Both are pre-existing open items, so premise (A) introduces *no independent open content*: it factors exactly into $A_2 + R_1$ with every number forced. The "free-bulk cloud" is gone; what remains is the same two identifications the programme already names — (A) is closed *conditionally* on them (a unification, honestly not an unconditional proof) [E]/[O].

And both residual identifications are pinned to their floor, so (A) has no independent open gate left ([`verification/v165_premise_a_closure.py`]). A_2 (net existence) is not open research but *assembly* of established mathematics: the 16 free Majoranas define a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic $c{=}8$ lattice net *is* $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125); the one TFPT-specific input — that the seam Calderón kernel is the free-fermion two-point function — is done because the DtN symbol is universally $|k|$ (v156/v157). So A_2 re-types [O] → [C] (closed by assembly). R_1 (the seam clock) reduces to `GATE.QGEO`: a μ_4-equivariant flat gapped generator on the carrier is *unique* ($= A_0^\star$, v115/v116), so its gap $(2/3)^6$ is forced once the seam generator lies in that class, and the residual is exactly one geometric identification — the seam collar boundary is the μ_4-punctured sphere $\mathbb{P}^1 \setminus \mu_4$ — whose cohomological half is already established (v137, $b_1{=}3{=}N_{\text{fam}}$). That is `GATE.QGEO`, an already-open realisation premise, *not* a new gate. By the No-Unit Theorem (v153) the irreducibles stay exactly $\{\pi, v_{\text{geo}}\}$ — closing (A) adds none. **Final accounting:** premise (A) = (A_2 assembly, [C]) + (R_1=`GATE.QGEO`, already open) + the $\{\pi, v_{\text{geo}}\}$ irreducible core (a theorem) — it carries *zero* independent open gates and ceases to be its own gate [E]/[O].

And the two structural residuals — net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the one irreducible open premise ([`verification/v175_net_existence_full_cone.py`]). The CAR second-quantisation functor $\Gamma(t){=}\bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the *complete* Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16}{=}65536$); by the exterior-power spectrum theorem every eigenvalue of $\Gamma(t)$ is a subset product of $\operatorname{spec}(t)$, so checking all 2^{16} products shows $\Gamma(t)$ is a *contraction* (full-cone RP), with eigenvalue-1 multiplicity $2^8{=}256$ (the wedges of the rank-8 K_Σ polarisation) and sub-leading eigenvalue exactly the one-particle gap $(2/3)^6$ — so full-cone RP/gap reduce *for every m* (not only $m \leq 6$, v161) to the one-particle contraction (Shale–Stinespring), *discharging the v160 scope premise by a theorem.* And A_2 is now an *assembled and verified* certificate: the $(E_8)_1$ character $E_4/\eta^8{=}1{+}248q{+}4124q^2{+}34752q^3{+}213126q^4{+}1057504q^5$ (level-1=248), the embedding 248=120+128, and the E_8 Cartan matrix even with det 1 (the unique rank-8 even unimodular lattice, v83) — the net *exists* by cited theorems. **Both residuals then collapse to the *same* input**: that the actual seam Calderón kernel is this gapped one-particle contraction on $\mathbb{P}^1 \setminus \mu_4$, i.e. `QGEO.REALIZE.01`, whose finite half is proven (v168). That realisation half is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]— the single irreducible structural premise; net existence and full-cone RP are [E].

The No-Unit Theorem — v_{geo} is an irreducible metrology primitive [E]/[O] ([`verification/v153_no_unit_theorem.py`])

v_{geo} is not an open gap. Every load-bearing compiler datum — $g_{\text{car}}, |\mu_4|, N_{\text{fam}}, \text{rank}\, E_8, \alpha^{-1}$, the determinant ladder $(3,4,8,20)$, $c_3 = \frac{1}{8\pi}$, φ_0^{ret} — is mass-dimension 0 and *invariant* under a unit rescaling $L \mapsto \lambda L$, whereas a mass / energy / $1/G$ carries nonzero dimension. A map from dimension-0 inputs to a dimension-$d{\neq}0$ output is simultaneously invariant and covariant: impossible unless the unit is introduced. **So a dimensionless boundary compiler provably cannot select an absolute scale** — v_{geo} is a *forced input*. The three apparent residuals then collapse onto the one unit, differing only in mass-dimension:

$$U_{\text{point}} \sim v_{\text{geo}}, \qquad \tfrac{1}{G} \sim v_{\text{geo}}^2, \qquad m/\mu = e^{3/4} \text{ (a pure ratio)},$$

so the irreducibles stay exactly $\{\text{one dimensional unit}, \pi\}$. *Re-typing:* v_{geo} is an *irreducible metrology primitive* (a declared unit, like choosing a metre), not a missing derivation.

Sharpened across gravity, and the final tally ([`verification/v364_vgeo_sharpen.py`]). After the parameter-free gravity (v358/v359/v361) the *gravity* sector adds no new scale either: the Einstein coefficient $1/c_3 = 8\pi$ and the Λ prefactor $(8\pi)^2 \cdot 48 c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\text{geo}}^2$ and $\Lambda \sim$ (dimensionless) v_{geo}^4 both reduce to v_{geo} times atoms. Hence v_{geo} is the *single* dimensionful input of the *complete* theory (matter and gravity), and the final tally is $\{v_{\text{geo}}, \pi\}$ with *zero* dimensionless dials — a $\sim$26 $\to$ 1 reduction against the Standard Model. ($1/G$ stays UV-sensitive (Sakharov, v68), so it is a metrology readout, not a clean prediction — consistent with v_{geo} being the anchor.)

Theorem G — the closure statement [O] (target)

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure μ_{QG}. Its Ward identities imply covariant conservation, and its low-curvature expansion is the TFPT $R + R^2$ seam action

$$f_R R_{\mu\nu} - \tfrac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu}\Box - \nabla_\mu \nabla_\nu) f_R = \bar{M}_{\text{Pl}}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: the local covariant field equation is parameter-free (v358/v359); Theorem G remains open only for the projective-limit ambient* measure μ_{QG} *(G6, the deepest item). G5 certified; the regime equation closed [`verification/v28_gravity_fR.py`].*

Research Contract — `CELEST.SEAM.01` (the celestial-holographic route)

Reader's note. A compact synthesis of this contract's executed work packages (WP1–WP5d, both WP5d stages, + the WP5e α and β stages) — the narrative arc from the μ_4 clock to the $(E_8)_1$ boundary shadow — is given as a dedicated section in `tfpt_3_e8_audit_bootstrap` ("The celestial route: from the μ_4 clock to the $(E_8)_1$ boundary shadow"); this contract remains the full technical reference (typing fence, kill tests, work-package statements).

Hypothesis (as corrected by WP1). *The seam is the glue-equivariant $\mathbb{Z}_4$-orbifold sector of the celestial chiral algebra on the A_3 ALE space $\mathbb{C}^2/\mathbb{Z}_4$; the μ_4 clock is the Kähler $U(2)$ phase* $\text{diag}(i, 1)$ *whose square is the deck group (spin bridge $\mathbb{Z}_8$, $8 = 2|\mu_4|$ — the $c_3 = 1/(8\pi)$ seam winding integer); and the associated twistorial bulk is the* self-dual *sector of TFPT gravity.* This is the fourth research contract, alongside (U_{wall}), (G_{metric}) and `CONTRACT.F.01`: a numbered chain of work packages with pre-registered kill tests, *not* a claim. The method is the celestial-chiral-algebra (CCA) construction of Bittleston–Homans–Sharma on Eguchi–Hanson space (`arXiv:2305.09451`, JHEP 09 (2023) 008 — the $\mathbb{Z}_2$ case, refined here to $\mathbb{Z}_4$ with a flat connection at infinity in the Kronheimer–Nakajima / heterotic-on-ALE sense), inside the celestial-holography programme (Strominger, PRL **127**, 221601: $w_{1+\infty}$; Costello–Paquette–Sharma, PRL **130**, 061602: Burns holography, the $SO(8)$ WZW$_4$ model; Bittleston–Sharma–Skinner, CMP **403** (2023): quantizing the non-linear graviton). The gauge-algebra admissibility gate is already passed in the literature: Costello's one-loop axionic-consistency analysis (`arXiv:2111.08879`) admits the exceptional algebras *including* E_8 for SDYM on twistor space (Okubo: no independent quartic Casimir). The spin bridge in the hypothesis is no longer a bookkeeping convention: on the 16-Majorana seam the NS deck implementation is *forced* to order 4 ($U^2 = (-1)^F$ exactly, nonsplit $\mathbb{Z}_4$) and the quarter-shift lifts to a canonical $\mathbb{Z}_8$ tower with $V^2 \propto U$ — "$8 = 2|\mu_4|$" fermionically explained ([`verification/v506_seam_clock_rigidity.py`], `SEAM.CLOCK.RIGIDITY.01`).

WP1, executed and verified: the E_8 μ_4-glue *is* the flat $\mathbb{Z}_4$ monodromy of the equivariant CCA on the A_3 ALE space — verdict B ([`verification/v492_celestial_z4_orbifold.py`], `CELEST.WP1.01`)

Everything in this box is exact arithmetic (sympy, no floats). **Inner grading / flat monodromy [E]:** the v128 glue grading ($240 = 52+64+60+64$, additive on all 6720 root pairs, dims with Cartan $(60, 64, 60, 64)$, $\mathfrak{g}_0 = D_5 \oplus A_3$ carrier $= 60$) is *inner*: $h = (2, 2, 2, 2, 2; 0^4)$ reads the glue class mod 4 on all 240 roots, so the glue is $\mathrm{Ad}(\exp(2\pi i h/4))$ — an order-4 torus element, i.e. a *flat* $\mathbb{Z}_4$ *monodromy* in the Kronheimer–Nakajima sense; the A_3-side detector $h' = (0^5; 1, 1, 1, -3)$ reads the *same* class, so the glue charge is the diagonal $(1, 1)$ of the discriminant group $\mathbb{Z}_4 \times \mathbb{Z}_4$ — the v92/v125 Lagrangian glue. **The spatial side [E]:** $\mathbb{C}^2/\mathbb{Z}_4$ under the deck $(z_1, z_2) \mapsto (iz_1, i^{-1}z_2)$ *is* the A_3 singularity $XY = Z^4$ (invariant ring exact; Hilbert series $(1+t^4)/((1-t^2)(1-t^4))$; Molien table verified). **The equivariant sector closes [E]:** the glue-equivariant SDYM(E_8) modes ($j + m \equiv 0 \bmod 4$) form a Lie subalgebra with graded dimensions $D[d] = 60(d+1)$ (even d) / $64(d+1)$ (odd d) — uniform per parity *only* because $\dim \mathfrak{g}_0 = \dim \mathfrak{g}_2 = 60$ and $\dim \mathfrak{g}_1 = \dim \mathfrak{g}_3 = 64$: the v128 dimensions are exactly the consistency condition. The zero modes ($d{=}0$) are $\mathfrak{g}_0$ alone — *the unbroken algebra on the orbifold is the carrier* $D_5 \oplus A_3$+Cartan $= 60$; the asymptotic density is $62/248 = 1/4 = 1/|\mathbb{Z}_4|$ (the orbifold index), and the $\mathbb{Z}_2$ halfway sector reproduces the Eguchi–Hanson towers $120/128 = SO(16)$ adjoint/spinor — μ_4 refines the sheet $\mathbb{Z}_2$ exactly as $4 = 2 \times 2$ (v125). **Character level [E]:** Θ_{E_8} splits into the four glue-coset thetas (shell $1 = (52, 64, 60, 64)$ re-derived from the *lattice*); the four sector characters Θ_{C_j}/η^8 sum *exactly* to $\chi_{(E_8)_1} = 1{+}248q{+}4124q^2{+}34752q^3$ (the v377 series); the sector conformal weights are $h_{D_5} = (0, \frac{5}{8}, \frac{1}{2}, \frac{5}{8})$, $h_{A_3} = (0, \frac{3}{8}, \frac{1}{2}, \frac{3}{8})$ — the v92/v125 discriminant form $(5x^2{+}3y^2)/8 \bmod 1$ — and the glue diagonal has *integer* $h = (0, 1, 1, 1)$: the v125 isotropy *is* the locality of the extension to $(E_8)_1$. **The critical correction [E] (why verdict B, not A):** the A_3 deck $\mathrm{diag}(i, i^{-1})$ induces on the celestial sphere $z = z_1/z_2$ only $z \mapsto -z$ (projective order 2, the sheet flip) — it is *not* the order-4 clock $z \mapsto iz$. The exact bridge: any $SU(2)$ lift of the clock has order 8 (spin group $\mathbb{Z}_8$, which would quotient to A_7), and (spin clock)^2 = the deck generator *exactly*, with $|\mathbb{Z}_8| = 8 = 2|\mu_4|$ — the $c_3 = 1/(8\pi)$ winding integer. The order-4 clock *is* realised, as the $U(2)$ phase $\mathrm{diag}(i, 1)$: projective order 4, fixed points $\{0, \infty\}$, free μ_4 orbit (all v180 clock properties), normalising the deck (so preserving the glue flat connection), with $\det = i$ rotating the holomorphic symplectic form by the μ_4 generator (a Kähler, *not* triholomorphic isometry — it rotates the twistor sphere by $\pi/2$). **Gravity side [E]:** clock-invariance selects from the 3-parameter versal A_3 family $XY = Z^4{+}a_2Z^2{+}a_1Z{+}a_0$ *exactly* the 1-parameter family $XY = Z^4{+}a_0$, whose four branch points are *one* μ_4 orbit with cross-ratio 2 — the seam marks (v179 clock orbit; v168/v214 cross-ratio $2 \Rightarrow j{=}1728$ pillowcase). **Negative controls [E]:** the false spatial action $\mathrm{diag}(i, i)$ is killed three ways ($\det = -1 \notin SL(2, \mathbb{C})$, no CY/ALE resolution; Poisson closure fails in 14/14 nonzero cases; conjugation symmetry broken), the false glue (60/64 blocks swapped) breaks additivity on 3112/6720 root pairs, and of all 24 relabelings exactly $\{\mathrm{id}, j \mapsto -j\} = \mathrm{Aut}(\mathbb{Z}_4)$ are additive — the grading is rigid. **Verdict [C]:** the celestial and glue $\mathbb{Z}_4$ structures match under *one* precisely named extra identification — clock^2 = deck on the sphere (the sheet-$\mathbb{Z}_2$ spin bookkeeping) — so WP1 lands at verdict B (match modulo one named identification), close to but honestly short of A. [E]/[C]

Typing and non-circularity. Admissible *inputs* are the finite seam/glue data already in the suite — the μ_4 Q-system glue (v92/v125), the graded hull (v128), the seam rigidity and pillowcase geometry (v168/v214), the Möbius clock (v180) — plus the WP1 [E] arithmetic ([`verification/v492_celestial_z4_orbifold.py`]). *Not* admissible as an input: the *continuum existence* of the holomorphic $(E_8)_1$ net on the seam — that is precisely the `SEAM.EQUIV.01` target, and assuming it would make the contract circular. *Scope fence:* the twistorial bulk of this contract is the *self-dual* sector only, never full Einstein gravity — the classical field-equation route (entanglement equilibrium, v358/v359) and the entropic-action bridge (`GRAV.ENTROPIC.ACTION.01`, v473) are separate, non-competing statements about the full equation. `SEAM.EQUIV.01` stays [O] throughout; nothing in this contract moves it.

Work packages.

- **WP1 (*done*, verdict B) [E]/[C]:** the structural $\mathbb{Z}_4$ arithmetic of the keybox above, executed and machine-verified as `CELEST.WP1.01` [`verification/v492_celestial_z4_orbifold.py`].
- **WP2 (*done*, verdict B) [E]/[C]:** the clock-invariant deformation theory, executed and machine-verified as `CELEST.WP2.01` [`verification/v493_celestial_wp2_clock_deformation.py`] (sympy exact, 47 checks). Both targets landed, *sharpened*: (i) a_0 is the pure *scale* of the seam configuration — the binary-quartic invariants of $w^2 = Z^4{+}a_0$ are $I = 12a_0$, $J = 0$ *identically*, so $j = 1728$ and the $\tau{=}i$ pillowcase shape (v168/v214) is *frozen* for every $a_0 \neq 0$ (no shape modulus survives clock-invariance; the negative controls $a_1 Z \Rightarrow j{=}0$ and $a_2 \neq 0 \Rightarrow j{=}1556068/81$ show the test has teeth); (ii) the three resolved spheres ($3 = \operatorname{rank} A_3 = N_{\mathrm{fam}}$) carry *exactly* the three nontrivial μ_4 characters $\{i, -1, -i\}$ under the clock, which acts on H_2 as a *Coxeter element* of $W(A_3)$ (char $x^3{+}x^2{+}x{+}1$, order $h(A_3){=}4{=}|\mu_4|{=}N_{\mathrm{fam}}{+}1$) and *is* the Picard–Lefschetz monodromy of the family ($a_0 \to e^{2\pi i}a_0$ gives the 4-cycle); the surviving deformation direction is the weight-1 χ_1-Fourier diagonal $(1, i, i^2)$ — *not* the naive invariant diagonal ($\det(A{-}1) = -4$: no invariant cycle) — and all three sphere volumes move in lockstep ($|\Pi_j| = \sqrt{2}\,t$). The BHS deformed-algebra pattern transfers from $\mathbb{Z}_2$ to $\mathbb{Z}_4$: the fibre bracket $-4\,\mathrm{Nambu}(XY{-}Z^4{-}a_0)$, anchored at the $a_0{=}0$ orbifold, closes with corrections exactly *linear* in a_0 at the $\mathbb{Z}_4$ wrap, conserving the μ_4 grade (no sector leak — the WP1 equivariant sector deforms consistently), with a_0 in the weight-0 slot of BHS's c^2. Verdict B via three named [C] identifications ($-4\,$Nambu as *the* $k{=}4$ CCA bracket; period = root difference; the seam-scale reading via clock^2 =deck); the full deformed twisted-sector OPEs and the $W(\mu)$-type scaling limit for $k{=}4$ stay **[O]**. The v216 [O]-residual is *typed*, not moved: given the order-4 clock, the deformation freezes the $(2,2,2,2)$ modulus to the square automatically — a relocation of the same order-4 carrier input, not a new derivation.
- **WP3 (*done*, verdict B) [E]/[C]:** the Green–Schwarz alignment test, executed and machine-verified as `CELEST.WP3.01` [`verification/v495_celestial_wp3_gs_alignment.py`] (exact Fraction/sympy, 25 checks). The arithmetic side is sharp: the Okubo coefficient $\mathrm{Tr}_{\mathrm{adj}}X^4/(\mathrm{Tr}_{\mathrm{adj}}X^2)^2 = 5/(2(\dim\mathfrak{g}{+}2))$ is *derived* as a polynomial identity for all 8 algebras on Costello's list (with a sharp $\mathfrak{sl}_4$ negative control: 5/32 vs 3/32), the closed form $\tilde\lambda^2 = 10h^{\vee 2}/(\dim{+}2) = h^\vee{+}6$ holds exactly across the Deligne series, and for E_8 the axion coefficient is *exactly* $\tilde\lambda = 6$ (unit-trace) resp. $\lambda_{\mathrm{fund}} = 1/10$ (adjoint-trace), so the GS coupling $\kappa = \lambda_{\mathfrak{g}}/(8\pi\sqrt{3})$ satisfies $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\mathrm{fam}}$ resp. 1/300 — exact anchor rationals, with the anchor decomposition $h^\vee{=}30{=}|\mathbb{Z}_2|N_{\mathrm{fam}}g_{\mathrm{car}}$, $\varphi(30){=}8{=}\,$rank, $\tilde\lambda{=}|\mathbb{Z}_2|N_{\mathrm{fam}}$, $\lambda_{\mathrm{fund}}{=}1/(|\mathbb{Z}_2|g_{\mathrm{car}})$ (a byproduct: the printed $\lambda^2_{\mathfrak{so}_8} = 3/2$ in Costello's appendix A is a factor-2 slip; his own weight content gives 3, Okubo-consistent). κ/c_3 itself is *irrational* ($2\sqrt{3}$): only the squared alignment is exact. **The look-elsewhere caveat is part of the result [C]:** the same squared-rational alignment holds for *all eight* algebras on the list (8/8 — the test as such has zero selective power), $\tilde\lambda$-integrality passes 2/8 (shared with $\mathfrak{sl}_3$), and the only $\mathfrak{e}_8$-selective single test is $g_{\mathrm{car}}{=}5 \mid h^\vee$ (1/8); the conjunction that isolates $\mathfrak{e}_8$ is post hoc, not preregistered. *Alignment survives; selectivity does not* — the c_3-connection of the celestial route is convention-level compatibility plus genuine λ-arithmetic, *not* $\mathfrak{e}_8$-selective evidence, and never a derivation of c_3 from celestial data.
- **WP4 (*done*, verdict B(ii)) [E]/[C]:** the type mismatch, executed head-on and machine-verified as `CELEST.WP4.01` [`verification/v496_celestial_wp4_character_shadow.py`] (exact integer/Fraction/sympy, 25 checks). The $(E_8)_1$ character is *not* a conformal block of the celestial $E_8[\mathbb{C}^2]$ S-algebra in its own (jet) grading — the obstruction is localised *three* independent ways: (a) the CP grading gives spin $1{-}d/2$, unbounded below, and 248 is never a jet-tower dimension ($60(d{+}1)/64(d{+}1)$, $d \le 100$); (b) the cumulative generator count is quadratic ($31s^2{+}92s{+}60$, $31 = k{+}h^\vee$), so the jet Fock grows as $n^{2/3}$ against the character's $n^{1/2}$ (Meinardus poles $s{=}2$ vs $s{=}1$; the ratio f_n/χ_n increases strictly for $n = 1..12$); (c) the level-2 null ideal of $(E_8)_1$

deletes *exactly* $27000 = 30^3 = (h^\vee)^3$ out of $\mathrm{Sym}^2(248) = 1{+}3875{+}27000$ (the character keeps $4124 = 1{+}248{+}3875$) — and has no jet analogue. **But the boundary/period reading holds exactly at the current stratum**: the zero-mode slice is the 60 vacuum-sector currents, the single-line jet slice cycles the four sector counts $(60, 64, 60, 64)$, one full μ_4 period of loop energies sums exactly to 248 (the single-particle level of the $248q$ stage), and the glue-diagonal weights $(0, 1, 1, 1)$ are integers — while the free loop Fock at level 1 counts $897266 \gg 248$, i.e. the rational truncation must be *imposed in a limit*. That is precisely the MMST scaling-limit shape of `SEAM.EQUIV.01` (v336/v449: the rational net exists in the *limit* of the lattice collar, not in the pre-limit algebra): WP4 is hereby *retyped* from "character as conformal block" to "character as boundary/limit shadow", and the constructive limit question (which limit, which topology, convergence) passes to WP5 — where WP5a has since answered the "which limit" part at the counting level (the coefficientwise χ_w limit below), WP5b has constructed the deleting operator, WP5c the limit state whose GNS kernel contains the ideal, WP5d the local-net legs (the α-stage two-interval index chain and the β-stage split + strong-additivity witnesses — all three KLM ingredients of complete rationality on the lattice), and WP5e-α/β have anchored *both* faces of the twistor uplift (the CFT-side prefactor + level pinning; the equivariant twistor skeleton with the index bridge and the rigid $32\,T_3$ residual), leaving the global BCOV quantisation (WP5e proper) as the remaining open piece.

- **WP5 (the summit) — subdivided; WP5a–WP5d (both WP5d stages) plus the WP5e α and β stages are landed.** The twistorial construction as a *second continuum route* for `SEAM.EQUIV.01` (Costello–Li grade), carrying the WP4 constructive-boundary-limit question. Subdivision WP5a–WP5e:
  - **WP5a (*done*) [E]/[C]:** "the null ideal from the limit", executed and machine-verified as `CELEST.WP5A.01` [`verification/v497_celestial_wp5a_null_ideal.py`] (exact integer/Fraction arithmetic, 34 checks). Three results make the WP4 limit reading precise. (i) *Stabilisation theorem*: the one-parameter family χ_w of graded Fock characters on the chiral jet generators (loop energy m, radial number r; $E_w = m + wr$ in quarter units $u = q^{1/4}$) *contains* the chiral jet grading as its $w{=}2$ member (generator counts $64, 120, 128, 180, 192, 240, 256, 300$ per degree), and its u^n coefficient equals the quarter-moded loop Fock for *all* $w \geq n{+}1$ — strictly larger for every $w \leq n$ (verified $n \leq 8$, $w \leq 10$): the WP4 "boundary-limit shadow" is a precise *coefficientwise* limit $\varepsilon = 1/w \to 0$ with explicit stabilisation threshold $w = n{+}1$, not a slice. (ii) *The null ideal is derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give $\mathrm{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$ with peeling residual exactly zero (independent Weyl-orbit cross-check); $27000 = 30^3 = (h^\vee)^3$ exactly; level-1 integrability ($\langle 2\theta, \theta^\vee\rangle = 4$, $\langle \omega, \theta^\vee\rangle = 2$) leaves the vacuum as the only level-1 primary ([C] Kac, used only through its counting consequence); level-2 quotient $31124 - 27000 = 4124 = 1 + 248 + 3875$ exactly. (iii) *Two routes, one number*: the μ_4 theta-split sector sum at $q^2 = (1036, 1024, 1040, 1024)$ gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space (q^3: 34752). Negative controls: $SO(16)_1$ through the same pipeline gives *four* components $(5304{+}1820{+}135{+}1)$, $5304 \neq 14^3$ (the $h^{\vee\,3}$ identity is not generic), quotient $2076 = \Theta_{D_8}/\eta^8$ at q^2 (the recipe validated on a second algebra), and block weights $(0, \frac{1}{2}, 1, 1)$ that can*not* fuse into one local character; only $P = 4 = |\mu_4|$ periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor. *Honest limit* [O]: the limit does *not* generate the truncation (the loop Fock counts $897266 \gg 248$ at integer level 1) — WP5a fixes the ideal's *size and location* quantitatively and gives the celestial route the same two-step shape as MMST (limit + maximal ideal); the operator content is exactly WP5b–e, of which WP5b–WP5d and the WP5e α and β stages are executed below.
  - **WP5b (*done*) [E]/[C]:** "the singular vector as an operator", executed and machine-

verified as `CELEST.WP5B.01` [`verification/v498_celestial_wp5b_singular_vector.py`] (exact integer/Fraction arithmetic, 53 checks, deterministic) — *success on the preregistered criterion*: the deleting object exists and is explicit. $|s\rangle = (E^{\theta}_{-1})^2|0\rangle$ (weight 2θ, level $2 = 8$ quarter units $= q^2$, an *integer* level) is constructed in a machine-built Chevalley/Frenkel–Kac basis (cocycle asymmetry on all 57600 pairs; the $[e_\alpha, e_{-\alpha}]$ sign *forced* by Jacobi, SGN $= -1$; κ derived, $\kappa(\theta^\vee, \theta^\vee) = 2$), and $J^a_1|s\rangle = 0$ is machine-verified for *all* 248 generators with the case classification 190 (first bracket) / 57 (second) / 1 ($a = F^\theta$, the central-term cancellation — the *only* case that sees the level k); plus $J^a_2|s\rangle = 0$, $E^a_0|s\rangle = 0$, exact weight and Shapovalov norm 0. *Level dial*: the F^θ_1 coefficient is $2(1-k)$ — nonzero at k=0 and k=2, zero exactly at k=1: without the central extension the deletion operator does not exist. μ_4 compatibility: glue class $j(\theta) = 1$ (machine-built h-adapted chamber, $\langle\theta, h\rangle = 5$, height $29 = h^\vee-1$), clock phase $i^{2j} = -1$, class$(2\theta) = 2$ sheet-even (WP5a-consistent) with the deleting θ-$\mathfrak{sl}_2$ crossing the sheet-odd classes $(1, 3)$; 8 quarters $= q^2$ via the per-period dictionary. The module it generates is *the* ideal: weight 2θ has multiplicity 1 in the level-2 Fock, and the direct $\mathfrak{g}_0$-lowering orbit BFS from $|s\rangle$ reproduces the Freudenthal multiplicities of $V(2\theta)$ exactly through depth 4 ($27000 = (h^\vee)^3$, quotient 4124; [C] Weyl complete reducibility beyond depth 4). Negative controls separate honestly: the level-1 current state and generic level-2 states are *not* singular; at k=2 the *cube* is (the $(E^\theta)^{k+1}$ pattern is generic Kac level mechanics); $SO(16)_1$ has the same singular vector but keeps three extra level-1 primaries ($h = \frac{1}{2}$ breaks one-block fusion) — the singular-vector mechanism is level-1 *generic*, the *one-block closure* is the E_8/μ_4-specific part. *Handover to WP5c* (answered below): in the twisted quarter-*slot* moding of the χ_w limit Fock, two sector-C_1 modes never sum to 8 quarters (minimum $6 = q^{3/2}$) — the per-*period* dictionary (v496), not the per-slot identification, carries $|s\rangle$ to q^2; exactly the GNS/limit-state question (kernel $\supseteq$ ideal) that WP5c answers.

- **WP5c (*done*) [E]/[C]:** "the GNS limit state", executed and machine-verified as `CELEST.WP5C.01` [`verification/v500_celestial_wp5c_gns_limit_state.py`] (exact integer/Fraction arithmetic, 35 checks, deterministic) — *success on the preregistered criterion*: the family ω_w exists and its limit has the null ideal in its GNS kernel. ω_w is the quasi-free state with loop sector = the affine k=1 vacuum n-point functions (via the machine-determined compact anti-involution $\theta(e_\alpha) = -e_{-\alpha}$, the unique anti-automorphism sign on all 61504 basis pairs) and radial sector = oscillator pairings x^{wr} (the exact Gibbs regulator, x-adic); *the WP5a threshold holds at the state level*: $\omega_w = \omega_\infty \bmod x^{N+1}$ for all $w \geq N+1$, sharp at $w = N$. The core is fully machine: the *complete* exact level-2 Gram — 31124 monomials in 9361 weight blocks — has rank 4124 exactly (the preregistered target), kernel $27000 = V(2\theta)$ *weight by weight* (Freudenthal cross-check on all 9361 weights), every block PSD, rank table per Weyl orbit $(0, 0, 1, 8, 44)$, level-1 rank 248 positive definite (the current layer survives). μ_4: the clock descends to GNS with level-2 rank split $(1036, 1024, 1040, 1024) = \Theta_{C_j}/\eta^8$ at q^2 — the WP5a two-routes identity at the *state* level — and $|s\rangle$ *is* the zero vector of GNS(ω_∞), resolving the WP5b twisted-slot tension. The KILL branch is probed and not triggered: a CCR obstruction shows *no* w-uniform state can damp the radial modes (the family formulation is *necessary*), and the wrong constructions are caught (family $E'_w = mw+r$ erases the 248 layer, $710955 \neq 248$; no damping keeps $897266 \neq 248$); k-dial and D_8 controls separate honestly (k=2: $\langle s|s\rangle = +4$, no ideal in the kernel; D_8: one block of four, $h = \frac{1}{2}$). [C] quasi-free extension to the full $*$-algebra; Kac positivity beyond level 2.
- **WP5d — both stages (*done*): α [E]/[C], β [E]/[C] (continuum uplift [O]):** "the local net", approached from the lattice. The α stage — the KLM *two-interval index* as the operational Haag-duality discriminator ($(E_8)_1 \Rightarrow \mu$=1, one sector; $SO(16)_1 \Rightarrow \mu$=4) — is executed and machine-verified as `CELEST.WP5D.01`

[verification/v501_celestial_wp5d_two_interval_index.py] (39 checks; Gaussian lattice machinery ED-validated to 10^{-15}, exact Fractions for the arithmetic). Measured on the 16-layer seam carrier (critical Majorana ring, $c_- = 8$): the fermionic two-interval mutual information is *extensive* (the μ=1 reference; c fit 0.5000, residual $\to 0$ with N), while the sector-summed (orbifold) prescription pays exactly one classical bit — the $\ln 2$ plateau at machine precision ($|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$ at N=512), so $[F : F_{\text{even}}] = 2$ and $\mu_{\text{gauged}} = 4$, the entropic twin of the v490 parity census (two independent lattice witnesses); the orbifold also breaks two-interval complementarity ($S(E) \neq S(E')$, the direct duality-failure witness), with the complementary-pair budget bounded by $\ln 4 = \ln \mu(SO(16)_1)$ as a *double-limit* statement. The condensation arithmetic is anchored at both measured ends: $\det \mathrm{Cartan}(D_5) \cdot \det \mathrm{Cartan}(A_3) = 16$, KLM/Longo–Rehren $16/4^2 = 4/2^2 = 1$, $\Sigma d_i^2 = (4, 4, 1)$, $\theta_v = 1$ exactly at $\nu = 2c_- = 16$ (rivals $\neq 1$) — so $\mu = 1$ after condensation and the preregistered KILL ("μ-offset $\neq 0$ after condensation") is *not* triggered; the ν=1 control shows a non-removable offset ($\theta_v \neq 1$: the discriminator has teeth), the trivial phase shows none, and the wrong sector sum loses the full $\ln 2$. [C] the continuum dictionary "offset = ln index" (KLM; Longo–Xu; Casini–Huerta–Magan–Pontello). **The β stage (*done*) [E]/[C] — split + strong additivity, the two remaining KLM legs** — is executed and machine-verified as `CELEST.WP5DB.01` [verification/v504_celestial_wp5d_klm_completeness.py] (37 checks; Gaussian lattice machinery ED-validated, exact GF2/integer algebra for the exact legs; the $U(1)$/Dirac number-conserving subnet as the preregistered negative control — the current-net analogue where strong additivity famously *fails* in the continuum). *Strong additivity is algebraically exact on the lattice*: with the shared boundary Majorana ("touching = share the point"), $\mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$ *exactly* (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32), while disjoint intervals give exactly *half* (rank 16/32, index 2 — the missing sector is odd $\otimes$ odd, the single $\ln 2$ bit of the α stage, localised at the split point); the $U(1)$/neutral algebras do *not* generate the union even with the shared site (gaps 2/10/52, growing). The entropic witness uses the honest discriminator *bounded vs divergent*: the $\mathbb{Z}_2$/orbifold touching deficit stays $< \ln 2$ with the Ising approach $(\ln 2 - \Delta_2) \sim N^{-p}$, $p = 0.2444$ vs $2\Delta_\mu = \frac{1}{4}$ (the naive "defect $\to 0$" expectation is *false* at the sharp lattice split — bounded $\Leftrightarrow$ finite index, Longo–Xu), while the $U(1)$ control *bursts* $\ln 2$ from $L = 128$ on and grows as $\frac{1}{2} \ln \mathrm{Var}\, Q_A$ (Klich–Levitov slope 0.10134 vs $1/\pi^2 = 0.10132$: infinite index). The split property is witnessed quantitatively: the coupling-block singular values fall exponentially at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0% over $d = 16..512$, $\sigma_1 \sim x^{0.5044}$ (vs $\frac{1}{2}$), trace norm summable — and the orbifold *inherits* the same ladder exactly at machine level (P_A flips $C \to -C$, σ_k identical to 8.3×10^{-17}; the even-bilinear coupling Gram is the second compound $\Lambda^2 C$ with singular values $\{\sigma_a \sigma_b\}$ — Longo heredity, [C]). The Pimsner–Popa bound is a one-line *identity*, $E(a) - a/2 = PaP/2$, so $\lambda = \frac{1}{2} = 1/[F : F_{\text{even}}]$ exactly, attained in exact integers ($16384 + 2048$ monomial sweeps, 0 violations; random pencil $\min \lambda = 0.5000000000$); the two-interval group gives $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ exactly, and the index is pinned by *two independent routes* ($e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\text{PP}}$; $e^{2\Delta_\infty} = 4 = 1/\lambda_{E_4} = \mu$); the $U(1)$ control has $\lambda = 1/(m+1) \to 0$. *With the α stage, all three KLM ingredients of complete rationality — split, strong additivity, finite μ — are now witnessed on the lattice.* The continuum uplift stays [O] and is honestly fenced: the theorem "finite-group orbifolds of completely rational nets are completely rational (hence split + strongly additive)" is *Xu's theorem* (algebraic orbifold CFTs) — cited, *not* claimed; the continuum proof for the concrete seam quotient net on the collar (v336/v449) and for the interacting condensed $(E_8)_1$ net remains WP5e / Costello–Li territory.

– **WP5e (subdivided): α stage (*done*) [E]/[C] — the CFT-side prefactor + level pinning; β stage (*done*) [E]/[C] — the equivariant anomaly ledger on twistor**

space, including the honest a_0 refutation; the global BCOV quantisation stays [O]. The α stage is executed and machine-verified as `CELEST.WP5E.ALPHA.01` [`verification/v502_celestial_wp5e_prefactor_level.py`] (exact sympy/Fraction arithmetic, 33 checks). Both exactly computable consistency anchors of the uplift hold. (i) *The $q^{-1/3}$ prefactor is exact μ_4 vacuum-energy bookkeeping*: the clock is *inner* ($h = (2,2,2,2,2;0^4)$ and the A_3 detector $h' = (0^5;1,1,1,-3)$ both read the glue class mod 4 on all 240 roots; $(h,h) = 20$, $(h',h') = 12$, sum 32), so the induced twist on the 8 torus bosons is $\theta = 0^8$ in *every* sector — a *shift* orbifold, not a rotation orbifold — and each sector carries the same $8\times(-\frac{1}{24}) = -\frac{1}{3} = -c/24$ (Hurwitz-ζ exact, $E_b(\theta) = -\frac{1}{24}+\theta(1-\theta)/4$); the sector ground exponents are $-\frac{1}{3}+(0,1,1,1)$ with $h_j =$ the discriminant form, an identity holding via spectral flow ($j^2(h,h)/32$, mod 1) *and exactly* via the $10+6 = 16 = 2^{g_{\mathrm{car}}-1}$ Majorana free fermions of the WP5d seam carrier (R–NS shift $n/16$: $\frac{5}{8},\frac{3}{8},1$); the four sector characters sum to $E_4/\eta^8 = q^{-1/3}(1,248,4124,34752)$; the rotation reading *fails* (the geometric A_3 deck gives $\frac{3}{16} \neq \frac{3}{8}$ — the WP1 "clock $\neq$ deck" lesson at the Casimir level). (ii) $k = 1$ *is forced three independent ways*: the current condition $h(J) = k = 1$ (the $\mathbb{Z}_4$ extension closes onto the E_8 adjoint $60+64+60+64 = 248$ only at spin 1); the conformal embedding ($D_5+A_3 \subset E_8$: $47k^2+219k-266 = 47(k-1)(k+266/47) = 0$; $D_8 \subset E_8$: $128k(1-k) = 0$); and the central charge $248k/(k+30) = 8 \Leftrightarrow 240(k-1) = 0$ (= the prefactor itself); plus the WP5b singular-vector dial ($\|(E^{\theta}_{-1})^m 0\|^2 = m!\,k(k-1)\cdots(k-m+1)$, singular vector exactly at grade $k+1$; $k = 1$ deletes $V(2\theta)$: $31124 - 27000 = 4124$). *Honest sharpening*: glue-h *integrality* $h(J^j;k) = k\,(0,1,1,1)$ holds for *all* $k = 1..8$ and fixes nothing — the originally conjectured integrality route is dead; the three other dials carry. Controls: $D_8/SO(16)_1$ has the *same* $q^{-1/3}$ prefactor but $h = (0,\frac{1}{2},1,1)$ (the discriminator is glue-h integrality, not the prefactor); $\mathbb{Z}_2$ or μ_4 *rotation* twists break the common prefactor; wrong $k \in \{2,3,4\}$ fails all five dials coherently. **The β stage (*done*) [E]/[C] — the equivariant anomaly ledger on twistor space** — is executed and machine-verified as `CELEST.WP5E.BETA.01` [`verification/v505_celestial_wp5e_beta_equivariant_ledger.py`] (exact sympy/Fraction arithmetic, 47 checks). The twistor side is pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop box anomaly on $\mathbb{C}^2/\mathbb{Z}_4$ with the μ_4-glue monodromy. (i) *The AB ledger is exact*: denominators $\det(1-g_j^{-1}) = (2,4,2)$ with Dedekind sum $5/4 = (|\mathbb{Z}_4|^2-1)/12$; equivariant characters $\mathrm{tr}_{g_j}(\mathrm{Ad}\,E_8) = (248,0,-8,0)$ by two independent routes; invariant average $60 =$ the carrier; Frobenius 61568. (ii) *Okubo per sector, the honest sharpening*: only the *invariant* sector is Okubo-quadratic ($Q^{(0)} = 36\langle x,x\rangle^2$, $36 = \tilde{\lambda}^2_{\mathfrak{e}_8}$ — the v495 identity re-derived through the glue grading); the twisted sectors carry irreducible T_5/T_3 content, and the AB-weighted sum cancels the D_5 quartic *exactly* while leaving the *rigid* residual $32\,T_3$ (no admissible reweighting fixes it; the graded GS exchange is rank-obstructed in sectors 1–3) — twisted-sector closed strings must carry the rest [O]. (iii) *The index bridge* (the centrepiece): $f(m) = \frac{1}{4}\sum_j (i^{jm}-1)/\det_j = \mathrm{ch}_2(T_m) = -(C^{-1})_{mm}/2$ *exactly* — the fixed-point ledger and the McKay/Kronheimer intersection ledger are the same arithmetic; the glue-bundle ch_2 defect is the integer -78 by *both* routes (classes $(-24,-30,-24)$). (iv) *The level dials say $k = 1$ geometrically*: the lattice current count is 240 at k=1 and exactly 0 at $k = 2,3,4$; the embedding residual $47k^2+219k-266 = (0,360,814,1362)$; $h(J;k) = k$ is integer for *all* k (integrality alone fixes nothing — honest, exactly as on the CFT side); one scale $\Rightarrow$ one level (v493 periods). (v) *The honest refutation*: the clock-invariant modulus $a_0 \in O(8)$ fills the BSS *graviton* slot $O(2)$, *not* the axion slot $O(-2)$ (Hamiltonian weights $X = -2 \neq +2 = Y$; mismatch $4 = |\mu_4|$, typed [C] as a coincidence) — "the theory brings its own GS axion as a_0" is *refuted*; what does match: the three $H^2(\mathrm{ALE})$ classes carry exactly the three twisted-sector Coxeter characters $\{i,-1,-i\}$ (bijection, no invariant class), so the bulk axion must come from the $O(-2)$ tower field itself (construction [O]). Controls: $\mathrm{diag}(i,i)$ breaks the ledger four independent ways; $SO(16)$ glue gives characters $(120,0,+120,0)$ and defect $-30 \neq -78$

with a *failing* bulk Okubo (independent quartic Casimir); $k = 2$ dies on the closure dial ($-31/48 \neq -\frac{1}{3}$) while naive integrality dials stay integer. The preregistered kill does *not* fire on the equivariant skeleton. **WP5e proper [O] (the global BCOV quantisation):** the BCOV/Kodaira–Spencer theory on $\mathbb{PT}/\mathbb{Z}_4$ itself (quantisation, non-fixed-point contributions, the twisted closed-string measure), the $O(-2)$ bulk-axion construction, the Costello–Paquette–Sharma level-from-flux mechanism, and the pairing of the rigid $32\,T_3$ twisted residual with the three sphere axions — the named remaining targets. Success: partition function $= E_4/\eta^8$ *including* the $q^{-1/3}$ prefactor *derived from the twistor side*; kill: an anomaly mismatch at the Costello–Li grade — neither the CFT-side dials of v502 nor the equivariant skeleton of v505 trigger it (all computable dials say $k = 1$), but the global quantisation is untested.

Honest status line: WP5a–WP5d (both WP5d stages) and the WP5e α and β stages are exact-arithmetic [E] constructions (plus ED-validated lattice witnesses in WP5d) with [C] named identifications ("radial decoupling = collar thickness $\to 0$"; Weyl complete reducibility beyond BFS depth 4; the Kac mechanics beyond level 2; the quasi-free extension; the KLM/Longo–Xu/CHMP index dictionary; Longo split heredity; Kac generation of the maximal submodule for $k \geq 3$; the WP5e-β dictionaries — $\mathrm{CS}(\rho)$ mod 1 = discriminant weight, $32 = (h,h)+(h',h')$, $-30 = -h^\vee$, the Ω-contraction); *WP5a–WP5d plus WP5e-α/β are landed — all three KLM ingredients of complete rationality carry lattice witnesses, and both faces of the inflow are anchored* — WP5e proper (the global BCOV quantisation) is the single remaining open piece, with the continuum uplift of the WP5d lattice witnesses honestly fenced as Xu's theorem (cited, not claimed) and the rigid $32\,T_3$ twisted residual + the sphere-axion pairing as the named remaining targets — `SEAM.EQUIV.01` stays [O] and nothing in WP5a–WP5e-β moves it.

Kill tests (pre-registered). (K1) the equivariant sector could have *contradicted* the 52+64+60+64 split (uniformity of $D[d]$ needs exactly $\dim \mathfrak{g}_j = (60, 64, 60, 64)$) — *survived by WP1 and WP2*: the false-glue control shows the test has teeth (3112/6720 violations), and WP2 adds the deformation side (a shape modulus surviving clock-invariance, or a_0-corrections leaking between μ_4 sectors, would have fired — neither occurred; the clock-variant controls fail in two independent ways each). **(K2)** if one-loop associativity of the celestial E_8 theory demanded extra fields that TFPT does not possess, the contract dies (Costello's axionic mechanism is the first pass: E_8 is admissible; the TFPT-specific field content is the open half). **(K3)** — *evaluated by WP3* ([`verification/v495_celestial_wp3_gs_alignment.py`]): the trigger ("no exact alignment between the GS coefficient and the c_3 normalisation") does *not* fire — exact alignment exists ($(\kappa/c_3)^2 = 12$ resp. $1/300$). But the look-elsewhere battery *demotes the scope of K3 itself*: the alignment format passes 8/8 across Costello's whole list, so surviving K3 certifies compatibility plus λ-arithmetic, *not* $\mathfrak{e}_8$-selective evidence — alignment survives, selectivity does not. **(K4)** — *evaluated by WP4* ([`verification/v496_celestial_wp4_character_shadow.py`]): K4 fires only against the *exact*-block reading (the character is provably not a conformal block in the jet grading); it does *not* degrade the contract to "E_8 is an admissible gauge algebra for celestial SDYM", because the sector arithmetic holds exactly — WP4 lands at the boundary-limit reading (verdict B(ii), the `SEAM.EQUIV.01` scaling-limit shape), with the constructive limit handed to WP5 — and made precise there by WP5a ([`verification/v497_celestial_wp5a_null_ideal.py`]: the coefficientwise χ_w limit with threshold $w = n+1$ and the derived ideal), with the deleting operator itself constructed by WP5b ([`verification/v498_celestial_wp5b_singular_vector.py`]: $|s\rangle$ explicit, level dial $2(1-k)$), the limit state by WP5c ([`verification/v500_celestial_wp5c_gns_limit_state.py`]: the ideal in the GNS kernel, level-2 rank 4124 exactly), the two-interval index chain by WP5d-α ([`verification/v501_celestial_wp5d_two_interval_index.py`]: μ-chain $16 \to 4 \to 1$ anchored at both measured ends), the split + strong-additivity legs by WP5d-β ([`verification/v504_celestial_wp5d_klm_completeness.py`]: all three KLM ingredients of complete rationality witnessed on the lattice; the continuum uplift is Xu's theorem, cited not claimed), the CFT-side anchors of the uplift itself by WP5e-α ([verification/v502_celestial_wp5e_prefactor_level.py]: the $q^{-1/3}$ prefactor as exact μ_4 vacuum-energy bookkeeping, $k = 1$ forced three ways), and the equivariant twistor skeleton by WP5e-β ([verification/v505_celestial_wp5e_beta_equivariant_ledger.py]: the index bridge $f(m) = \mathrm{ch}_2(T_m)$ exact, glue defect -78 two routes, geometric $k = 1$ dials, the rigid $32\,T_3$ residual, and the honest a_0 refutation — the kill branch does not fire on the skeleton), leaving the global BCOV quantisation (WP5e proper) as the single remaining open piece.

Honest success estimate. The probability of *full* success (WP5e: a second continuum route to SEAM.EQUIV.01) remains low — it is constructive holography at the Costello–Li grade. The *partial*-findings expectation is now realised and *exceeded*: WP1–WP4, all five WP5 constructive milestones (both WP5d stages) and the WP5e α and β stages are executed with sharp structural results (verdicts B, B, B, B(ii); WP5a–WP5d passed; WP5e-α pins the CFT side, WP5e-β the equivariant twistor skeleton; *WP5e proper — the global BCOV quantisation — is the single remaining open piece*) — the WP1 critical correction (clock $\neq$ deck; clock^2 = deck), the WP2 sharpenings (the clock *is* the Picard–Lefschetz/Coxeter monodromy of the seam deformation; a_0 pure scale with the $\tau{=}i$ shape frozen), the WP3 non-selective alignment (exact λ-arithmetic, honestly typed by the 8/8 look-elsewhere table), the WP4 boundary-limit localisation of the character mismatch (three exactly-located obstructions, one surviving limit structure), the WP5a null-ideal derivation (stabilisation threshold $w = n{+}1$; $27000 = (h^\vee)^3$ derived; quotient = sector sum = 4124; $SO(16)_1$ contrast), the WP5b singular-vector construction (the deletion operator explicit, case tally 190/57/1, level dial $2(1{-}k)$, one-block closure typed E_8/μ_4-specific), the WP5c limit state (the ideal in the GNS kernel, the full 9361-block Gram at rank 4124, the family proved *necessary* by a CCR obstruction), the WP5d-α two-interval index (the $\ln 2$ plateau at 10^{-15}, the μ-chain $16 \to 4 \to 1$ measured at both ends), the WP5d-β KLM completion (strong additivity exact on the lattice with the shared boundary point; the bounded-vs-divergent entropic discriminator with the Ising $\frac{1}{4}$-exponent approach against the diverging $U(1)$ control; the elliptic-nome split ladder with exact orbifold inheritance; Pimsner–Popa $\lambda = \frac{1}{2}$ and $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ with integer attainment — all three KLM ingredients witnessed, Xu's continuum theorem cited not claimed), the WP5e-α prefactor/level pinning (the $q^{-1/3}$ prefactor as exact μ_4 vacuum bookkeeping; $k = 1$ forced three ways; the glue-integrality route honestly retired), and the WP5e-β equivariant ledger (the index bridge $f(m) = \mathrm{ch}_2(T_m)$ exact with the -78 defect; the rigid $32\,T_3$ twisted residual; the honest a_0 refutation with the three sphere classes filling the three twisted axion slots) — exactly the kind of sharp, falsifiable refinement the contract format is for. Status: CELEST.SEAM.01 [O] (research contract, WP5e proper the single remaining open milestone); CELEST.WP1.01 [E]/[C] (executed); CELEST.WP2.01 [E]/[C] (executed); CELEST.WP3.01 [E]/[C] (executed); CELEST.WP4.01 [E]/[C] (executed); CELEST.WP5A.01 [E]/[C] (executed); CELEST.WP5B.01 [E]/[C] (executed); CELEST.WP5C.01 [E]/[C] (executed); CELEST.WP5D.01 [E]/[C] (executed, α stage); CELEST.WP5DB.01 [E]/[C] (executed, β stage — the continuum uplift stays [O], Xu's theorem cited); CELEST.WP5E.ALPHA.01 [E]/[C] (executed, α stage — the CFT-side prefactor + level pinning); CELEST.WP5E.BETA.01 [E]/[C] (executed, β stage — the equivariant anomaly ledger; the global BCOV quantisation itself stays [O]); SEAM.EQUIV.01 untouched, stays [O].

Certifiability and recommended order

Building block	Machine	Hardness
$W_{\rm wall}$ enumeration ($C_U^{(1)}$)	Lean / Python / Wolfram	easy (*done*)
D_4 fixed-locus polynomial ($C_U^{(2)}$)	Sage / Mathematica	medium
selector uniqueness ($C_U^{(3)}$)	interval arithmetic	medium
SNF, determinants, spectra	Lean	easy–medium
principal-symbol ellipticity (G1)	xAct / Cadabra / Lean	medium
gap inequality (G5)	Lean	easy (*done*)
operator-norm bound (G5)	analysis + numerics	hard
projective limit (G6)	not primarily machine	very hard
Ward identities (G7)	BV/BRST formal + checks	hard

Recommended order.

$$\boxed{\text{Selector-triangle pairings} \;\to\; v_{\rm geo} \text{ scale anchor} \;\to\; G_{\rm net} \text{ inclusion theorem}}$$

The flavor interface (historically $U_{\rm wall}$) is now reduced to the *selector-triangle pairings* (the dual normal pair (d, n) forcing R columnwise, v136/v139); it is finite, falsifiable and the cheapest visible upgrade. The $v_{\rm geo}$ scale anchor is the one dimensionful input (same nature as $1/G$). $G_{\rm net}$ — the index-4 seam-net inclusion certifying the metric sector — is more fundamental but a *programme*, not a single proof: heavy machinery, with the projective limit the genuine summit. $F_{\rm transfer}$ (source→pole / relic / cosmology) is the remaining downstream interface.

$F_{\rm transfer}$ **is a typed functor, not a bag of open topics.** It is standard physics fed TFPT *source* data, $F_{\rm transfer} = F_{\rm observable} \circ F_{\rm threshold} \circ F_{\rm RG}$, with exactly four interfaces:

Interface	Map	Status	Source data → observable
$F_{\rm pole}$	source→pole masses	[E] / [C]	Koide $53/54 = a^T(R+Q)\mathbf{1}/(2\,\mathbf{1}^T Ra)$ is an exact operator factor [E] (the missing Sheet-Diamond corner); the pole-transfer interpretation is [C] ([`verification/v183_koide_f_corner_transfer.py`])
$F_{\rm Boltzmann}$	CP source→ η_B	[C]	$\tilde m_1 = m_3/A_\Lambda$ anchors the washout, but $M_1 = M_R\varphi_0^4$ only relocates the free input (M_R is the seesaw scale, not a compiler power), so η_B stays a *sharper scenario*, not a closure ([`verification/v184_etaB_anchored_boltzmann.py`])
$F_{\rm relic}$	$(f_a, m_a, \theta_i, N_{\rm DW}) \to \Omega_a$	[C]	the determinant-line axion can be all of DM, but only on the $\theta_i \approx 170°$ hilltop where the relic is exponentially sensitive — a viable scenario ([`verification/v185_axion_relic_solver.py`])
$F_{\rm QCD}$	$(\alpha_s, b_3, \Lambda_{\rm QCD}, \text{lattice}) \to m_p/m_e$	[O]	a QCD matching contract ($b_3 = -7$ a carrier output), *not* a compiler number — the rejected 1836 near-formula is the discipline working

A machine guard enforces the typing: the four families must always carry a [C]/[O] marker and must never be promoted to a primitive [E] compiler prediction ([`verification/v187_ftransfer_laws.py`]). Exact *algebraic* sub-parts (the Koide factor, $b_3 = -7$) may be [E]; the *physical* prediction never is.

The functor contract `CONTRACT.F.01`, four axioms ([`verification/v213_ftransfer_functor.py`] **).** Beyond the typing guard, the functor itself satisfies four checkable structural axioms, each instance a *discrete* compiler kernel [E] composed with an *external* continuous solver [C]: **(1) μ_4-deck equivariance** — the discrete kernels are carrier/deck invariants $\{|\mu_4|, N_{\rm fam}, g_{\rm car}, A_\Lambda\}$, and the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ *is* the μ_4-deck transfer eigenvalue (v82/v54); **(2) Plücker preservation** — the dimensionless readouts are Plücker readings ($53 = a^T(R+Q)\mathbf{1}$, $\mathrm{Pl}(F) = (1, 22, 30)$; $\|\mathrm{Pl}(K)\|_1 = 11$); **(3) positivity/stochasticity** — $\mathrm{spec}\, T = \{1, (2/3)^6, (1/3)^6\}$ is positive (RP/OS),

$\kappa_f\in(0,1)$; **(4) external modules explicit** — the continuous content (pole transfer, the flavored Boltzmann network, the finite-T $\chi(T)$ ODE, the lattice C_N) is named and external, never a compiler number. So `CONTRACT.F.01` is the third research contract alongside ($U_{\rm wall}$) and ($G_{\rm metric}$): a typed functor, *not* a bag of open topics — but a consolidation, *not* a closure.

The discrete→dynamic lens: `F_transfer` is the readout end of the one principle ([`verification/v303_ftransfer_dynamics.py`], `FR.DYNAMICS.01`). The typing above is sharpened by a *dynamical* reading. The main-branch mechanism is a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update $m\leftarrow(A+2I)/4\,m$ → the E_8 marks, gap $6\ln(3/2)$). All four `F_transfer` instances share that *same shape*: [E] **F_pole** (Koide) *is* the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier $F'=(2/3)^6=\lambda_2$, the seam-clock subleading eigenvalue, so $|F'|<1 \Rightarrow$ a unique attractor at the *seam* rate (v82); [C] **F_Boltzmann** (η_B) is a gapped positive contraction (washout $\kappa_f\in(0,1)$ toward the unique balance, an H-theorem); [C] **F_relic** (axion) freezes at an adiabatic fixed point (the comoving number relaxes to a constant); [E] **F_QCD** (m_p/m_e) is the 1-loop RG flow with the carrier $b_3=-7$, whose Gaussian UV fixed point is the attractor (asymptotic freedom, $\Lambda_{\rm QCD}$ generated). *Honest boundary:* only F_pole has the seam rate $(2/3)^6$ exactly; the other three share the *shape* with external rates (thermal, cosmological, RG). So `F_transfer` is not a bolt-on but the downstream manifestation of the one discrete→dynamic principle — the compiler fixes the discrete seed + integer kernel, a gapped flow carries it to the observable — and the theory's simplicity is preserved precisely because the external rates are honestly fenced (v187), not because they reduce to the seam. (Even F_pole's flow time is non-integer, v93: the relaxation is the *continuous* semigroup $e^{t\log F}$ generated by the discrete operator — exactly the diffusion-semigroup reading of the main branch.)

The single-flow reduction (`DYN.TRANSFER.UNIVERSAL.01`). The dynamic lens is sharpened to a precise reduction ([`verification/v425_dyn_transfer_universal.py`]): the shared "shape" is literally the *one* native flow of the theory — the seam modular/recovery semigroup (v238/v240, gap $6\ln(3/2)$) — restricted to each sector, extending the static universal-gap principle (v383) to a dynamic one. [E] the v99 Koide generator $\dot q=(\Delta/N_{\rm fam})(q-2)(q-5)$ linearised at the attractor $q=2$ has rate $-\Delta$, so the time-1 contraction is $(2/3)^6$ (the recovery eigenvalue, $q=5$ unstable); [E] the Boltzmann washout is that same contraction, with the CP phase $4\pi/3$ native (v320); [E] the QCD rate is $b_3=-7$. So the transfer *dynamics* is native; only the *anchors* stay external — the IR pole scale and M_R are $v_{\rm geo}$-class, C_p is the lattice $O(1)$, and $F_{\rm relic}$'s cosmological initial condition θ_i is the one genuine wall. A reduction, *not* internalisation: no new premise, and the firewall (v187) holds — the observable stays [C]. So the visible residual is one geometric postulate (`QGEO.SYM.01`), one unit gauge ($v_{\rm geo}$), and this one typed transfer functor — everything else is out-of-sample data.

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

Modular theory and algebraic quantum field theory.

- M. Takesaki, *Tomita's Theory of Modular Hilbert Algebras and its Applications*, Lecture Notes in Math. **128**, Springer (1970).
- J. J. Bisognano, E. H. Wichmann, *On the duality condition for a Hermitian scalar field / for quantum fields*, J. Math. Phys. **16** (1975) 985–1007; **17** (1976) 303–321.
- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.
- S. Doplicher, R. Haag, J. E. Roberts, *Local observables and particle statistics* I, II, Comm. Math. Phys. **23** (1971) 199–230; **35** (1974) 49–85.

- R. Haag, *Quantum field theories with composite particles and asymptotic conditions*, Phys. Rev. **112** (1958) 669–673; D. Ruelle, *On the asymptotic condition in quantum field theory*, Helv. Phys. Acta **35** (1962) 147–163.
- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

Conformal nets, vertex operator algebras and lattice constructions.

- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, `arXiv:2107.13834`, `doi:10.1007/s00220-022-04521-8`. (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur → Virasoro convergence, $c = \frac{1}{2}$, and the smeared WZW-current Theorem 5.1. The internal TFPT code-token "MMST" and the legacy claim-ids `SEAM.MMST.*` / filenames `v*_mmst_*` denote *this* Osborne–Stottmeister theorem.)
- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, `arXiv:2010.11121`, `doi:10.1007/s00220-021-04152-5`; and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)
- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, `arXiv:2407.18222` (2024).
- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, `arXiv:2506.01008` (2025); *Representations of conformal nets associated with infinite-dimensional groups*, `arXiv:2508.07109` (2025).
- P. Naaijkens, D. Penneys, B. Wallick, *(commuting-projector reflection positivity / LTO-RP)*, `arXiv:2605.10693`.

Lattices and quadratic forms.

- J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der math. Wiss. **290**, Springer (1999).
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Singularities and geometry.

- E. Brieskorn, *Beispiele zur Differentialtopologie von Singularitäten*, Invent. Math. **2** (1966) 1–14.
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- H. Casini, M. Huerta, R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **05** (2011) 036, `arXiv:1102.0440`.

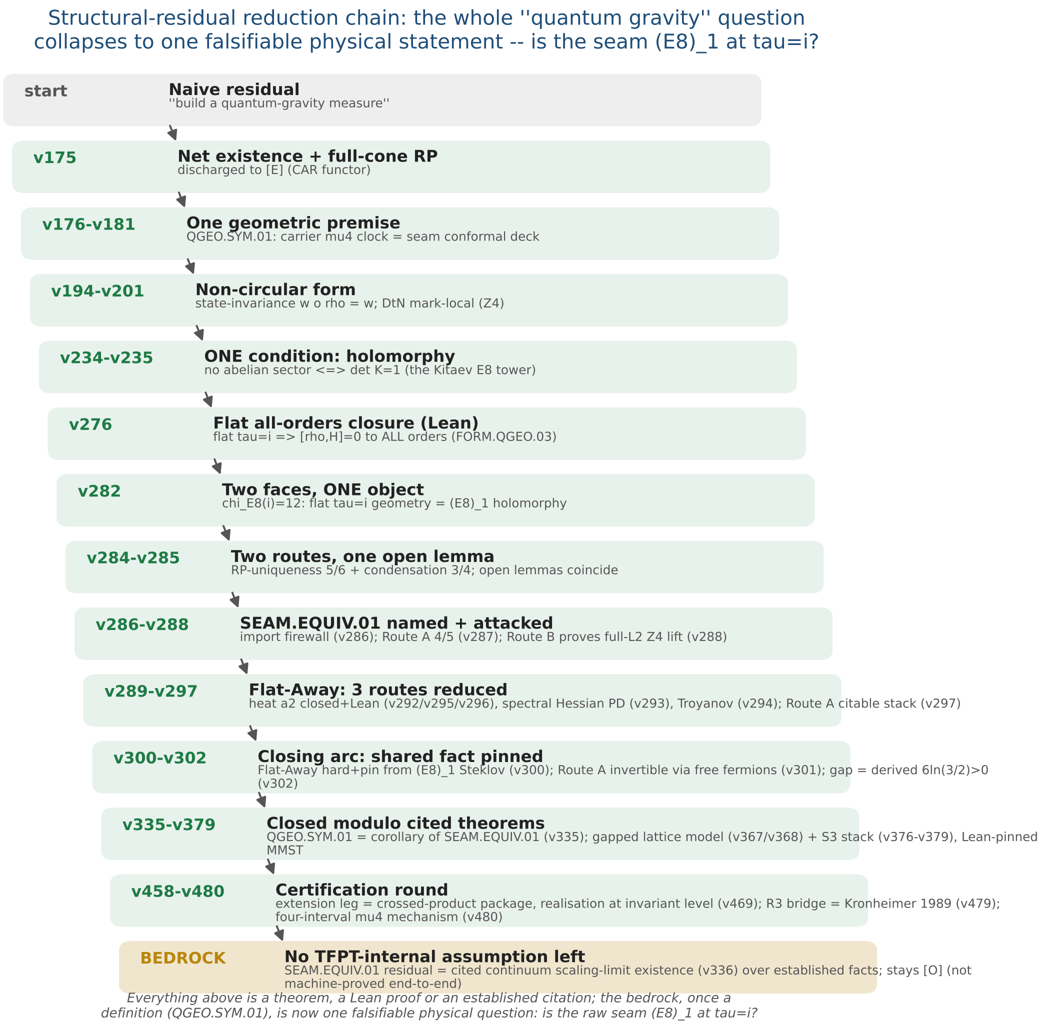

Figure 3: The structural-residual reduction chain (v175 → the certification round). The naive "build a quantum-gravity measure" shrinks, one machine-checked step at a time, from a vague measure to one falsifiable physical statement — *is the raw seam* $(E_8)_1$ *at* $\tau = i$*?* (`SEAM.EQUIV.01`; its conformal-deck face `QGEO.SYM.01` is a corollary, v335). The closing arc (v276–v302) pins the shared fact and derives the gap $6\ln(3/2) > 0$; the certification round discharges the extension leg to the crossed-product package (v469), the R3 graph→geometry bridge to Kronheimer 1989 (v479) and exhibits the four-interval μ_4 mechanism (v480). Every step above the bedrock is a theorem, a Lean proof or an established citation; the residual — the cited continuum scaling-limit existence (v336) — stays honestly [O].

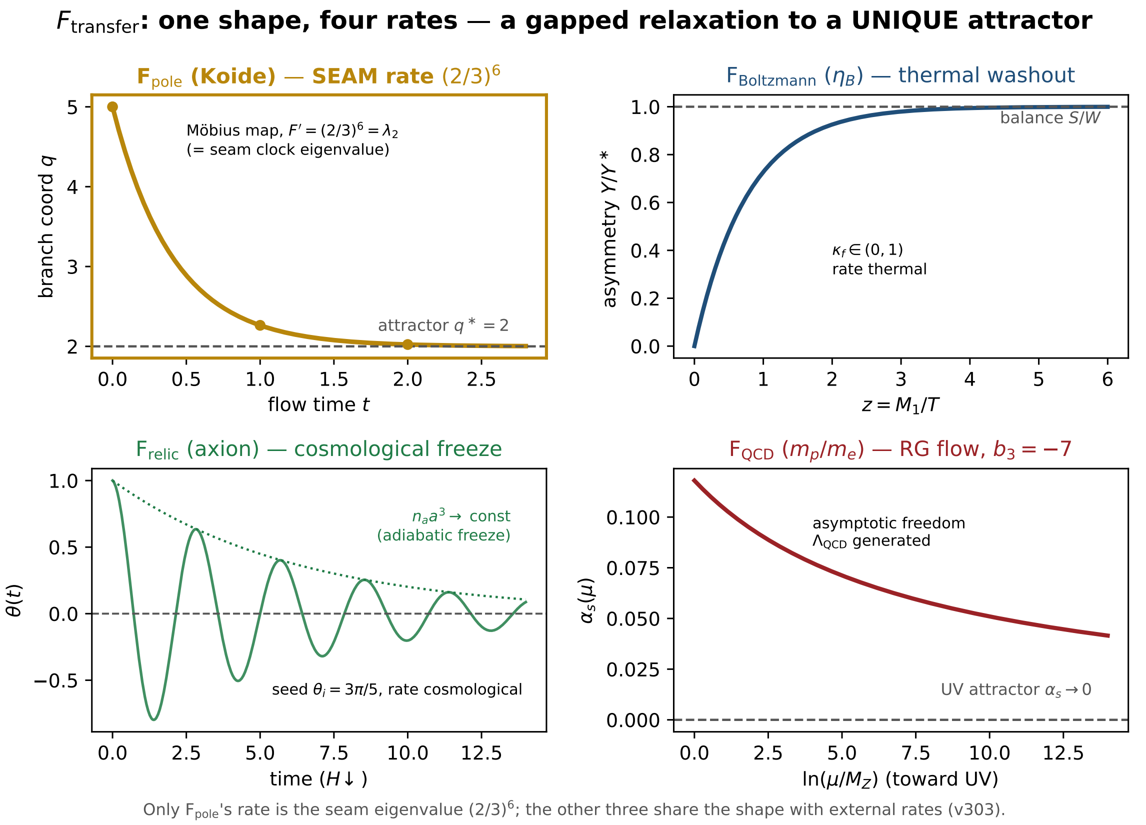

Figure 4: **The discrete→dynamic lens on `F_transfer` ([`verification/v303_ftransfer_dynamics.py`]).** All four transfer instances share *one* shape — a gapped, positivity-preserving relaxation to a *unique* attractor, the same Perron-Frobenius / H-theorem / RG-fixed-point structure as the main-branch update $m \leftarrow (A+2I)/4\,m$ → the E_8 marks. *Only* F_{pole} (Koide, gold) runs it at the *seam* rate: its Möbius multiplier is $F'=(2/3)^6=\lambda_2$, the seam-clock subleading eigenvalue. $F_{Boltzmann}$ (η_B, thermal washout), F_{relic} (axion, cosmological freeze) and F_{QCD} (m_p/m_e, the RG flow to the Gaussian UV fixed point with the carrier $b_3{=}-7$) share the *shape* with *external* rates. So `F_transfer` is the readout end of the one principle, not a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam.